
Curvature-Calibrated Exploration: Matrix-Free GGN Widths and Conditional Regret Bounds

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Abstract

Curvature surrogates in differentiable-model bandits change exploration widths, but greater fidelity to full curvature need not improve online decisions. We use the full generalized Gauss–Newton (GGN)/Fisher operator as a reference, computing its widths with curvature–vector products and conjugate gradients (CG) without materializing a $d \times d$ matrix. For Gaussian rewards and squared loss on a certified region with uniformly Lipschitz parameter gradients, a transfer theorem separates linearization, centering, feature and operator transfer, CG error, and realized width complexity. Predictable Welford and residual statistics instantiate its nonlinear certificates with $O(d)$ additional path state. When a fixed tangent subspace is supplied before interaction, the relevant widths reduce exactly to an r -dimensional projected Gram. Additional bounded-path, width, residual, optimizer, transfer, and CG premises then yield $O(r\sqrt{T}\log T) + O(T/\sqrt{W})$; this is a conditional projected-feature corollary, not a neural-network training guarantee. A growing-window theorem retains an explicit burn-in, fixed-subspace excitation, certificate factor H_T , and linearization term E_T ; its simplified $T^{1-q/2}$ bound and $T^{1+3q/2}$ width-solve sample-CVP count require those terms not to dominate. Experiments are a reference-operator stress test, not a performance demonstration. A linear-Gram witness, a 3,600-run scaling grid, a mechanism map, and a matched benchmark show no uniform fidelity-based regret ordering in the tested regimes; diagonal and last-layer policies often have lower regret.

1 Introduction

Contextual bandits are a standard framework for sequential decision-making under partial feedback, and increasingly the reward predictor is a large differentiable parametric model—a pretrained network for text, vision, or multimodal data composed with a reward head (Zhou et al., 2020; Riquelme et al., 2018). The central algorithmic question is how to build an exploration bonus that faithfully reflects epistemic uncertainty over such a model’s predictions. We analyze the Gaussian/squared-loss curvature model with sub-Gaussian noise and uniformly Lipschitz parameter gradients on a known certified region. The closed sublinear rate further requires a supplied low-rank tangent path; pretrained and foundation models motivate the operator, but unrestricted neural training is not covered.

In the *white-box* setting—gradient and curvature–vector-product access to the model—the canonical bonus is the GGN/Fisher-linearized predictive variance

$$s_t^2(a) = \mathbf{g}_t(a)^\top \mathbf{C}_t^{-1} \mathbf{g}_t(a), \quad (1)$$

where $\mathbf{g}_t(a) = \nabla_{\boldsymbol{\theta}} \mu_{\boldsymbol{\theta}_t}(\mathbf{x}_t, a)$ is the gradient feature (Jacobian of the mean reward) and $\mathbf{C}_t$ is the damped Gauss–Newton (GGN) curvature at the current estimate $\boldsymbol{\theta}_t$. Exact full-history GGN/Fisher widths are costly at large d or t : dense materialization uses $O(d^2)$ state, while matrix-free CG replays history at every iteration. NeuralUCB retains the full neural-gradient Gram in its theoretical algorithm but diagonalizes it in experiments (Zhou et al., 2020); neural-linear and scalable Laplace methods instead use last-layer or structured curvature approximations (Riquelme et al., 2018; Ritter et al., 2018; Daxberger et al., 2021). Outside online bandits, implicit Jacobian/Fisher products and CG have already been used to compute full-network linearized predictive variances without forming the Gram matrix (Maddox et al., 2021). Each curvature surrogate changes the *spectrum* of $\mathbf{C}_t$ and hence the value of $s_t^2(a)$. The spectral change does not by itself imply a change in the selected action, and therefore in regret; whether it does is an empirical question, which we make

measurable and study directly.

We use full curvature as a reference operator in a falsification study. Common-trajectory diagnostics isolate surrogate-induced width distortion, while independently executed policies test whether those distortions survive adaptive feedback. We do not attribute a uniform fidelity–regret ordering claim to prior work; instead we use it as a deliberately strong diagnostic null hypothesis. A controlled reversal rejects that hypothesis, whereas agreement does not establish superiority. Because executed policies follow different histories, their regret contrasts are descriptive rather than causal operator effects. We adapt the established matrix-free predictive-variance primitive to online contextual exploration: curvature–vector products (CVPs), available through automatic differentiation, let us solve $\mathbf{C}_t \mathbf{u} = \mathbf{g}_t(a)$ by conjugate gradients (CG) *without forming $\mathbf{C}_t$* : only $O(d)$ additional curvature-state vectors (a constant number of CG vectors) are maintained and no $d \times d$ curvature matrix or per-sample gradient vector is stored—though exact full-history curvature–vector products still require retaining or replaying the historical raw examples to recompute their Jacobian products, and model parameters, optimizer state, activations, replay data, and any frozen-feature parameter checkpoints are not counted in this $O(d)$ curvature state. Because the damped GGN is symmetric positive definite, CG converges in exact arithmetic. All CG convergence and residual-to-energy statements below use exact arithmetic unless a verified rounding-error enclosure is supplied. Building on this, we propose **Curvature-Calibrated UCB (CC-UCB)**, a bandit algorithm whose exploration bonus is the predictive variance under the *uncompressed* GGN/Fisher metric, accessed entirely through implicit CG solves.

The analysis tracks predictable certificate requirements for each approximation source and constructs them for the nonlinear path terms. The Gaussian/squared-loss theorem (Theorem 1) handles a general predictable SPD curvature sequence with a one-sided transfer factor, a per-round CG factor, and a realized width complexity Λ_T^C . Its exact determinant identity accounts for endpoint growth and determinant-destroying refreshes but does not itself bound either term. More importantly, the previously external nonlinear inputs are computable: Welford parameter moments and observed collection residuals upper-bound feature transfer and centering, while an analytic Taylor schedule supplies the confidence radius (Lemma 7 and Corollary 2). These schedules are conservative and need known smoothness constants and a certified optimizer residual. For a supplied fixed-rank tangent path satisfying the stated width-dependent smoothness, residual, and optimizer envelopes, we close information

gain and give an explicit $O(r\sqrt{T}\log T) + O(T/\sqrt{W})$ rate; a persistently excited growing window gives a conditional regret and width-solve sample-CVP accounting bound (Corollary 9 and Theorem 3). Neither result certifies unrestricted fine-tuning.

Contributions. (1) A realized-width regret reduction with one-sided feature/operator transfer and explicit centering, linearization, and CG terms (Theorem 1 and Lemma 11). (2) A conditional projected-Gram rate for a supplied tangent subspace and a separate growing-window bound with its burn-in and certificate terms exposed (Corollary 9 and Theorem 3). (3) Predictable $O(d)$ additional-state nonlinear transfer and centering certificates (Lemma 7 and Corollary 2). (4) An online adaptation of matrix-free full-GGN/Fisher widths via residual-checked CG/CVP solves, plus an off-diagonal linear-Gram witness relevant to the GGN metric (Proposition 26). (5) Executed nonlinear, mechanism, baseline, and systems audits that separate online outcomes from post-hoc diagnostics (Section 5).

2 Problem Setup

We present the problem setup for the *Gaussian/squared-loss* instantiation, which is the setting analyzed in Theorem 1. An extension to exponential-family likelihoods is discussed in Appendix H.

Protocol. At each round $t = 1, \dots, T$ the learner observes a context $\mathbf{x}_t \in \mathcal{X}$ and a finite action set $\mathcal{A}_t$. It selects $a_t \in \mathcal{A}_t$ and observes a stochastic reward

$$r_t = \mu^*(\mathbf{x}_t, a_t) + \eta_t, \quad \eta_t \text{ is } \sigma\text{-sub-Gaussian,} \quad (2)$$

where $\mu^* : \mathcal{X} \times \mathcal{A} \rightarrow \mathbb{R}$ is the unknown mean reward. The goal is to minimize cumulative regret $R_T = \sum_{t=1}^T [\mu^*(\mathbf{x}_t, a_t^*) - \mu^*(\mathbf{x}_t, a_t)]$, where $a_t^* = \arg \max_{a \in \mathcal{A}_t} \mu^*(\mathbf{x}_t, a)$. We assume $|\mathcal{A}_t| \leq A_{\max}$ deterministically for all t .

Reward model and parameterization convention.

We fix a pretrained initialization $\theta_0 \in \mathbb{R}^d$ and parameterize the model by the *displacement* $\theta \in \mathbb{R}^d$ from it: the deployed predictor is $\mu_{\theta_0+\theta}(\mathbf{x}, a)$, and we abbreviate $\mu_{\theta}(\mathbf{x}, a) := \mu_{\theta_0+\theta}(\mathbf{x}, a)$ throughout. The initial displacement is $\theta_1 = \mathbf{0}$ (the deployed model starts at the pretrained weights θ_0). With this convention the ridge penalty below is a penalty on the *displacement from the pretrained weights*, not a zero-centered prior on the raw weights, and the realizability radius S bounds $\|\theta^*\| = \|\theta_{\text{orig}}^* - \theta_0\|$, the distance of the true parameter from the pretrained initialization. Define

the regularized loss at round t :

$$\mathcal{L}_t(\boldsymbol{\theta}) := \frac{1}{2\sigma^2} \sum_{s=1}^{t-1} (r_s - \mu_{\boldsymbol{\theta}}(\mathbf{x}_s, a_s))^2 + \frac{\lambda}{2} \|\boldsymbol{\theta}\|^2. \quad (3)$$

We write $\boldsymbol{\theta}_t$ for the parameter returned by the warm-started optimizer on $\mathcal{L}_t$. Let $\mathcal{D}_{t-1} := \{(\mathbf{x}_s, a_s, r_s) : s < t\}$ and $N_t := |\mathcal{D}_{t-1}| = t - 1$. Equation (3) is a *full-history* objective. The curvature replay set $\mathcal{S}_t \subseteq \{1, \dots, t-1\}$ and its size $m_t := |\mathcal{S}_t|$ affect $\mathcal{C}_t$, but do not replace $\mathcal{D}_{t-1}$ in (3). Thus a generic full-batch evaluation of $\nabla \mathcal{L}_t$ replays N_t observations even when $m_t \ll N_t$. In convex settings (e.g. a frozen backbone with a linear head), under exact optimization this coincides with the global minimizer; in nonconvex settings $\boldsymbol{\theta}_t$ may be a local minimizer or approximate stationary point. The gap between $\boldsymbol{\theta}_t$ and an idealized global minimizer is absorbed by the action-scaled centering certificate (Assumption 4).

Gradient features and curvature matrices. Define the *gradient feature* (Jacobian of the mean reward) at the current parameter estimate:

$$\mathbf{g}_t(\mathbf{x}, a) := \nabla_{\boldsymbol{\theta}} \mu_{\boldsymbol{\theta}_t}(\mathbf{x}, a) \in \mathbb{R}^d. \quad (4)$$

The algorithm uses the *relinearized curvature matrix*

$$\mathbf{C}_t = \lambda \mathbf{I} + \frac{1}{\sigma^2} \sum_{s=1}^{t-1} \mathbf{g}_s(\mathbf{x}_s, a_s) \mathbf{g}_s(\mathbf{x}_s, a_s)^\top, \quad (5)$$

the *damped GGN curvature* for the Gaussian/squared-loss model at the current parameter $\boldsymbol{\theta}_t$. (For this likelihood, the GGN coincides with the true Fisher information; see the remark below.) While $\mathbf{C}_t$ is the natural object for computing predictive variances at $\boldsymbol{\theta}_t$, it re-evaluates all past Jacobians at $\boldsymbol{\theta}_t$ and therefore does *not* admit a sequential rank-one update—all summands change when $\boldsymbol{\theta}_t$ changes.

For the regret analysis we introduce the *frozen-feature Gram matrix*:

$$\bar{\mathbf{C}}_t := \lambda \mathbf{I} + \frac{1}{\sigma^2} \sum_{s=1}^{t-1} \mathbf{g}_s(\mathbf{x}_s, a_s) \mathbf{g}_s(\mathbf{x}_s, a_s)^\top, \quad (6)$$

where each $\mathbf{g}_s(\mathbf{x}_s, a_s) = \nabla_{\boldsymbol{\theta}} \mu_{\boldsymbol{\theta}_s}(\mathbf{x}_s, a_s)$ is evaluated at the parameter $\boldsymbol{\theta}_s$ that was current when the sample was collected. This matrix satisfies the recursive update

$$\bar{\mathbf{C}}_{t+1} = \bar{\mathbf{C}}_t + \sigma^{-2} \mathbf{g}_t(\mathbf{x}_t, a_t) \mathbf{g}_t(\mathbf{x}_t, a_t)^\top, \quad (7)$$

and, critically, each realized design vector $\mathbf{g}_s(\mathbf{x}_s, a_s)$ is measurable with respect to the pre-reward sigma-algebra $\mathcal{G}_s = \mathcal{H}_s \vee \sigma(a_s)$ of (69), which is the filtration used in the self-normalized bound. Both the additive update (7) and this measurability are structural requirements of the self-normalized martingale bound and the

elliptic potential lemma invoked in Section 4. The algorithm computes UCB widths using the algorithmic curvature $\mathcal{C}_t$ (via CG; the exact case is $\mathcal{C}_t = \mathbf{C}_t$), but the confidence analysis is conducted in terms of $\bar{\mathbf{C}}_t$; the one-sided optimism-transfer factor of Assumption 5, certified by Lemma 11, provides the bridge.

3 Method: Curvature-Calibrated Exploration

3.1 CC-UCB Algorithm

For a predictable SPD oracle $\mathcal{C}_t \succeq \lambda \mathbf{I}$, set $\omega_t = \bar{\beta}_t + \bar{\psi}_t$ for the original nonlinear center. Algorithm 1 is the analyzed full-enumeration policy. The same fixed curvature operator is reused across separate per-action CG solves; a solution for one action is not reused as the solution for another action.

Algorithm 1 Curvature-Calibrated UCB (CC-UCB)

- 1: Observe context $\mathbf{x}_t$ and finite action set $\mathcal{A}_t$.
 - 2: Construct a fixed predictable SPD oracle $\mathbf{v} \mapsto \mathcal{C}_t \mathbf{v}$ with $\mathcal{C}_t \succeq \lambda \mathbf{I}$.
 - 3: **for** each $a \in \mathcal{A}_t$ **do**
 - 4: Compute $\hat{\mu}_t(a) = \mu_{\boldsymbol{\theta}_t}(\mathbf{x}_t, a)$ and $\mathbf{g}_t(a) = \nabla_{\boldsymbol{\theta}} \mu_{\boldsymbol{\theta}_t}(\mathbf{x}_t, a)$.
 - 5: If $\mathbf{g}_t(a) \neq \mathbf{0}$, solve $\mathcal{C}_t \mathbf{u} = \mathbf{g}_t(a)$ separately by CG until the certified tolerance holds; otherwise set $\tilde{\mathbf{u}} = \mathbf{0}$.
 - 6: Set $\tilde{s}_t^2(a) = \mathbf{g}_t(a)^\top \tilde{\mathbf{u}}$.
 - 7: **end for**
 - 8: Select and play $a_t \in \arg \max_{a \in \mathcal{A}_t} \{\hat{\mu}_t(a) + \omega_t \sqrt{u_t(a)/(1 - \bar{\varepsilon}_t)} \sqrt{\tilde{s}_t^2(a)}\}$.
 - 9: Observe r_t , append $(\mathbf{x}_t, a_t, r_t)$, and update $\boldsymbol{\theta}_{t+1}$.
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The oracle is accessed only through matrix–vector products. At each round the agent computes, for every candidate action a , the gradient feature $\mathbf{g}_t(a) = \nabla_{\boldsymbol{\theta}} \mu_{\boldsymbol{\theta}_t}(\mathbf{x}_t, a)$ and the *curvature-calibrated variance proxy*, the generic algorithmic width,

$$s_{\mathcal{C}_t}^2(a) = \mathbf{g}_t(a)^\top \mathcal{C}_t^{-1} \mathbf{g}_t(a). \quad (8)$$

Under the GGN/Fisher-linearized Gaussian approximation with covariance $\mathcal{C}_t^{-1}$, the quantity $s_{\mathcal{C}_t}^2(a)$ is the metric-dependent predictive variance of the linearized reward at $(\mathbf{x}_t, a)$. The *exact* full-history instantiation of CC-UCB takes $\mathcal{C}_t = \mathbf{C}_t$, the relinearized curvature (5); weighted replay, subsampling, sliding windows, periodic refreshes, and randomized sketches are special implementations of $\mathcal{C}_t$ (§A.3), covered by the theory only when the corresponding one-sided transfer factor is certified.

Because $\mathcal{C}_t \succ 0$ by construction, conjugate gradients (CG) is guaranteed to converge on the system $\mathcal{C}_t \mathbf{u} =$

$\mathbf{g}_t(a)$, provided the operator is held *fixed* across all iterations of the solve (the randomized sketch/subsample is drawn into Ω_t before action selection; §2). For each action with $\mathbf{g}_t(a) \neq \mathbf{0}$ we run CG to the per-action relative energy-norm error

$$\varepsilon_{t,a} := \frac{\|\mathbf{u}_{t,a} - \tilde{\mathbf{u}}_{t,a}\|_{\mathcal{C}_t}}{\|\mathbf{u}_{t,a}\|_{\mathcal{C}_t}}, \quad \mathbf{u}_{t,a} := \mathcal{C}_t^{-1} \mathbf{g}_t(a), \quad (9)$$

yielding the computable proxy $\tilde{s}_t^2(a) = \mathbf{g}_t(a)^\top \tilde{\mathbf{u}}_{t,a}$; if $\mathbf{g}_t(a) = \mathbf{0}$ we set $\tilde{s}_t^2(a) = 0$ and skip the solve (leaving the residual ratio $\|r_I\|/\|\mathbf{g}\|$ undefined), and by convention put $\varepsilon_{t,a} := 0$ in this case, so that $\varepsilon_{t,a}$ is defined for every $a \in \mathcal{A}_t$ and (58) holds trivially there. Lemma 17 shows $\tilde{s}_t^2(a)$ approximates $s_{\mathcal{C},t}^2(a)$ within a multiplicative $(1 \pm \varepsilon_{t,a})$ factor. To ensure the UCB bonus *upper-bounds* the confidence width $\omega_t \tilde{s}_t(a)$ despite the approximation, the algorithm inflates the width by $\sqrt{u_t(a)/(1 - \bar{\varepsilon}_t)}$ (line 8), where $\bar{\varepsilon}_t \in (0, 1)$ is a *per-round* known tolerance enforced uniformly over all actions,

$$\max_{a \in \mathcal{A}_t} \varepsilon_{t,a} \leq \bar{\varepsilon}_t < 1, \quad (10)$$

and $u_t(a) \geq 1$ is the one-sided transfer factor (Assumption 5).

4 Theoretical Analysis

We state a regret bound for CC-UCB (Algorithm 1, with full action enumeration $K=|\mathcal{A}_t|$) under the Gaussian/squared-loss setup of Section 2 and Assumptions 1 to 5. The analysis does not constitute a full exponential-family or generalized linear bandit proof; the extension in Appendix H is a curvature-definition extension only. The analysis handles a *general predictable SPD algorithmic curvature sequence* $\mathcal{C}_t$ (38). Confidence is proved for the frozen-feature Gram $\bar{\mathbf{C}}_t$ (Eq. 6), which has the sequential-update and predictability properties required by the self-normalized martingale bound; the *one-sided* optimism-transfer factor $u_t(a)$ (Assumption 5) bridges from $\bar{\mathbf{C}}_t$ to $\mathcal{C}_t$ in the exploration direction only (no lower drift sandwich); the CG approximation enters through the per-round factor α_t ; and the nonmonotonicity of $\mathcal{C}_t$ is retained in the realized width complexity. Appendix Appendix A.5 gives an exact determinant bookkeeping identity, not a primitive sublinear bound.

The theorem is a modular transfer result. The path statistics below instantiate its nonlinear inputs from pre-action quantities; smallness still requires a restricted parameter path and is not claimed for unrestricted fine-tuning.

Core conditions and links. Assume conditionally σ -sub-Gaussian noise, $\|\mathbf{g}_t(\mathbf{x}_s, a)\| \leq G$, realizability

by $\|\boldsymbol{\theta}^*\| \leq S$, and a local expansion with round- t remainder at most $\varepsilon_{\text{lin}}(t)$. For the original center assume the action-scaled discrepancy is at most $\bar{\psi}_t \bar{s}_t(a)$. The operator $\mathcal{C}_t$ is predictable, SPD, satisfies $\mathcal{C}_t \succeq \lambda \mathbf{I}$ and $\mathcal{C}_1 = \lambda \mathbf{I}$, and remains fixed throughout every CG solve. Define

$$\begin{aligned} \bar{s}_t^2(a) &= \mathbf{g}_t(a)^\top \bar{\mathbf{C}}_t^{-1} \mathbf{g}_t(a), & s_{\mathcal{C},t}^2(a) &= \mathbf{g}_t(a)^\top \mathcal{C}_t^{-1} \mathbf{g}_t(a), \\ \tilde{s}_t^2(a) &= \mathbf{g}_t(a)^\top \tilde{\mathbf{u}}_t(a). \end{aligned} \quad (11)$$

The analysis requires only the one-sided transfer and uniform CG relations

$$\begin{aligned} \bar{s}_t^2(a) &\leq u_t(a) s_{\mathcal{C},t}^2(a), \\ (1 - \bar{\varepsilon}_t) s_{\mathcal{C},t}^2(a) &\leq \tilde{s}_t^2(a) \leq (1 + \bar{\varepsilon}_t) s_{\mathcal{C},t}^2(a), \end{aligned} \quad (12)$$

for every candidate action, where $u_t(a) \geq 1$, $\bar{\varepsilon}_t < 1$, and $\alpha_t = \sqrt{(1 + \bar{\varepsilon}_t)/(1 - \bar{\varepsilon}_t)}$. Sufficient feature-drift and approximate-operator constructions are given in Lemma 11 and Theorem 24; no lower spectral transfer is needed.

Dynamic width complexity. The realized quantity used below is

$$\Lambda_T^{\mathcal{C}} := \sum_{t=1}^T \log(1 + \sigma^{-2} s_{\mathcal{C},t}^2(a_t)). \quad (13)$$

The bounded-width cap gives

$$\sum_{t=1}^T s_{\mathcal{C},t}^2(a_t) \leq (\sigma^2 + G^2/\lambda) \Lambda_T^{\mathcal{C}}. \quad (14)$$

For a general predictable nonmonotone SPD sequence, we do not derive a primitive sublinear bound on $\Lambda_T^{\mathcal{C}}$. Appendix Appendix A.5 gives an exact determinant bookkeeping identity, but its upper bound is informative only when both the terminal log determinant and determinant-decreasing refresh charge are independently controlled, for example under the rank and spectral-floor hypotheses of Lemma 15.

4.1 Regret Bound

Let $\mathcal{E}_{\text{conf}}$ denote the simultaneous self-normalized confidence event and let $\mathcal{E}_{\text{cert}}$ collect the predictable linearization, centering, transfer, condition-number, CG, and randomized-operator certificates. The appendix defines each component and its failure allocation; write their total failure probability as δ_{cert} .

Theorem 1 (Dynamic transfer regret bound for CC-UCB (Gaussian/squared-loss)). *Under the conditions above and Assumptions 1 to 5, suppose $|\mathcal{A}_t| \leq A_{\text{max}}$, the optimal action is present, and Algorithm 1 enumerates all actions using a predictable schedule $\bar{\beta}_t \geq \beta_t$ and uniform per-round CG tolerance. Let $E_T =$*

$\sum_{t=1}^T \varepsilon_{\text{lin}}(t)$, $\omega_t = \bar{\beta}_t + \bar{\psi}_t$, $\alpha_t = \sqrt{(1 + \bar{\varepsilon}_t)/(1 - \bar{\varepsilon}_t)}$, and $S_T = \sum_{t=1}^T \alpha_t^2 u_t(a_t) \omega_t^2$. On $\mathcal{E}_{\text{conf}} \cap \mathcal{E}_{\text{cert}}$,

$$R_T \leq 2\sqrt{(\sigma^2 + G^2/\lambda) \Lambda_T^C S_T} + 2E_T, \quad (15)$$

with probability at least $1 - \delta - \delta_{\text{cert}}$. Deterministic or almost-sure certificates contribute zero to δ_{cert} . For the corrected center, the same statement holds without Assumption 4 after replacing ω_t by $\bar{\beta}_t$.

In particular, the bound is sublinear whenever $\Lambda_T^C S_T = o(T^2)$ and $E_T = o(T)$.

The exact-curvature instantiation is $\mathcal{C}_t = \mathbf{C}_t$ with $u_t = (1 + \bar{\chi}_t)^2$ from Lemma 11; approximate operators are covered through Theorem 24. The proof is in Appendix D.

Proof sketch. Self-normalized confidence plus the centering and linearization terms gives $|\mu^*(a) - \mu_{\theta_t}(a)| \leq \omega_t \bar{s}_t(a) + \varepsilon_{\text{lin}}(t)$. The one-sided relation and CG lower bound make the computed bonus optimistic, so instantaneous regret is at most $2w_t(a_t) + 2\varepsilon_{\text{lin}}(t)$. The CG upper bound gives $w_t(a_t) \leq \alpha_t \omega_t \sqrt{u_t(a_t) s_{C,t}(a_t)}$. Summing, applying Cauchy–Schwarz only once, and using (14) yields (15); hence time-varying factors remain inside S_T rather than being replaced by horizon maxima.

The corrected center replaces ω_t by $\bar{\beta}_t$ and removes the centering certificate, but requires exact or certified frozen-feature access; the formal statement and implementation discussion are in Corollary 5.

4.2 Online nonlinear certificates

At the start of round t , let $n_t = t - 1$, and let $\bar{\theta}_{< t}$ and $J_{< t} = \sum_{s < t} \|\theta_s - \bar{\theta}_{< t}\|^2$ be maintained by a vector Welford update. With collection residuals $c_s = \mu_{\theta_s}(x_s, a_s) - r_s$, define

$$Q_t = J_{< t} + n_t \|\theta_t - \bar{\theta}_{< t}\|^2, \quad R_t^{\text{col}} = \sum_{s < t} c_s^2. \quad (16)$$

Both are pre-action quantities and $Q_t = \sum_{s < t} \|\theta_t - \theta_s\|^2$ exactly. If prediction gradients are L_g -Lipschitz on the certified region, set

$$\bar{\chi}_t = \frac{L_g \sqrt{Q_t}}{\sigma \sqrt{\lambda}}, \quad u_t = \kappa_{+,t} (1 + \bar{\chi}_t)^2, \quad (17)$$

where $\kappa_{+,t} = 1$ for exact full curvature and for an unrescaled current-parameter subset, and otherwise is a supplied one-sided operator certificate. If $\|\nabla \mathcal{L}_t(\theta_t)\| \leq$

ζ_t , an observable centering schedule is

$$\begin{aligned} \bar{\psi}_t &= \frac{\zeta_t + \bar{M}_t}{\sqrt{\lambda}}, \\ \bar{M}_t &= \sigma^{-2} \left[L_g \sqrt{R_t^{\text{col}}} Q_t + \frac{3}{2} G L_g Q_t + \frac{1}{2} L_g^2 Q_t^3 \right]. \end{aligned} \quad (18)$$

For a known radius- R trust region, the final numerator term improves to $L_g^2 R Q_t$. Finally, for an L_μ -Lipschitz prediction gradient define $\bar{e}_t^{\text{lin}} = \frac{L_\mu}{2} (S + \|\theta_t\|)^2$, $\bar{F}_t = \sum_{s < t} (\bar{e}_s^{\text{lin}})^2$, and

$$\begin{aligned} \bar{\beta}_t &= \sqrt{\hat{\gamma}_{t-1} + 2 \log(1/\delta)} + \sqrt{\lambda} S + \sigma^{-1} \sqrt{\bar{F}_t}, \\ \hat{\gamma}_{t-1} &= \sum_{s < t} \log(1 + q_s), \quad q_s = \frac{u_s(a_s) \bar{s}_s^2(a_s)}{\sigma^2 (1 - \bar{\varepsilon}_s)}. \end{aligned} \quad (19)$$

Corollary 2 (Operational CC-UCB). *Under the structural hypotheses of Theorem 1, with its certificate inputs instantiated by valid smoothness constants, the optimizer residual bound ζ_t , uniform CG certificates, and a valid $\kappa_{+,t}$, (16)–(19) are predictable and satisfy every nonlinear input of the theorem. Thus Theorem 1 holds with E_T replaced by $\bar{E}_T = \sum_{t \leq T} \bar{e}_t^{\text{lin}}$ and the observable CG upper bound on Λ_T^C from Corollary 16.*

Proof. Lemma 7 proves $\bar{\chi}_t \geq \chi_t$, $\bar{\psi}_t \geq \psi_t$, $\bar{F}_t \geq F_t$, and $\bar{E}_T \geq E_T$. The one-sided operator relation and Corollary 21 give u_t and $\hat{\gamma}_{t-1} \geq \gamma_{t-1}$, hence $\bar{\beta}_t \geq \beta_t$. All quantities use the summary through round $t - 1$ and the already formed θ_t ; uniform candidate-action CG accuracy is checked before selection. Substitution in Theorem 1 and Corollary 16 proves the claim. $\square$

The proof, filtration-order pseudocode, and a scalar-tanh specialization with $G \leq B_\phi$ and $L_\mu = L_g \leq 4B_\phi^2/(3\sqrt{3})$ are in Lemma 7 and Corollary 10.

For a width parameter W , suppose $\|\theta_t\|, \|\theta^*\| \leq R$, $L_\mu \leq c_\mu/\sqrt{W}$, $L_g \leq c_g/\sqrt{W}$, $\zeta_t \leq \zeta_0/\sqrt{W}$, and $R_t^{\text{col}} \leq C_R t P_T$ for a supplied $P_T \geq 1$. Then

$$\begin{aligned} \bar{E}_T &\leq \frac{2c_\mu R^2 T}{\sqrt{W}}, \quad \bar{F}_{T+1} \leq \frac{4c_\mu^2 R^4 T}{W}, \\ \bar{\chi}_t &\leq \frac{2c_g R \sqrt{t-1}}{\sigma \sqrt{\lambda W}}. \end{aligned} \quad (20)$$

and Corollary 8 gives an explicit trust-region bound on $\bar{\psi}_t$. If a supplied fixed active subspace gives rank bound $r \geq 1$, define

$$\Gamma_{T,r} = r_T \log \left(1 + \frac{T G^2}{r_T \lambda \sigma^2} \right), \quad r_T = \min\{r, T\}. \quad (21)$$

and set $\Gamma_{0,r} := 0$ for the round-one confidence schedule. The bound r must be known before selection; a terminal realized rank is only post-hoc. Then $\gamma_T \leq \Gamma_{T,r}$ and

Corollary 9 gives the fully explicit conditional projected-Gram bound

$$R_T \leq 2\alpha_I \sqrt{\rho_+/\rho_-} \Omega_{T,r} \sqrt{(\sigma^2 + G^2/\lambda)T\Gamma_{T,r}} + 2\bar{E}_T, \quad (22)$$

where $x_T = 2c_g R \sqrt{T-1}/(\sigma\sqrt{\lambda W})$, $\rho_{\pm} = 1 \pm (2x_T + x_T^2)$, and $\Omega_{T,r}$ is displayed with every $T, W, r, P_T, \lambda, \sigma$ dependence in the appendix. At fixed r and problem constants, $W = \Omega(T^2 P_T)$ makes the path inflation bounded and gives $R_T = O(r\sqrt{T} \log T) + O(T/\sqrt{W})$. This is a low-rank tangent-space or parameter-efficient adaptation regime, not generic full-network training.

Theorem 3 (Conditional growing-window bound under fixed-subspace excitation). *Adopt the hypotheses, events, and valid predictable schedules of Theorem 1, let $m_t = \min\{t-1, \lceil t^q \rceil\}$ for $q \in (0, 1]$, and let $\mathcal{C}_t$ be the unrescaled current-parameter Gram over the last m_t observations. Thus Proposition 23 gives $\kappa_{+,t} = 1$ and $u_t = (1 + \bar{\chi}_t)^2$. Suppose candidate gradients lie in a fixed subspace U , and let P_U be its orthogonal projector. For a supplied burn-in index $\tau_w \in \{1, \dots, T+1\}$, assume the predictable window-excitation condition*

$$\mathcal{C}_t \succeq \lambda \mathbf{I} + \kappa_w m_t P_U$$

holds for a known $\kappa_w > 0$ and every $t \geq \tau_w$. Then, on the same event of probability at least $1 - \delta - \delta_{\text{cert}}$ as Theorem 1,

$$\begin{aligned} \Lambda_T^{\mathcal{C}} &\leq L_q(T; \tau_w), \\ L_q(T; \tau_w) &:= (\tau_w - 1) \log \left(1 + \frac{G^2}{\lambda \sigma^2} \right) \\ &\quad + \frac{G^2}{\sigma^2} \sum_{t=\tau_w}^T \frac{1}{\lambda + \kappa_w m_t}, \quad (23) \\ L_q(T; \tau_w) &= \begin{cases} O(\tau_w + T^{1-q}), & q < 1, \\ O(\tau_w + \log T), & q = 1. \end{cases} \end{aligned}$$

For $\tau_w = 1$, more explicitly, $L_q(T; 1) \leq (G^2/\sigma^2)[\lambda^{-1} + \kappa_w^{-1} h_q(T)]$, where, for $T \geq 2$, $h_q(T) = 1 + ((T-1)^{1-q} - 1)/(1-q)$ for $q < 1$ and $h_1(T) = 1 + \log(T-1)$, while $h_q(1) := 0$ in both cases. Writing $r_U = \dim U$, the excitation premise can first hold nontrivially only once $m_t \geq r_U$, because $\text{rank}(\mathcal{C}_t - \lambda \mathbf{I}) \leq m_t$; thus τ_w includes the required rank- r_U burn-in. If $H_T = \max_{t \leq T} \alpha_t \sqrt{u_t(a_t)} \omega_t$, then

$$R_T \leq 2H_T \sqrt{(\sigma^2 + G^2/\lambda)TL_q(T; \tau_w)} + 2E_T. \quad (24)$$

If $H_T = \tilde{O}(1)$, $E_T = \tilde{O}(T^{1-q/2})$, and the burn-in obeys $\tau_w = \tilde{O}(T^{1-q})$ for $q < 1$ (or $\tau_w = \tilde{O}(1)$ for $q = 1$), then (24) simplifies to $R_T = \tilde{O}(T^{1-q/2})$; otherwise its displayed L_q and E_T terms remain. This is conditional on the fixed subspace, excitation, and all confidence,

transfer, optimizer, and CG certificates above. Certified worst-case width-solve work is $O(KT^{1+3q/2} \log(1/\bar{\varepsilon}))$ sample-CVPs when a fixed uniform target $\bar{\varepsilon} \in (0, 1)$ is enforced; this count excludes model optimization, candidate-gradient computation, and certificate construction.

Proof. Before τ_w , $\mathcal{C}_t \succeq \lambda \mathbf{I}$ bounds each logarithm in $\Lambda_T^{\mathcal{C}}$ by $\log(1 + G^2/(\lambda \sigma^2))$. Thereafter, for $g_t(a) \in U$, inversion of the excitation inequality gives $s_{\mathcal{C},t}^2(a) \leq G^2/(\lambda + \kappa_w m_t)$; summing $\log(1+x) \leq x$ proves (23). Since $S_T \leq TH_T^2$, substitution in Theorem 1 proves (24). The condition-number bound $\kappa(\mathcal{C}_t) \leq 1 + m_t G^2/(\lambda \sigma^2)$ makes CG use $O(\sqrt{m_t} \log(1/\bar{\varepsilon}))$ iterations, each replaying m_t samples; summing $Km_t^{3/2}$ proves the cost claim. $\square$

Under the hypotheses of Corollary 9, if the window policy uses the predictable rank envelope $\Gamma_{t-1,r}$ instead of the realized $\hat{\gamma}_{t-1}$ in its confidence radius, then $H_T \leq \alpha_I(1 + x_T)\Omega_{T,r}^{\text{win}}$, where $\Omega_{T,r}^{\text{win}}$ is the explicit $\Omega_{T,r}$ in Corollary 9 with $A_{\text{inf}} = 1$. Hence (24) contains no uncontrolled information-gain placeholder. The excitation premise is restrictive and standard in persistently excited identification; contextual diversity alone does not imply it. For $\dim U = r > 1$ one must choose τ_w after a rank- r burn-in, as the formal early charge records. The predictable rank schedule in Corollary 9 makes H_T polylogarithmic under the same bounded-path conditions. Illustrative achieved upper-bound exponents are reported in Appendix A.4; they are not a Pareto frontier or a lower bound. When U is supplied, the projected Gram in (36) can instead be solved directly.

5 Experiments

All regret rows are independently executed policies. Common random numbers are used only where pairing is valid. Exact teacher and dense replay quantities are post-action audits; common-trajectory operator comparisons never report causal regret. Tuning and evaluation seeds are disjoint. The derived full-grid aggregate retains seed-level checkpoints and complete hash provenance; release scope and missing raw tiers are stated in Appendix J.

The grid contains three ambient dimensions, three active ranks, eight methods, and 50 fixed seeds. For the column-orthonormal embedding E_d ,

$$(E_d g)^\top (\lambda I_d + E_d A_t E_d^\top)^{-1} (E_d g) = g^\top (\lambda I_r + A_t)^{-1} g.$$

Thus non-diagonal widths are ambient-invariant in exact arithmetic. Empirically, the maximum cross- d regret range is zero; the dynamic-width range is 5.7×10^{-14} for non-diagonal methods and 2.2501×10^3 for the

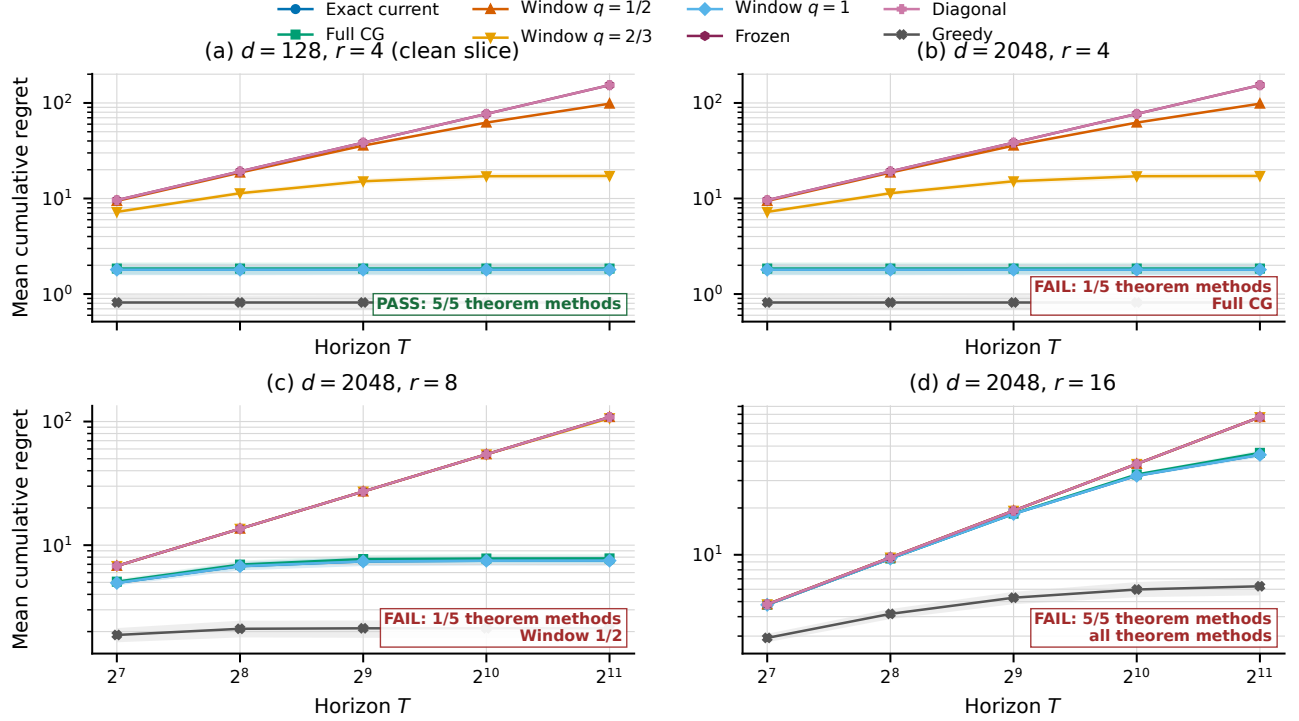


Figure 1: Finite-horizon theorem-scaling diagnostic. The first panel is the event-clean $d = 128, r = 4$ reference; the remaining panels use $d = 2048$. Curves are 50-seed means and bands are pointwise percentile-bootstrap 95% intervals at five preregistered prefixes. Panel labels report the conjunction of float64 theorem-event fields for exact, CG, and the three window methods. Greedy is uncertified. No exponent comparison tests Theorem 3 unless every premise and audit field for that method/rank cell passes.

embedding-dependent diagonal method. Ambient-CVP runtime is excluded. Every scaling method uses the same full-history optimizer from (3); m_t changes only the curvature operator. The implementation also constructs full-history dense audit objects for window rows, so its runtime is not a measurement of the window-only cost model.

The policy predeclares $\zeta_t = 10^{-6}$ and stops after at most 40 optimizer iterations; it does not enforce that envelope. A failed field therefore means the executed radius does not instantiate the theorem on that round, so the path is an uncertified diagnostic. Five-point slopes and fit residuals are in Table 4.

Full CG is a one-step equivalence test: each right-hand side is an active coordinate and the Gram is diagonal in that basis. Width errors are about 10^{-15} , yet near-tie perturbations can change an action and the subsequent adaptive trajectory. The rotated-SPD audit in Appendix C, not this scaling row, tests nontrivial Krylov convergence.

5.1 Predictable nonlinear execution

We first use the genuinely nonlinear Gaussian model $\mu_\theta(x, a) = \tanh(\phi(x, a)^\top \theta)$ with $d = 6$, four bounded actions, $T = 100$, $\|\theta^*\| \leq 0.35$, and a radius-0.5 projected optimizer. The analytic constants in Corollary 10, exact full-objective gradient norm, the path statistics in (16), and the analytic condition bound $1 + (t - 1)G^2/(\lambda\sigma^2)$ are computed before action selection. No teacher, future reward, exact χ_t , exact ψ_t , or exact Taylor error enters the policy. A fixed protocol is evaluated on 50 seeds (160–209); an eight-cell one-factor sensitivity grid is confined to ten tuning seeds.

Every original- and corrected-center trajectory passes every recorded transfer, centering, linearization, information, CG, confidence, optimism, and regret event: zero observed failures over 5,000 rounds per center. Mean regret is 7.86 [7.66, 8.07] for the original center and 5.69 [5.46, 5.92] for the corrected center. The schedules are deliberately conservative: at the final round, original χ_t averages 0.0216 versus $\bar{\chi}_t = 4.89$, and ψ_t averages 0.0520 versus $\bar{\psi}_t = 128.1$. The mean observable RHS/regret ratios are 6,956.7 and 1,328.9,

Cell (d/r)	Exact	CG	$q = 1/2$	$q = 2/3$	$q = 1$	Frozen	Diag.	Greedy
128/4	1.80	1.85	98.30	17.23	1.80	153.60	153.60	0.82
2048/4	1.80	1.85	98.30	17.23	1.80	153.60	153.60	0.82
2048/8	7.50	7.82	108.61	106.44	7.50	108.61	108.61	2.13
2048/16	43.95	45.24	76.80	76.80	43.95	76.80	76.80	6.27

Theorem-event status and failed round-event categories				
Method	$d = 128, r = 4$	$d = 2048, r = 4$	$d = 2048, r = 8$	$d = 2048, r = 16$
Exact current	PASS	PASS	PASS	FAIL [$O=49, P_e=46, P_\lambda=45$]
Full CG	PASS	FAIL [$O=1$]	PASS	FAIL [$O=49, P_e=46, P_\lambda=45$]
Window $q = 1/2$	PASS	PASS	FAIL [$O=1$]	FAIL [$O=49, P_e=46, P_\lambda=45$]
Window $q = 2/3$	PASS	PASS	PASS	FAIL [$O=49, P_e=46, P_\lambda=45$]
Window $q = 1$	PASS	PASS	PASS	FAIL [$O=49, P_e=46, P_\lambda=45$]

PASS means that all recorded float64 theorem-event fields passed for that method and cell.
Failure counts are round-event fields: O is optimizer residual, P_e is ψ -excitation, and P_λ is ψ -lambda.

Table 1: Preregistered scaling summary. Panel (a) reports terminal mean regret for every executed method. Panel (b) decomposes failed float64 theorem-event fields by method: O is the prescribed optimizer-residual envelope, P_e is the path bound using the excitation floor, and P_λ is the path bound using damping. Pass means zero observed failures for the displayed fields, not a verified numerical enclosure. Any nonzero entry precludes interpreting that cell’s slope as a test of the corresponding conditional theorem.

respectively, so the bounds are numerically vacuous. Because the residual and dense audit checks are ordinary float64 point computations, these executions are classified *post-hoc theorem-event verified with fully predictable schedules*, not verified-enclosure theorem certified.

5.2 Reference-operator falsification and matched baselines

An eight-cell phase map runs seven operators on 30 independent policies per cell and replays the exact-full path for detailed width/rank diagnostics. Full CG matches dense full regret; diagonal has lower regret in all eight cells, block in six, while full beats window and stale refresh in all eight and rank-3 Lanczos in seven. Thus fidelity does not order regret. The separate linear-Gram witness changes the diagonal policy’s action, but is existential rather than a dominance result (Figure 3 and table 6).

The balanced nonlinear benchmark uses separate tuning and 30 evaluation seeds; local NeuralUCB is the historical-time full-gradient-Gram comparison, alongside NeuralTS, NeuralLinear, last-layer, diagonal, linear, greedy, and context-free baselines. These are local matched implementations, not verified reproductions. At $T = 200$, CC-UCB full GGN-CG has regret 21.94 [20.64, 23.23]; diagonal has 16.11 [14.62, 17.61], LinUCB 10.58 [9.82, 11.33], NeuralLinear 10.01 [9.33, 10.68], and frozen last-layer UCB 7.44 [6.90, 7.99]. Context-free controls have regret 90.35 and 89.40; the sanity check passes, but the experiment is negative evidence for full-curvature superiority.

5.3 Audits and systems scope

All 280 bounded linear traces pass float64 checks, but lack verified enclosures; their bounds are vacuous. Synthetic CPU operators reach $d = 8192$, but use unmatched stored-feature primitives. Small-model policies likewise materialize Jacobians for dense audits, validating algebra rather than on-the-fly $O(d)$ Krylov state. The defining autodiff JVP/VJP benchmark was not run because its Buck PyTorch dependency does not configure; no method-speed, trained-model, or accelerator claim follows.

Real-data baseline check. Covertypes fails its baseline check: context-free controls beat every contextual method at every horizon (Appendix C.1).

6 Conclusion

CC-UCB supplies a matrix-free full-GGN/Fisher reference with explicit transfer, CG, and width terms. Its only closed rate is a conditional projected-Gram corollary for a supplied subspace, where full-dimensional CG is unnecessary; it is not an unrestricted neural-training result. Empirically, the operator tests width and action-order preservation: the tanh bound is vacuous, the scaling grid exposes failed envelopes, and full curvature is not uniformly preferred. Covertypes fails its baseline check.

Limitations. Theory assumes Gaussian squared loss, enumeration, realizability, smoothness, and an optimizer certificate; approximate operators and screening need further certificates. Float64 checks lack enclosures, bounds are vacuous, autodiff timing is absent, and adaptive comparisons are non-causal.

References

- Yasin Abbasi-Yadkori, Dávid Pál, and Csaba Szepesvári. Improved algorithms for linear stochastic bandits. *Advances in Neural Information Processing Systems*, 24, 2011.
- Shipra Agrawal and Navin Goyal. Thompson sampling for contextual bandits with linear payoffs. In *International Conference on Machine Learning (ICML)*, pages 127–135, 2013.
- Jock A. Blackard. Covertypes [dataset]. UCI Machine Learning Repository, 1998. DOI: 10.24432/C50K5N. Distributed under CC BY 4.0.
- Ha Manh Bui, Enrique Mallada, and Anqi Liu. Variance-aware linear UCB with deep representation for neural contextual bandits. In *Proceedings of the 28th International Conference on Artificial Intelligence and Statistics*, volume 258 of *Proceedings of Machine Learning Research*, pages 4213–4221. PMLR, 2025. URL <https://proceedings.mlr.press/v258/bui25a.html>.
- Peter G. Chang, Gerardo Durán-Martín, Alexander Y. Shestopaloff, Matt Jones, and Kevin Murphy. Low-rank extended Kalman filtering for online learning of neural networks from streaming data. In *Proceedings of the 2nd Conference on Lifelong Learning Agents*, volume 232 of *Proceedings of Machine Learning Research*, 2023. URL <https://proceedings.mlr.press/v232/chang23a.html>.
- Erik Daxberger, Agustinus Kristiadi, Alexander Immer, Runa Eschenhagen, Matthias Bauer, and Philipp Hennig. Laplace Redux—effortless Bayesian deep learning. In *Advances in Neural Information Processing Systems*, volume 34, pages 20089–20103, 2021.
- Rohan Deb, Yikun Ban, Shiliang Zuo, Jingrui He, and Arindam Banerjee. Contextual bandits with online neural regression. In *The Twelfth International Conference on Learning Representations*. OpenReview.net, 2024. URL <https://openreview.net/forum?id=5ep85sakT3>.
- Gerardo Durán-Martín, Aleya Kara, and Kevin Murphy. Efficient online Bayesian inference for neural bandits. In *International Conference on Artificial Intelligence and Statistics (AISTATS)*, pages 6002–6021, 2022.
- Avishek Ghosh, Sayak Ray Chowdhury, and Aditya Gopalan. Misspecified linear bandits. In *AAAI Conference on Artificial Intelligence*, pages 3761–3767, 2017.
- Diego Granziol and Khurshid Juarev. Hessian spectral analysis at foundation model scale. *arXiv preprint arXiv:2602.00816*, 2026.
- David Gross and Vincent Nesme. Note on sampling without replacing from a finite collection of matrices. *arXiv preprint arXiv:1001.2738*, 2010.
- Wassily Hoeffding. Probability inequalities for sums of bounded random variables. *Journal of the American Statistical Association*, 58(301):13–30, 1963.
- Parnian Kassraie and Andreas Krause. Neural contextual bandits without regret. In *International Conference on Artificial Intelligence and Statistics (AISTATS)*, pages 240–278, 2022.
- Pang Wei Koh and Percy Liang. Understanding black-box predictions via influence functions. In *International Conference on Machine Learning (ICML)*, pages 1885–1894, 2017.
- Agustinus Kristiadi, Matthias Hein, and Philipp Hennig. Being Bayesian, even just a bit, fixes overconfidence in ReLU networks. In *International Conference on Machine Learning (ICML)*, pages 5436–5446, 2020.
- Ilya Kuzborskij, Leonardo Cella, and Nicolò Cesa-Bianchi. Efficient linear bandits through matrix sketching. In *International Conference on Artificial Intelligence and Statistics (AISTATS)*, pages 177–185, 2019.
- Tor Lattimore, Csaba Szepesvári, and Gellért Weisz. Learning with good feature representations in bandits and in RL with a generative model. In *International Conference on Machine Learning (ICML)*, pages 5662–5670, 2020.
- David J. C. MacKay. A practical Bayesian framework for backpropagation networks. *Neural Computation*, 4(3):448–472, 1992.
- Wesley Maddox, Shuai Tang, Pablo Moreno, Andrew Gordon Wilson, and Andreas Damianou. Fast adaptation with linearized neural networks. In *International Conference on Artificial Intelligence and Statistics (AISTATS)*, pages 2737–2745, 2021.
- James Martens. New insights and perspectives on the natural gradient method. *Journal of Machine Learning Research*, 21(146):1–76, 2020.
- James Martens and Roger Grosse. Optimizing neural networks with Kronecker-factored approximate curvature. In *International Conference on Machine Learning (ICML)*, pages 2408–2417, 2015.
- Razvan Pascanu and Yoshua Bengio. Revisiting natural gradient for deep networks. *arXiv preprint arXiv:1301.3584*, 2013.
- Barak A. Pearlmutter. Fast exact multiplication by the Hessian. *Neural Computation*, 6(1):147–160, 1994.
- Carlos Riquelme, George Tucker, and Jasper Snoek. Deep Bayesian bandits showdown: An empirical

-
- comparison of Bayesian deep networks for Thompson sampling. In *International Conference on Learning Representations (ICLR)*, 2018.
- Hippolyt Ritter, Aleksandar Botev, and David Barber. A scalable Laplace approximation for neural networks. In *International Conference on Learning Representations (ICLR)*, 2018.
- Sudeep Salgia. Provably and practically efficient neural contextual bandits. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 29800–29844. PMLR, 2023. URL <https://proceedings.mlr.press/v202/salgia23a.html>.
- Nicol N. Schraudolph. Fast curvature matrix-vector products for second-order gradient descent. *Neural Computation*, 14(7):1723–1738, 2002.
- William R. Thompson. On the likelihood that one unknown probability exceeds another in view of the evidence of two samples. *Biometrika*, 25(3–4):285–294, 1933.
- Joel A. Tropp. An introduction to matrix concentration inequalities. *Foundations and Trends in Machine Learning*, 8(1–2):1–230, 2015.
- Shashanka Ubaru, Jie Chen, and Yousef Saad. Fast estimation of $\text{tr}(f(a))$ via stochastic Lanczos quadrature. *SIAM Journal on Matrix Analysis and Applications*, 38(4):1075–1099, 2017.
- Michal Valko, Nathan Korda, Rémi Munos, Ilias Flaounas, and Nello Cristianini. Finite-time analysis of kernelised contextual bandits. In *Uncertainty in Artificial Intelligence (UAI)*, pages 654–663, 2013.
- Dongxie Wen, Hanyan Yin, Xiao Zhang, Peng Zhao, Lijun Zhang, and Zhewei Wei. Revisiting matrix sketching in linear bandits: Achieving sublinear regret via dyadic block sketching. In *International Conference on Learning Representations (ICLR)*, 2026.
- Pan Xu, Zheng Wen, Handong Zhao, and Quanquan Gu. Neural contextual bandits with deep representation and shallow exploration. In *International Conference on Learning Representations (ICLR)*, 2022a.
- Pan Xu, Hongkai Zheng, Eric Mazumdar, Kamyar Azizzadenesheli, and Anima Anandkumar. Langevin Monte Carlo for contextual bandits. In *International Conference on Machine Learning (ICML)*, pages 24830–24850, 2022b.
- Weitong Zhang, Dongruo Zhou, Lihong Li, and Quanquan Gu. Neural Thompson sampling. In *International Conference on Learning Representations (ICLR)*, 2021.
- Dongruo Zhou, Lihong Li, and Quanquan Gu. Neural contextual bandits with UCB-based exploration. In *International Conference on Machine Learning (ICML)*, pages 11492–11502, 2020.

A Deferred main-text derivations

This appendix collects the formal statements, proofs, and derivations deferred from the compact main text; all cross-referenced results are stated here.

For the sequential frozen-feature Gram, the usual information gain is

$$\gamma_T := \log \frac{\det(\bar{\mathbf{C}}_{T+1})}{\det(\lambda \mathbf{I})} = \sum_{t=1}^T \log(1 + \sigma^{-2} \bar{s}_t^2(a_t)). \quad (25)$$

Lemma 4 (Rank and statistical-dimension information closure). *For $A \succeq 0$, $r \geq \text{rank}(A)$, and $q_A = \|A\|_{\text{op}}/\lambda$,*

$$\log \det(\mathbf{I} + A/\lambda) \leq \min \left\{ r \log \left(1 + \frac{\text{tr } A}{r\lambda} \right), k(q_A) d_{\text{eff}, \lambda}(A) \right\}, \quad (26)$$

where $d_{\text{eff}, \lambda}(A) = \text{tr}[A(A + \lambda \mathbf{I})^{-1}]$, $k(0) = 1$, and $k(q) = (1 + q) \log(1 + q)/q$ for $q > 0$; the rank term is defined as zero when $r = 0$ (which forces $A = 0$). Consequently, if $\text{rank}(\bar{\mathbf{C}}_{T+1} - \lambda \mathbf{I}) \leq r$, then $\gamma_T \leq \Gamma_{T,r}$ from (21).

Proof. Writing $x_i = \lambda_i(A)/\lambda$, concavity of $\log(1 + x)$ after padding the spectrum with zeros gives the rank–trace term. Also

$$\log(1 + x) = \frac{(1 + x) \log(1 + x)}{x} \frac{x}{1 + x}.$$

The first factor is increasing because its derivative is $(x - \log(1 + x))/x^2 \geq 0$; bounding it at q_A and summing proves the statistical-dimension term. For the frozen Gram, $\text{tr}(\bar{\mathbf{C}}_{T+1} - \lambda \mathbf{I}) \leq TG^2/\sigma^2$, which gives (21) (with $r_T = \min\{r, T\}$). $\square$

The second term is useful only with a predictable or deterministic envelope on $d_{\text{eff}, \lambda}$; a realized value is post-hoc. It is not the empirical trace effective rank $\text{tr}(A)/\|A\|_{\text{op}}$, which does not in general control log determinant.

The dynamic-potential and CG-observable consequences of Theorem 1 are

$$\begin{aligned} R_T &\leq 2\sqrt{(\sigma^2 + G^2/\lambda)\Gamma_T^{\text{dyn}} S_T} + 2E_T, \\ R_T &\leq 2\sqrt{(\sigma^2 + G^2/\lambda)\widehat{\Lambda}_T^{\mathcal{C}} S_T} + 2E_T. \end{aligned} \quad (27)$$

Corollary 5 (Corrected-center variant, no centering certificate). *Replace the predictive center in Algorithm 1 by*

$$\widehat{\mu}_t^{\text{corr}}(a) := \mu_{\theta_t}(\mathbf{x}_t, a) + \mathbf{g}_t(a)^\top (\widehat{\theta}_t^{\text{lin}} - \theta_t) \quad (28)$$

and use $\omega_t = \bar{\beta}_t$. Under the hypotheses of Theorem 1 except Assumption 4, the bounds (15)–(27) hold.

Proof. Local linearization gives $\mu^*(\mathbf{x}_t, a) - \widehat{\mu}_t^{\text{corr}}(a) = \mathbf{g}_t(a)^\top (\theta^* - \widehat{\theta}_t^{\text{lin}}) + \rho_t$. Lemma 19 bounds the first term by $\beta_t \bar{s}_t(a)$ and $|\rho_t| \leq \varepsilon_{\text{lin}}(t)$, which is the first step of the regret proof with $\omega_t = \bar{\beta}_t$; all later steps are unchanged. $\square$

Remark (Cost of the corrected center). *Exact evaluation requires a solve against $\bar{\mathbf{C}}_t$. A constant number of $O(d)$ CG vectors is insufficient by itself: one additionally needs stored historical frozen gradients, raw replay data with the historical parameter checkpoints needed to reconstruct them, a frozen-feature matrix–vector oracle, an explicit low-dimensional representation, or a certified sketch. Maintaining $\bar{\mathbf{C}}_t$ explicitly costs $O(d^2)$ memory. Approximate frozen-feature access introduces an error that must be returned to the centering certificate unless separately controlled.*

A.1 Assumptions

Assumption 1 (Sub-Gaussian noise). *The noise η_t is conditionally σ -sub-Gaussian given the context and action: $\mathbb{E}[\exp(s\eta_t) \mid \mathcal{G}_t] \leq \exp(s^2\sigma^2/2)$ for all $s \in \mathbb{R}$, where $\mathcal{G}_t := \mathcal{H}_t \vee \sigma(a_t)$ is the pre-reward sigma-algebra of (69), available immediately after the action is selected and before the reward is revealed.*

Assumption 2 (Bounded gradient features). For a constant $G > 0$,

$$\sup_{t \leq T} \sup_{s \leq t} \sup_a \|\nabla_{\boldsymbol{\theta}} \mu_{\boldsymbol{\theta}_t}(\mathbf{x}_s, a)\| \leq G.$$

This bounds both the current-round features $\mathbf{g}_t(\mathbf{x}_t, a)$ and the replayed features $\mathbf{g}_t(\mathbf{x}_s, a_s)$ appearing in the curvature matrix (5), as well as the frozen features $\mathbf{g}_s(\mathbf{x}_s, a_s)$ in (6).

Assumption 3 (Local linearization). There is a known convex parameter region Θ containing $\boldsymbol{\theta}^*$ and every realized $\boldsymbol{\theta}_t$, on which $\nabla_{\boldsymbol{\theta}} \mu_{\boldsymbol{\theta}}(\mathbf{x}, a)$ is L_μ -Lipschitz uniformly over candidate $(\mathbf{x}, a)$. Thus

$$\mu_{\boldsymbol{\theta}}(\mathbf{x}, a) = \mu_{\boldsymbol{\theta}_t}(\mathbf{x}, a) + \mathbf{g}_t(\mathbf{x}, a)^\top (\boldsymbol{\theta} - \boldsymbol{\theta}_t) + \rho_t(\mathbf{x}, a, \boldsymbol{\theta}), \quad (29)$$

where the remainder satisfies $|\rho_t(\mathbf{x}, a, \boldsymbol{\theta})| \leq \frac{L_\mu}{2} \|\boldsymbol{\theta} - \boldsymbol{\theta}_t\|^2$ for $\boldsymbol{\theta} \in \Theta$. Evaluating at $\boldsymbol{\theta} = \boldsymbol{\theta}^*$ gives a time-varying linearization error

$$\varepsilon_{\text{lin}}(t) := \sup_{a \in \mathcal{A}_t} |\rho_t(\mathbf{x}_t, a, \boldsymbol{\theta}^*)| \leq \frac{L_\mu}{2} \|\boldsymbol{\theta}^* - \boldsymbol{\theta}_t\|^2. \quad (30)$$

For Theorem 1 to give a sublinear guarantee, we require the cumulative linearization error $E_T := \sum_{t=1}^T \varepsilon_{\text{lin}}(t) = o(T)$; see Theorem 1 for the precise dependence. Concretely, $E_T = O(1)$ when $\|\boldsymbol{\theta}^* - \boldsymbol{\theta}_t\| = O(1/t)$ (since $\varepsilon_{\text{lin}}(t) = O(1/t^2)$ and $\sum t^{-2} < \infty$; achievable in convex or strongly convex regimes), and $E_T = O(\sqrt{T})$ when $\|\boldsymbol{\theta}^* - \boldsymbol{\theta}_t\| = O(t^{-1/4})$. In practice this is enforced by restricting fine-tuning depth, using trust-region updates, or freezing the backbone (which renders the last-layer model convex or near-linear with small L_μ). These rate examples are illustrative only. The present paper does not prove convergence of the nonconvex optimization path $\{\boldsymbol{\theta}_t\}$ to $\boldsymbol{\theta}^*$; a sublinear guarantee from this bound therefore requires this term to be controlled by an external stability or convergence argument.

Assumption 4 (Action-scaled centering certificate). Lemma 6 below proves the geometric bound $|\mathbf{g}_t(\mathbf{x}_t, a)^\top (\boldsymbol{\theta}_t - \hat{\boldsymbol{\theta}}_t^{\text{lin}})| \leq \bar{s}_t(a) \psi_t$ for every $a \in \mathcal{A}_t$, where

$$\psi_t := \frac{\zeta_t}{\sqrt{\lambda}} + \|M_t\|_{\bar{\mathbf{C}}_t^{-1}}, \quad (31)$$

$\zeta_t \geq 0$ is an $\mathcal{H}_t$ -measurable optimizer gradient-residual bound ($\|\nabla \mathcal{L}_t(\boldsymbol{\theta}_t)\|_2 \leq \zeta_t$) and M_t is the gradient mismatch vector defined there. We assume there is an $\mathcal{H}_t$ -measurable upper bound $\bar{\psi}_t \geq \psi_t$ (the centering certificate) such that, for every action $a \in \mathcal{A}_t$,

$$|\mathbf{g}_t(\mathbf{x}_t, a)^\top (\boldsymbol{\theta}_t - \hat{\boldsymbol{\theta}}_t^{\text{lin}})| \leq \bar{\psi}_t \bar{s}_t(a). \quad (32)$$

The discrepancy is thus action-scaled: it multiplies the confidence width $\bar{s}_t(a)$ rather than entering additively. The certificate absorbs the same three sources as before—optimizer residual ζ_t , replayed-Jacobian drift inside M_t , and higher-order nonlinear terms—but now inside a single per-round scalar ψ_t . In the convex-in-parameters regime (frozen backbone with a linear reward head), under exact optimization $\hat{\boldsymbol{\theta}}_t^{\text{lin}} = \boldsymbol{\theta}_t$ and $\bar{\psi}_t = 0$. The corrected-center variant of Corollary 5 removes $\bar{\psi}_t$ entirely, at the cost of an explicit frozen-feature solve.

Remark (Status of Assumption 4). Lemma 7 supplies a predictable valid $\bar{\psi}_t$ from the optimizer residual, Welford path moments, and observed residual energy. We do not prove that this conservative quantity is small outside a restricted path. It enters the exploration term through $\omega_t = \bar{\beta}_t + \bar{\psi}_t$ (Equation (60)), not as an additive $O(T)$ term.

The next lemma gives the primitive geometric bound underlying Assumption 4, expressed through the certifiable quantity ψ_t .

Lemma 6 (Primitive upper bound for prediction discrepancy). Let $z_s = (\mathbf{x}_s, a_s)$, $\mathbf{g}_s = \nabla_{\boldsymbol{\theta}} \mu_{\boldsymbol{\theta}_s}(z_s)$, $\mathbf{g}_{s,t} = \nabla_{\boldsymbol{\theta}} \mu_{\boldsymbol{\theta}_t}(z_s)$, and $\delta_{s,t} = \mathbf{g}_{s,t} - \mathbf{g}_s$. Define

$$\bar{\mu}_{s,t} := \mu_{\boldsymbol{\theta}_s}(z_s) + \mathbf{g}_s^\top (\boldsymbol{\theta}_t - \boldsymbol{\theta}_s), \quad e_{s,t} := \mu_{\boldsymbol{\theta}_t}(z_s) - \bar{\mu}_{s,t}, \quad \bar{r}_{s,t} := \bar{\mu}_{s,t} - r_s.$$

Let $\bar{\mathcal{L}}_t$ be the frozen-feature linearized ridge objective whose minimizer is $\hat{\boldsymbol{\theta}}_t^{\text{lin}}$, and define

$$M_t := \frac{1}{\sigma^2} \sum_{s < t} [\bar{r}_{s,t} \delta_{s,t} + e_{s,t} \mathbf{g}_s + e_{s,t} \delta_{s,t}].$$

Assumption	Status	Use	Checkability	Experimental instantiation
Conditional sub-Gaussian noise	Standard	Self-normalized confidence	Distributional	Gaussian generator
Realizability, $\ \theta^*\ \leq S$	Restrictive	Bias and Taylor schedules	Not generally testable	Known synthetic teacher
Bounded gradients	Standard on compact sets	Width and condition bounds	Analytic or monitored	Analytic tanh bound
L_μ, L_g on convex Θ	Restrictive	Linearization, transfer, centering	Analytic for tanh; otherwise hard	Global tanh Hessian bound and projection
Optimizer residual ζ_t	Algorithmic	Centering certificate	Pre-action gradient norm	Exact full-objective gradient norm
Operator and CG certificates	Algorithmic	Optimism and solve error	Residual/Loewner proof required	Analytic condition bound; float64 audit
Residual-energy envelope	Rate-only	Near-linear scaling	Observable after collection	Reported, not assumed by policy
Fixed active rank r	Restrictive	Closes γ_T	Supplied subspace, not terminal audit	Constructed active embedding
Window excitation K_w	Restrictive	Closes Λ_T^C	Analytic construction or assumed	Cyclic construction after burn-in

Table 2: Assumption ledger. Analytic and pre-action quantities are policy available; teacher comparisons and exact whitened quantities are post-hoc only.

If the nonlinear optimizer satisfies $\|\nabla \mathcal{L}_t(\theta_t)\|_2 \leq \zeta_t$, then for every candidate action $a \in \mathcal{A}_t$,

$$\left| \mathbf{g}_t(a)^\top (\theta_t - \hat{\theta}_t^{\text{lin}}) \right| \leq \bar{s}_t(a) \left(\frac{\zeta_t}{\sqrt{\lambda}} + \|M_t\|_{\bar{\mathbf{C}}_t^{-1}} \right), \quad \bar{s}_t(a) := \sqrt{\mathbf{g}_t(a)^\top \bar{\mathbf{C}}_t^{-1} \mathbf{g}_t(a)}.$$

The left-hand side is exactly the geometric bound $|\mathbf{g}_t(a)^\top (\theta_t - \hat{\theta}_t^{\text{lin}})| \leq \bar{s}_t(a) \psi_t$ of Assumption 4 with $\psi_t = \zeta_t/\sqrt{\lambda} + \|M_t\|_{\bar{\mathbf{C}}_t^{-1}}$ (Equation (31)). Consequently, since $\|M_t\|_{\bar{\mathbf{C}}_t^{-1}} \leq \|M_t\|_2/\sqrt{\lambda}$, a valid centering certificate is

$$\bar{\psi}_t := \frac{\zeta_t + \|M_t\|_2}{\sqrt{\lambda}} \geq \psi_t.$$

If, in addition, on a fixed event, $\theta \mapsto \nabla_{\theta} \mu_{\theta}(z)$ is L_g -Lipschitz, $|\bar{r}_{s,t}| \leq B_{t,\delta}$ for all $s < t$, and $\Delta_{s,t} := \|\theta_t - \theta_s\|$, then

$$\|M_t\|_2 \leq \frac{1}{\sigma^2} \sum_{s < t} \left[B_{t,\delta} L_g \Delta_{s,t} + \frac{G L_g}{2} \Delta_{s,t}^2 + \frac{L_g^2}{2} \Delta_{s,t}^3 \right].$$

Proof. The linearized ridge objective satisfies $\nabla \bar{\mathcal{L}}_t(\hat{\theta}_t^{\text{lin}}) = 0$ and, by its quadratic form,

$$\nabla \bar{\mathcal{L}}_t(\theta_t) = \bar{\mathbf{C}}_t(\theta_t - \hat{\theta}_t^{\text{lin}}).$$

Thus $\theta_t - \hat{\theta}_t^{\text{lin}} = \bar{\mathbf{C}}_t^{-1} \nabla \bar{\mathcal{L}}_t(\theta_t)$. Using $\mu_{\theta_t}(z_s) - r_s = \bar{r}_{s,t} + e_{s,t}$ and $\mathbf{g}_{s,t} = \mathbf{g}_s + \delta_{s,t}$, the gradients of the nonlinear loss and the frozen-feature linearized loss satisfy

$$\nabla \mathcal{L}_t(\theta_t) = \nabla \bar{\mathcal{L}}_t(\theta_t) + M_t, \quad \text{so} \quad \nabla \bar{\mathcal{L}}_t(\theta_t) = \nabla \mathcal{L}_t(\theta_t) - M_t.$$

Therefore, by Cauchy–Schwarz in the $\bar{\mathbf{C}}_t$ -norm,

$$|\mathbf{g}_t(a)^\top (\theta_t - \hat{\theta}_t^{\text{lin}})| \leq \bar{s}_t(a) \|\nabla \bar{\mathcal{L}}_t(\theta_t)\|_{\bar{\mathbf{C}}_t^{-1}} \leq \bar{s}_t(a) \left(\|\nabla \mathcal{L}_t(\theta_t)\|_{\bar{\mathbf{C}}_t^{-1}} + \|M_t\|_{\bar{\mathbf{C}}_t^{-1}} \right).$$

Since $\bar{\mathbf{C}}_t \succeq \lambda \mathbf{I}$, $\|\nabla \mathcal{L}_t(\theta_t)\|_{\bar{\mathbf{C}}_t^{-1}} \leq \zeta_t/\sqrt{\lambda}$, which gives the geometric inequality $|\mathbf{g}_t(a)^\top (\theta_t - \hat{\theta}_t^{\text{lin}})| \leq \bar{s}_t(a) (\zeta_t/\sqrt{\lambda} + \|M_t\|_{\bar{\mathbf{C}}_t^{-1}}) = \bar{s}_t(a) \psi_t$, i.e. the bound of (31). For the certificate, $\bar{\mathbf{C}}_t \succeq \lambda \mathbf{I}$ also gives $\|M_t\|_{\bar{\mathbf{C}}_t^{-1}} \leq \|M_t\|_2/\sqrt{\lambda}$, so $\psi_t \leq (\zeta_t + \|M_t\|_2)/\sqrt{\lambda} =: \bar{\psi}_t \geq \psi_t$, the proposed certificate. For the final display, Lipschitzness gives $\|\delta_{s,t}\| \leq L_g \Delta_{s,t}$ and the Taylor remainder bound $|e_{s,t}| \leq (L_g/2) \Delta_{s,t}^2$; applying these together with $\|\mathbf{g}_s\| \leq G$ (Assumption 2) and $|\bar{r}_{s,t}| \leq B_{t,\delta}$ term by term in the definition of M_t gives the claimed bound on $\|M_t\|_2$. $\square$

Lemma 7 ($O(d)$ -state path certificates). *Suppose $\nabla_{\theta}\mu_{\theta}(z)$ is L_g -Lipschitz and bounded by G on a convex region containing the realized iterates. Before round t , let $n_t = t - 1$, let $\bar{\theta}_{<t}$ and $J_{<t} = \sum_{s<t} \|\theta_s - \bar{\theta}_{<t}\|^2$ summarize past iterates, and let $R_t^{\text{col}} = \sum_{s<t} c_s^2$ with $c_s = \mu_{\theta_s}(z_s) - r_s$. At $t = 1$ use the initialization $\bar{\theta}_{<1} = \mathbf{0}$, $J_{<1} = Q_1 = R_1^{\text{col}} = 0$. Then, for all t ,*

$$Q_t := J_{<t} + n_t \|\theta_t - \bar{\theta}_{<t}\|^2 = \sum_{s<t} \|\theta_t - \theta_s\|^2$$

and the following $\mathcal{H}_t$ -measurable quantities are valid:

$$\bar{\chi}_t = \frac{L_g}{\sigma\sqrt{\lambda}} \sqrt{Q_t} \geq \chi_t, \quad (33)$$

$$\bar{M}_t = \frac{1}{\sigma^2} \left[L_g \sqrt{R_t^{\text{col}} Q_t} + \frac{3GL_g}{2} Q_t + \frac{L_g^2}{2} Q_t^{3/2} \right] \geq \|M_t\|_2. \quad (34)$$

If $\|\theta_s\|, \|\theta_t\| \leq R$, the final term in (34) may be replaced by $L_g^2 R Q_t$. If a nonnegative $\mathcal{H}_t$ -measurable optimizer certificate satisfies $\|\nabla \mathcal{L}_t(\theta_t)\| \leq \zeta_t$, then

$$\bar{\psi}_t = (\zeta_t + \bar{M}_t)/\sqrt{\lambda} \geq \psi_t.$$

If prediction gradients are L_{μ} -Lipschitz on a convex region containing θ_t and θ^* and $\|\theta^*\| \leq S$, then

$$\bar{\epsilon}_t^{\text{lin}} := \frac{L_{\mu}}{2} (S + \|\theta_t\|)^2 \geq \epsilon_{\text{lin}}(t).$$

Consequently $\bar{F}_t = \sum_{s<t} (\bar{\epsilon}_s^{\text{lin}})^2 \geq F_t$ and $\bar{E}_T = \sum_{t \leq T} \bar{\epsilon}_t^{\text{lin}} \geq E_T$.

Proof. The parallel-axis identity follows after writing $\theta_t - \theta_s = (\theta_t - \bar{\theta}_{<t}) + (\bar{\theta}_{<t} - \theta_s)$; the cross term sums to zero. Lemma 11 and $\|\nabla \mu_{\theta_t}(z_s) - \nabla \mu_{\theta_s}(z_s)\| \leq L_g \|\theta_t - \theta_s\|$ give (33). Let $d_{s,t} = \|\theta_t - \theta_s\|$. In Lemma 6,

$$|\bar{r}_{s,t}| \leq |c_s| + G d_{s,t}, \quad \|\delta_{s,t}\| \leq L_g d_{s,t}, \quad |e_{s,t}| \leq (L_g/2) d_{s,t}^2.$$

Thus each summand defining M_t has norm at most

$$L_g |c_s| d_{s,t} + \frac{3GL_g}{2} d_{s,t}^2 + \frac{L_g^2}{2} d_{s,t}^3.$$

Cauchy–Schwarz gives $\sum |c_s| d_{s,t} \leq \sqrt{R_t^{\text{col}} Q_t}$, while $\sum d_{s,t}^2 = Q_t$ and $\sum d_{s,t}^3 \leq Q_t^{3/2}$, proving (34). Under the trust region, $d_{s,t} \leq 2R$, so $\sum d_{s,t}^3 \leq 2R Q_t$. The centering claim follows from Lemma 6 and $\bar{\mathbf{C}}_t \succeq \lambda \mathbf{I}$. Finally Taylor’s theorem and $\|\theta^* - \theta_t\| \leq S + \|\theta_t\|$ prove the remainder, F_t , and E_T claims. $\square$

Filtration-order implementation. At round t , the summaries through $t - 1$, θ_t , and the certificate ζ_t are available before selection. The learner computes $Q_t, \bar{\chi}_t, u_t, \bar{M}_t, \bar{\psi}_t, \bar{\epsilon}_t^{\text{lin}}$ and β_t , then solves every candidate system to the uniform tolerance and selects a_t . After observing r_t , it adds c_t^2 to R_{t+1}^{col} , adds θ_t with the Welford update

$$d = \theta_t - \bar{\theta}_{<t}, \quad \bar{\theta}_{\leq t} = \bar{\theta}_{<t} + d/t, \quad J_{\leq t} = J_{<t} + d^{\top}(\theta_t - \bar{\theta}_{\leq t}),$$

and updates $\bar{F}_{t+1}$ and $\hat{\gamma}_t$. It then forms θ_{t+1} and certifies the nonnegative bound ζ_{t+1} before the next action. No current reward enters the current score.

In frozen linear-head models Q_t may be nonzero while $L_g = L_{\mu} = 0$; then $\bar{\chi}_t = \bar{M}_t = \bar{\epsilon}_t^{\text{lin}} = 0$, and exact optimization also gives $\bar{\psi}_t = 0$.

Corollary 8 (Bounded-path near-linear rate). *Fix T and suppose for every $t \leq T$*

$$\|\theta_t\|, \|\theta^*\| \leq R, \quad L_{\mu} \leq c_{\mu}/\sqrt{W}, \quad L_g \leq c_g/\sqrt{W}, \quad \zeta_t \leq \zeta_0/\sqrt{W}, \quad R_t^{\text{col}} \leq C_R t P_T,$$

where $P_T \geq 1$ is an explicitly supplied residual-energy factor. For exact current-parameter curvature and a uniform CG tolerance $\bar{\epsilon}_t \leq \bar{\epsilon} < 1$, set

$$x_T = \frac{2c_g R \sqrt{T-1}}{\sigma \sqrt{\lambda W}}, \quad \epsilon_{\text{drift}, T} = 2x_T + x_T^2, \quad \rho_{\pm} = 1 \pm \epsilon_{\text{drift}, T}.$$

If $\rho_- > 0$, then $\rho_- \bar{\mathbf{C}}_t \preceq \mathbf{C}_t \preceq \rho_+ \bar{\mathbf{C}}_t$ for all $t \leq T$. Writing

$$\alpha_I = \sqrt{\frac{1+\bar{\varepsilon}}{1-\bar{\varepsilon}}}, \quad A_{\inf} = \alpha_I^2 \frac{\rho_+}{\rho_-},$$

the observable information schedule satisfies $\gamma_T \leq \hat{\gamma}_T \leq A_{\inf} \gamma_T$, and

$$R_T \leq 2\alpha_I \sqrt{\rho_+/\rho_-} \omega_{\max,T}^{\text{op}} \sqrt{(\sigma^2 + G^2/\lambda)T\gamma_T} + 2\bar{E}_T \quad (35)$$

holds with

$$\omega_{\max,T}^{\text{op}} := \max_{t \leq T} (\bar{\beta}_t + \bar{\psi}_t)$$

and

$$\begin{aligned} \omega_{\max,T}^{\text{op}} &\leq \sqrt{A_{\inf} \gamma_T + 2 \log(1/\delta)} + \sqrt{\lambda} R + \frac{2c_\mu R^2 \sqrt{T}}{\sigma \sqrt{W}} \\ &\quad + \frac{A_T}{\sqrt{\lambda W}} + \frac{B_T}{\sqrt{\lambda W}}, \\ A_T &= \zeta_0 + \sigma^{-2} \left[2c_g R \sqrt{C_R T(T-1)P_T} + 6Gc_g R^2(T-1) \right], \\ B_T &= 4c_g^2 R^3(T-1)/\sigma^2. \end{aligned}$$

In addition, (20) holds and

$$\bar{\psi}_t \leq \frac{1}{\sqrt{\lambda}} \left\{ \frac{\zeta_0}{\sqrt{W}} + \frac{1}{\sigma^2} \left[\frac{2c_g R \sqrt{C_R t(t-1)P_T} + 6Gc_g R^2(t-1)}{\sqrt{W}} + \frac{4c_g^2 R^3(t-1)}{W} \right] \right\}.$$

Proof. The trust region gives $Q_t \leq 4R^2(t-1)$; substitution in Lemma 7 proves all displayed path bounds. Corollary 12 gives the two-sided operator relation. The CG sandwich and $u_t \leq (1+x_T)^2 = \rho_+$ imply that every logarithm in $\hat{\gamma}_T$ is at most $\log(1 + A_{\inf} \sigma^{-2} \bar{s}_t^2(a_t))$. For $A \geq 1$, $\log(1 + Ax) \leq A \log(1 + x)$, proving the information comparison. Substitution into Corollary 29 gives the regret display. $\square$

Corollary 9 (Conditional projected-Gram rate for a supplied subspace). *Under Corollary 8, additionally suppose a subspace $U \subseteq \mathbb{R}^d$ with $\dim U \leq r$ is fixed and supplied before interaction, and every candidate or replayed tangent feature through round T lies in U . Hence $\text{rank}(\bar{\mathbf{C}}_t - \lambda \mathbf{I}) \leq r$ for every $t \leq T+1$. Let $r_U = \dim U$, let $Q_U \in \mathbb{R}^{d \times r_U}$ have orthonormal columns spanning U , and define*

$$\mathbf{z}_{t,s} = Q_U^\top \mathbf{g}_t(\mathbf{x}_s, a_s), \quad \mathbf{z}_t(a) = Q_U^\top \mathbf{g}_t(\mathbf{x}_t, a).$$

Then the current-GGN width reduces exactly to

$$\mathbf{g}_t(a)^\top \mathbf{C}_t^{-1} \mathbf{g}_t(a) = \mathbf{z}_t(a)^\top \left(\lambda I_{r_U} + \sigma^{-2} \sum_{s < t} \mathbf{z}_{t,s} \mathbf{z}_{t,s}^\top \right)^{-1} \mathbf{z}_t(a), \quad (36)$$

with the analogous identity for $\bar{\mathbf{C}}_t$ using $Q_U^\top \mathbf{g}_s(\mathbf{x}_s, a_s)$. Thus a supplied basis permits an exact dense $r_U \times r_U$ Gram with $O(r_U^2)$ curvature state. Given tangent coordinates, forming the current Gram costs $O((t-1)r_U^2)$, factorization $O(r_U^3)$, and each width query $O(r_U^2)$; current-parameter replay, model training, and gradient generation can still be d -dimensional. Full-dimensional CG is not needed in this corollary's covered regime.

Define $\Gamma_{T,r}$ by (21). Then (22) holds with

$$\begin{aligned} \Omega_{T,r} &:= \sqrt{A_{\inf} \Gamma_{T,r} + 2 \log(1/\delta)} + \sqrt{\lambda} R + \frac{2c_\mu R^2 \sqrt{T}}{\sigma \sqrt{W}} + \frac{A_T}{\sqrt{\lambda W}} + \frac{B_T}{\sqrt{\lambda W}}, \\ A_T &= \zeta_0 + \sigma^{-2} \left[2c_g R \sqrt{C_R T(T-1)P_T} + 6Gc_g R^2(T-1) \right], \\ B_T &= 4c_g^2 R^3(T-1)/\sigma^2, \end{aligned}$$

and $A_{\text{inf}}, \rho_{\pm}$ are exactly those in Corollary 8. In particular, for fixed r and fixed problem constants, if $W \geq CT^2P_T$ for a constant C satisfying the explicit thresholds in (37), then

$$R_T = O(r\sqrt{T} \log T) + O(T/\sqrt{W}).$$

This rate is conditional on the ex ante subspace U and every bounded-path, width, residual, optimizer, transfer, and solver premise of Corollary 8. It neither learns U nor covers unrestricted representation learning or end-to-end neural-network training; it is an effective projected-feature closure, not a full-dimensional computational advantage for CC-UCB.

Proof. Extend Q_U to an orthogonal basis of $\mathbb{R}^d$. In that basis $\mathbf{C}_t$ is block diagonal with the $r_U \times r_U$ matrix in (36) and λI_{d-r_U} ; the query has zero coordinates in the second block. This proves (36), and the same argument proves the frozen-Gram identity. Lemma 4 gives $\gamma_T \leq \Gamma_{T,r}$. Every occurrence of γ_T in Corollary 8 is monotone, so direct substitution gives the display. Under the stated width scaling, x_T , the centering terms, and the linearization terms are bounded, while $\Gamma_{T,r} = O(r \log T)$. Hence $\Omega_{T,r} = O(\sqrt{r \log T} + 1)$ and the leading product is $O(r\sqrt{T} \log T)$. $\square$

For explicit targets $\bar{\epsilon}_{\text{drift}} \in (0, 1)$, $\Psi, b_F, \eta_E > 0$, put $x_* = \sqrt{1 + \bar{\epsilon}_{\text{drift}}} - 1$. The sufficient conditions

$$\begin{aligned} W &\geq \frac{4c_g^2 R^2 (T-1)}{\lambda \sigma^2 x_*^2}, & W &\geq \frac{4c_\mu^2 R^4 T}{\sigma^2 b_F^2}, & W &\geq \frac{4c_\mu^2 R^4}{\eta_E^2}, \\ W &\geq \max \left\{ \frac{4A_T^2}{\lambda \Psi^2}, \frac{2B_T}{\sqrt{\lambda} \Psi} \right\} \end{aligned} \quad (37)$$

respectively ensure bounded drift, $\sqrt{\bar{F}_{T+1}}/\sigma \leq b_F$, $\bar{E}_T/T \leq \eta_E$, and $\bar{\psi}_T \leq \Psi$. Thus the centering requirement has leading sufficient scaling $W = \Omega(T^2 P_T)$; these conditions do not bound γ_T .

Corollary 10 (Nonlinear tanh-link Gaussian bandit). *For $\mu_\theta(x, a) = \tanh(\phi(x, a)^\top \theta)$ with $\|\phi(x, a)\| \leq B_\phi$, one may take*

$$G = B_\phi, \quad L_\mu = L_g = \frac{4}{3\sqrt{3}} B_\phi^2.$$

On any certified trust region, Corollary 2 therefore gives an executable nonlinear Gaussian-reward CC-UCB policy using only the online path statistics, a pre-action optimizer-residual certificate, and the stated operator/CG certificates.

Proof. $\nabla_\theta \mu = \text{sech}^2(\phi^\top \theta) \phi$, so its norm is at most B_ϕ . The Hessian is $-2 \tanh(z) \text{sech}^2(z) \phi \phi^\top$. For $y = |\tanh z|$, $2y(1-y^2)$ is maximized at $y = 1/\sqrt{3}$ with value $4/(3\sqrt{3})$, proving both Lipschitz constants. Apply Lemma 7 and Corollary 2. $\square$

Predictable algorithmic curvature under certified transfer. The analysis is conducted for a *generic predictable SPD algorithmic curvature sequence* $\{\mathcal{C}_t\}_{t \geq 1}$ satisfying

$$\mathcal{C}_t \text{ is symmetric,} \quad \mathcal{C}_t \succeq \lambda \mathbf{I} \succ 0, \quad \mathcal{C}_t \text{ is } \mathcal{H}_t\text{-measurable.} \quad (38)$$

This is the operator whose inverse the algorithm actually applies (via CG) to form widths. The exact full-history result is the special case $\mathcal{C}_t = \mathbf{C}_t$ (relinearized curvature (5)); approximate, subsampled, windowed, refreshed, or sketched operators $\tilde{\mathbf{C}}_t$ are also represented by $\mathcal{C}_t$, but are covered only when the required transfer certificate below is available. Write the algorithmic width

$$s_{\mathcal{C},t}^2(a) := \mathbf{g}_t(a)^\top \mathcal{C}_t^{-1} \mathbf{g}_t(a). \quad (39)$$

Assumption 5 (One-sided optimism transfer). *For every round $t \leq T$ there is an $\mathcal{H}_t$ -measurable factor $u_t(a) \geq 1$ such that, for every candidate action $a \in \mathcal{A}_t$,*

$$\bar{s}_t^2(a) \leq u_t(a) s_{\mathcal{C},t}^2(a). \quad (40)$$

This is the *only* spectral relation the main theorem needs, and it is one-sided: it dominates the predictable frozen-feature width $\bar{s}_t(a)$ (the object of the self-normalized confidence bound) by an inflated algorithmic width. No *lower* Loewner factor on $\mathcal{C}_t$ is required for optimism. The factor may be action-dependent; when a uniform per-round u_t satisfies (40) for every action we write u_t and state the action-dependent theorem with $u_t(a)$. For the exact-curvature instantiation $\mathcal{C}_t = \mathbf{C}_t$, the whitened relative-drift lemma below certifies (40) with the uniform factor $u_t = (1 + \bar{\chi}_t)^2$; for an approximate operator, the composition of Lemma 11 with a one-sided spectral bound $\hat{\mathbf{C}}_t \preceq \kappa_{+,t} \mathbf{C}_t$ gives $u_t = \kappa_{+,t}(1 + \bar{\chi}_t)^2$ (Theorem 24).

The following lemma supplies the exact-curvature transfer factor through a *whitened, relative* feature-drift bound: the drift is measured in the $\bar{\mathbf{C}}_t^{-1}$ metric and accumulated in ℓ_2 , and the resulting one-sided bound needs *no* smallness condition.

Lemma 11 (Whitened relative feature-drift bound). *Let $\mathbf{g}_s := \nabla_{\boldsymbol{\theta}} \mu_{\boldsymbol{\theta}_s}(\mathbf{x}_s, a_s)$, $\mathbf{g}_{s,t} := \nabla_{\boldsymbol{\theta}} \mu_{\boldsymbol{\theta}_t}(\mathbf{x}_s, a_s)$, $\delta_{s,t} := \mathbf{g}_{s,t} - \mathbf{g}_s$, and $\Delta_{s,t} := \|\boldsymbol{\theta}_t - \boldsymbol{\theta}_s\|$. Using the symmetric (principal) square root $\bar{\mathbf{C}}_t^{1/2}$, define the $d \times (t-1)$ whitened column matrices*

$$\mathbf{G}_t := \left[\bar{\mathbf{C}}_t^{-1/2} \mathbf{g}_s / \sigma \right]_{s < t}, \quad \mathbf{D}_t := \left[\bar{\mathbf{C}}_t^{-1/2} \delta_{s,t} / \sigma \right]_{s < t}, \quad \chi_t := \|\mathbf{D}_t\|_{\text{op}}. \quad (41)$$

Then:

- (a) $\mathbf{G}_t \mathbf{G}_t^\top = \mathbf{I} - \lambda \bar{\mathbf{C}}_t^{-1} \preceq \mathbf{I}$, hence $\|\mathbf{G}_t\|_{\text{op}} \leq 1$.
- (b) $\bar{\mathbf{C}}_t^{-1/2} (\mathbf{C}_t - \bar{\mathbf{C}}_t) \bar{\mathbf{C}}_t^{-1/2} = \mathbf{G}_t \mathbf{D}_t^\top + \mathbf{D}_t \mathbf{G}_t^\top + \mathbf{D}_t \mathbf{D}_t^\top$, so $\|\bar{\mathbf{C}}_t^{-1/2} (\mathbf{C}_t - \bar{\mathbf{C}}_t) \bar{\mathbf{C}}_t^{-1/2}\|_{\text{op}} \leq 2\chi_t + \chi_t^2$.
- (c) (**One-sided, no smallness condition.**) For every $\chi_t \geq 0$,

$$\mathbf{C}_t \preceq (1 + \chi_t)^2 \bar{\mathbf{C}}_t, \quad \text{hence} \quad \bar{s}_t^2(a) \leq (1 + \chi_t)^2 s_t^2(a) \quad \forall a. \quad (42)$$

A predictable certificate $\bar{\chi}_t \geq \chi_t$ therefore satisfies Assumption 5 in the exact case $\mathcal{C}_t = \mathbf{C}_t$ with the uniform factor $u_t = (1 + \bar{\chi}_t)^2$.

- (d) (**Whitened ℓ_2 primitive bound.**)

$$\chi_t \leq \|\mathbf{D}_t\|_F = \left[\sigma^{-2} \sum_{s < t} \|\delta_{s,t}\|_{\bar{\mathbf{C}}_t^{-1}}^2 \right]^{1/2} \leq \frac{L_g}{\sigma \sqrt{\lambda}} \left[\sum_{s < t} \|\boldsymbol{\theta}_t - \boldsymbol{\theta}_s\|^2 \right]^{1/2}, \quad (43)$$

the last step under the L_g -Lipschitz-Jacobian assumption $\|\delta_{s,t}\| \leq L_g \Delta_{s,t}$.

Proof. (a) $\mathbf{G}_t \mathbf{G}_t^\top = \sigma^{-2} \sum_{s < t} \bar{\mathbf{C}}_t^{-1/2} \mathbf{g}_s \mathbf{g}_s^\top \bar{\mathbf{C}}_t^{-1/2} = \bar{\mathbf{C}}_t^{-1/2} (\sigma^{-2} \sum_{s < t} \mathbf{g}_s \mathbf{g}_s^\top) \bar{\mathbf{C}}_t^{-1/2} = \bar{\mathbf{C}}_t^{-1/2} (\bar{\mathbf{C}}_t - \lambda \mathbf{I}) \bar{\mathbf{C}}_t^{-1/2} = \mathbf{I} - \lambda \bar{\mathbf{C}}_t^{-1}$. Since $\lambda \bar{\mathbf{C}}_t^{-1} \succeq 0$, $\mathbf{G}_t \mathbf{G}_t^\top \preceq \mathbf{I}$, and $\|\mathbf{G}_t\|_{\text{op}}^2 = \|\mathbf{G}_t \mathbf{G}_t^\top\|_{\text{op}} \leq 1$.

(b) From $\mathbf{g}_{s,t} = \mathbf{g}_s + \delta_{s,t}$, $\mathbf{g}_{s,t} \mathbf{g}_{s,t}^\top - \mathbf{g}_s \mathbf{g}_s^\top = \mathbf{g}_s \delta_{s,t}^\top + \delta_{s,t} \mathbf{g}_s^\top + \delta_{s,t} \delta_{s,t}^\top$. Conjugating each summand by $\bar{\mathbf{C}}_t^{-1/2}$ and weighting by σ^{-2} , the three groups sum exactly to $\mathbf{G}_t \mathbf{D}_t^\top$, $\mathbf{D}_t \mathbf{G}_t^\top$, and $\mathbf{D}_t \mathbf{D}_t^\top$. By the triangle inequality and $\|\mathbf{G}_t\|_{\text{op}} \leq 1$, $\|\mathbf{G}_t \mathbf{D}_t^\top + \mathbf{D}_t \mathbf{G}_t^\top + \mathbf{D}_t \mathbf{D}_t^\top\|_{\text{op}} \leq 2\|\mathbf{G}_t\|_{\text{op}} \|\mathbf{D}_t\|_{\text{op}} + \|\mathbf{D}_t\|_{\text{op}}^2 \leq 2\chi_t + \chi_t^2$.

(c) The matrix $\mathbf{W}_t := \bar{\mathbf{C}}_t^{-1/2} (\mathbf{C}_t - \bar{\mathbf{C}}_t) \bar{\mathbf{C}}_t^{-1/2}$ is symmetric with $\|\mathbf{W}_t\|_{\text{op}} \leq 2\chi_t + \chi_t^2$, hence $\mathbf{W}_t \preceq (2\chi_t + \chi_t^2) \mathbf{I}$. Congruence by $\bar{\mathbf{C}}_t^{1/2}$ gives $\mathbf{C}_t - \bar{\mathbf{C}}_t \preceq (2\chi_t + \chi_t^2) \bar{\mathbf{C}}_t$, i.e. $\mathbf{C}_t \preceq (1 + \chi_t)^2 \bar{\mathbf{C}}_t$; no bound on χ_t is used. Both $\mathbf{C}_t, \bar{\mathbf{C}}_t \succ 0$, so inversion reverses the order: $\mathbf{C}_t^{-1} \succeq (1 + \chi_t)^{-2} \bar{\mathbf{C}}_t^{-1}$. The quadratic form at $\mathbf{g}_t(a)$ gives $s_t^2(a) \geq (1 + \chi_t)^{-2} \bar{s}_t^2(a)$, i.e. $\bar{s}_t^2(a) \leq (1 + \chi_t)^2 s_t^2(a)$. Since $x \mapsto (1 + x)^2$ is increasing on $[0, \infty)$, replacing χ_t by any $\bar{\chi}_t \geq \chi_t$ preserves the inequality, giving $u_t = (1 + \bar{\chi}_t)^2$.

(d) $\chi_t = \|\mathbf{D}_t\|_{\text{op}} \leq \|\mathbf{D}_t\|_F$, and $\|\mathbf{D}_t\|_F^2 = \sum_{s < t} \|\bar{\mathbf{C}}_t^{-1/2} \delta_{s,t} / \sigma\|^2 = \sigma^{-2} \sum_{s < t} \delta_{s,t}^\top \bar{\mathbf{C}}_t^{-1} \delta_{s,t} = \sigma^{-2} \sum_{s < t} \|\delta_{s,t}\|_{\bar{\mathbf{C}}_t^{-1}}^2$. Since $\bar{\mathbf{C}}_t^{-1} \preceq \lambda^{-1} \mathbf{I}$, $\|\delta_{s,t}\|_{\bar{\mathbf{C}}_t^{-1}}^2 \leq \|\delta_{s,t}\|^2 / \lambda \leq L_g^2 \Delta_{s,t}^2 / \lambda$; summing and taking square roots gives (43). $\square$

Corollary 12 (Two-sided feature sandwich). *If additionally $\varepsilon_t := 2\chi_t + \chi_t^2 < 1$, then*

$$(1 - \varepsilon_t) \bar{\mathbf{C}}_t \preceq \mathbf{C}_t \preceq (1 + \varepsilon_t) \bar{\mathbf{C}}_t. \quad (44)$$

Proof. From part (b), $\mathbf{W}_t \preceq -(2\chi_t + \chi_t^2) \mathbf{I} = -\varepsilon_t \mathbf{I}$; congruence by $\bar{\mathbf{C}}_t^{1/2}$ gives $\mathbf{C}_t \succeq (1 - \varepsilon_t) \bar{\mathbf{C}}_t$, and the upper bound is (42) with $(1 + \chi_t)^2 \leq 1 + \varepsilon_t$ when $\varepsilon_t < 1$ (indeed $(1 + \chi_t)^2 = 1 + \varepsilon_t$ exactly). $\square$

The main theorem uses only the one-sided bound (42), which holds for *any* $\chi_t \geq 0$; the smallness condition $\varepsilon_t < 1$ is needed only for the two-sided sandwich (44), which in turn is used by the explicit near-linear corollary and by any optional analysis that explicitly assumes a two-sided sandwich. An unrescaled current-parameter window is handled by Proposition 23 through a one-sided relation and does not require this condition. The binding quantity is the *whitened ℓ_2 accumulation* $\chi_t \leq \|\mathbf{D}_t\|_F$ of replayed-Jacobian drift, measured in the $\bar{\mathbf{C}}_t^{-1}$ metric—a relative, whitened ℓ_2 bound that can be sharper than an ℓ_1 sum normalized only by λ when the drift occurs in well-observed directions (it credits directions the design has already explored), though it does not dominate uniformly. Together, Lemmas 6 and 11 identify exact whitened diagnostic targets. The operational upper bounds in Lemma 7 replace those targets by the predictable path statistics Q_t and R_t , the certified optimizer residual ζ_t , and analytic smoothness constants. Thus $\bar{\psi}_t$ and $u_t = (1 + \bar{\chi}_t)^2$ require only $O(d)$ additional state; the exact whitened quantities remain post-hoc diagnostics used to measure the conservatism of these certificates.

Positive definiteness of both $\mathbf{C}_t$ and $\bar{\mathbf{C}}_t$ follows from $\lambda > 0$ and the outer-product form; it is not an independent assumption. All theoretical results hold under these explicit assumptions; we do *not* claim a regret guarantee for the full nonlinear model outside the local-linearization regime.

A.2 Novelty Relative to Prior Neural Bandits

NeuralUCB/NeuralTS use historical-time full-gradient Grams in their stated algorithms, while NeuralLinear and last-layer Laplace restrict uncertainty to a learned representation or final layer (Zhou et al., 2020; Zhang et al., 2021; Riquelme et al., 2018; Kristiadi et al., 2020). Matrix-free full-network predictive-variance solves also predate this work (Maddox et al., 2021). CC-UCB’s narrower adaptation is to replay every historical Jacobian at the current parameter, separate the predictable frozen confidence Gram from that algorithmic operator, and expose centering, drift, transfer, and residual-checked CG terms. We claim neither the CG/CVP primitive nor a prior assertion of uniform full-metric superiority as new.

The one-sided whitened result in Lemma 11 is the main technical step: optimism transfer needs no lower spectral comparison. The determinant formula is bookkeeping rather than a standalone rate, and the supplied- U closure is exactly the projected Gram in (36), not a concrete neural-training theorem. Appendix B compares NTK, regression, EKF/LO-FI, structured Laplace, and sketching alternatives.

A.3 Computable Curvature Operator

Applying $\mathbf{C}_t \mathbf{v} = \lambda \mathbf{v} + \sigma^{-2} \sum_{s < t} \mathbf{g}_t(\mathbf{x}_s, a_s) [\mathbf{g}_t(\mathbf{x}_s, a_s)^\top \mathbf{v}]$ *naively* requires one curvature–vector product (CVP) per historical sample per CG step—cost $O(It)$ CVPs per round. Each CVP evaluates one Jacobian–vector product (forward mode) followed by one vector–Jacobian product (reverse mode) through the network, at roughly the cost of one gradient evaluation (one JVP plus one VJP)¹; we do not store per-sample gradient features $\mathbf{g}_s$, only a constant number of $O(d)$ CG vectors for CG, with CVPs computed on the fly. We describe three strategies that limit the per-step cost to $|\mathcal{S}_t|$ CVPs:

(i) Subsampled GGN (worst-case spectral baseline). At each round draw a uniform subsample $\mathcal{S}_t \subset \{1, \dots, t-1\}$ of size n (without replacement) and set $w_s = (t-1)/n$ to obtain the rescaled operator. The following lemma is a worst-case uniform spectral guarantee.

Lemma 13 (Worst-case uniform spectral guarantee for subsampled GGN). *Fix round t and set $m = t-1$. Let $Y_s = \sigma^{-2} \mathbf{g}_t(\mathbf{x}_s, a_s) \mathbf{g}_t(\mathbf{x}_s, a_s)^\top$ for $s = 1, \dots, m$, so that $\mathbf{C}_t = \lambda \mathbf{I} + V$ with $V = \sum_{s=1}^m Y_s$. Each Y_s is PSD with $\|Y_s\|_{\text{op}} \leq G^2/\sigma^2 =: B$. For $\delta \in (0, 1)$ and $1 \leq n \leq m$, draw $\mathcal{S}_t$ of size n uniformly without replacement from $\{1, \dots, m\}$ and define $\hat{V} = \frac{m}{n} \sum_{s \in \mathcal{S}_t} Y_s$ and $\hat{\mathbf{C}}_t = \lambda \mathbf{I} + \hat{V}$. Write $L := \log(2d/\delta)$.*

(i) Deviation bound. *With probability $\geq 1 - \delta$,*

$$\|\hat{V} - V\|_{\text{op}} \leq mB \sqrt{\frac{2L}{n}} + \frac{4mBL}{3n}. \quad (45)$$

¹Implementations based on double-backward Hessian–vector products cost closer to two gradient-equivalents per CVP; the constant does not affect the $O(\cdot)$ accounting.

(ii) Sufficient sample size. For any $\varepsilon \in (0, 1)$, if

$$n \geq 8L \left[\frac{m^2 B^2}{\varepsilon^2 \lambda^2} + \frac{m B}{3 \varepsilon \lambda} \right], \quad (46)$$

then $\Pr[\|\hat{V} - V\|_{\text{op}} > \varepsilon \lambda] \leq \delta$ and consequently $(1-\varepsilon)\mathbf{C}_t \preceq \hat{\mathbf{C}}_t \preceq (1+\varepsilon)\mathbf{C}_t$.

Proof. The $\{Y_s\}_{s=1}^m$ are fixed PSD self-adjoint matrices (deterministic conditional on the history $\mathcal{F}_{t-1}$ and the current parameter θ_t). The displayed proof is for *with-replacement* (i.i.d.) sampling. The without-replacement version follows from the domination argument of Gross and Nemes (2010), building on Hoeffding’s reduction (Hoeffding, 1963): the matrix moment-generating function of a sum sampled without replacement from a fixed finite collection of self-adjoint matrices is dominated by that of the corresponding i.i.d. sum, so every tail bound derived from the matrix Laplace-transform method, including the matrix Bernstein inequality of Tropp (2015, Thm. 6.1.1), carries over with the same constants. In our setting the $\{Y_s\}$ are *fixed* PSD matrices (deterministic conditional on $\mathcal{F}_{t-1}$ and θ_t) and only the index set $\mathcal{S}_t$ is random, so the domination argument applies directly. We therefore state and prove the bound for i.i.d. draws $s_1, \dots, s_n$ uniform on $\{1, \dots, m\}$; the without-replacement guarantee is at least as strong. Define the centered random matrices $Z_k = (m/n)Y_{s_k} - V/n$ (mean zero, independent). The range satisfies $\|Z_k\|_{\text{op}} \leq (m/n)B + (1/n)\|V\| \leq 2mB/n =: R$ (using $\|V\| \leq mB$). For the variance parameter, since $Y_s \preceq B\mathbf{I}$,

$$\mathbb{E}[Z_k Z_k^\top] \preceq \frac{m^2}{n^2} \mathbb{E}[Y_{s_1}^2] = \frac{m}{n^2} \sum_{s=1}^m Y_s^2 \preceq \frac{m}{n^2} B V \preceq \frac{m^2 B^2}{n^2} \mathbf{I},$$

so $v := \|\sum_{k=1}^n \mathbb{E}[Z_k Z_k^\top]\| \leq m^2 B^2/n$. By the matrix Bernstein inequality (Tropp, 2015, Thm. 6.1.1), $\Pr[\|\hat{V} - V\| \geq t] \leq 2d \exp(-t^2/(2v + 2Rt/3))$. Choosing $t_0 = \sqrt{2vL} + \frac{2}{3}RL$ makes the matrix-Bernstein tail probability at most δ . Substituting $v \leq m^2 B^2/n$ and $R = 2mB/n$ gives $t_0 \leq mB\sqrt{2L/n} + \frac{4mBL}{3n}$, yielding (45). For (46), require each summand in (45) to be $\leq \varepsilon\lambda/2$: $mB\sqrt{2L/n} \leq \varepsilon\lambda/2$ gives $n \geq 8m^2 B^2 L/(\varepsilon^2 \lambda^2)$, and $4mBL/(3n) \leq \varepsilon\lambda/2$ gives $n \geq 8mBL/(3\varepsilon\lambda)$. Summing the two conditions yields (46); this is a conservative sufficient condition (using $a + b$ rather than $\max\{a, b\}$), not a sharp threshold. The multiplicative sandwich follows: $\|\hat{V} - V\| \leq \varepsilon\lambda$ gives $\mathbf{C}_t - \varepsilon\lambda\mathbf{I} \preceq \hat{\mathbf{C}}_t \preceq \mathbf{C}_t + \varepsilon\lambda\mathbf{I}$. Since $\mathbf{C}_t \succeq \lambda\mathbf{I}$, we have $\varepsilon\lambda\mathbf{I} \preceq \varepsilon\mathbf{C}_t$, so $\mathbf{C}_t - \varepsilon\lambda\mathbf{I} \succeq (1-\varepsilon)\mathbf{C}_t$ and $\mathbf{C}_t + \varepsilon\lambda\mathbf{I} \preceq (1+\varepsilon)\mathbf{C}_t$. $\square$

With $B = G^2/\sigma^2$, the right-hand side of the sufficient condition (46) is $O(L \cdot [m^2 G^4/(\varepsilon^2 \lambda^2 \sigma^4) + mG^2/(\varepsilon\lambda\sigma^2)])$. The first (variance) term dominates unless $\varepsilon\lambda\sigma^2 \gtrsim mG^2$, so this worst-case sufficient threshold has a *quadratic leading dependence* on $m = t-1$ (it is a sufficient upper bound on the required sample size, not an exact or necessary sample complexity). The condition (46) is conservative and sufficient, not necessary, with worst-case leading dependence quadratic in m . If its threshold exceeds the population size $m = t-1$, it provides no nontrivial subsampling guarantee for $n < m$. Taking $n = m$ recovers the full population exactly for the separate reason that every summand is included.

This is a worst-case baseline guarantee, too conservative for practical use at large t . In particular, the bound above does not by itself justify computational savings at large t ; it should be read as a worst-case spectral guarantee rather than as a practical scaling result. Subsampling reduces computation ($n \ll m$) only when m is moderate or $\varepsilon\lambda$ is large relative to mB ; for large t the full history may be needed under the uniform Bernstein bound. Under additional spectral structure, leverage-score sampling or Frequent Directions sketches may yield useful approximations with smaller state. For linear bandits, Kuzborskij et al. (2019) give regret bounds in terms of the Frequent Directions spectral-tail error. Translating such structure into a uniform GGN sandwich and a concrete buffer or sketch size is left open. The Bernstein guarantee is not the mechanism by which we expect practical savings at scale. Practical savings require small buffers treated heuristically, spectral decay/effective dimension, ridge-leverage sampling, Frequent Directions, or empirical verification of the spectral sandwich. We leave structure-aware sampling and empirical spectral certification as future work. To obtain a spectral sandwich that holds simultaneously for all $t \leq T$, union-bound over t (replacing δ by δ/T). Theorem 24 then exposes the one-sided transfer factor and realized width complexity as operator-dependent terms along the realized path; it does not identify a standalone causal regret inflation.

(ii) Windowed curvature (unrescaled, current parameter). Maintain a sliding window $\mathcal{S}_t \subseteq \{1, \dots, t-1\}$ of the most recent n interactions with unit weights ($w_s=1$), *re-evaluating the Jacobians at the current parameter θ_t* . This is a nonnegative-weighted *subset* of the same current-parameter outer products, so $\hat{\mathbf{C}}_t \preceq \mathbf{C}_t$ (Proposition 23);

Theorem 24 then applies with the *deterministic* approximate-to-full factor $\kappa_{+,t}=1$ and the full-to-frozen factor $u_t = (1 + \bar{\chi}_t)^2$ —no two-sided sandwich is required, though the overall guarantee remains conditional on that transfer certificate. The cost of the smaller operator is not hidden: it surfaces as larger realized widths $s_{\mathcal{C},t}^2(a_t)$ and hence a larger dynamic width complexity $\Lambda_T^{\mathcal{C}}$. The same holds for any current-parameter downweighted subset with $w_s \in [0, 1]$ (including an unrescaled subsample). A *stale* window that reuses Jacobians from an older parameter is *not* covered; see (iii).

(iii) Periodic refresh (heuristic; stale operator). Recompute the curvature operator every M rounds on a cached buffer and reuse it for M consecutive actions. Because the reused operator evaluates Jacobians at an *older* parameter, it is no longer a subset of the current-parameter outer products, so the $\hat{\mathbf{C}}_t \preceq \mathbf{C}_t$ argument of Proposition 23 does not apply and the stale operator can both over- and under-estimate $\mathbf{C}_t$. This remains a heuristic: Theorem 24 covers it only when a one-sided factor $\kappa_{+,t}$ (an explicit bound $\hat{\mathbf{C}}_t \preceq \kappa_{+,t}\mathbf{C}_t$) is separately verified, which we evaluate empirically. The rescaled subsample of (i), with weights $w_s = (t-1)/n > 1$, is likewise not covered by Proposition 23; it instead uses the two-sided matrix Bernstein bound of Lemma 13.

A.4 Computational Cost Model

Primitive assumptions. CC-UCB requires a single computational primitive: the *curvature–vector product* (CVP) $\mathbf{v} \mapsto \mathbf{C}_t \mathbf{v}$, which decomposes into per-sample operations $\mathbf{v} \mapsto \mathbf{g}_t(\mathbf{x}_s, a_s) [\mathbf{g}_t(\mathbf{x}_s, a_s)^\top \mathbf{v}]$. Each such operation consists of:

1. a Jacobian–vector product (JVP) $\mathbf{v} \mapsto \mathbf{g}_t^\top \mathbf{v}$ (forward-mode AD, cost $\approx 1 \times$ forward pass);
2. a vector–Jacobian product (VJP) $\alpha \mapsto \alpha \mathbf{g}_t$ (reverse-mode AD, cost $\approx 1 \times$ backward pass).

Sharded autodiff can distribute JVP/VJP evaluation, but its communication pattern and sample parallelism are implementation-dependent and are not benchmarked here. For $m_t := |\mathcal{S}_t|$, one matrix-free oracle application consists of m_t sample-CVPs plus $O(d)$ damping and accumulation work. Thus $I_{t,a}$ CG recurrences for action a use $I_{t,a}m_t$ sample-CVPs; an explicit final check of the original-system residual $\mathbf{g}_t(a) - \mathbf{C}_t \tilde{\mathbf{u}}$ uses another m_t . The sequential Krylov state is $O(d)$ ($O(Kd)$ if right-hand sides are batched), excluding model and optimizer state, activations, and raw replay. Caching an explicit $m_t \times d$ Jacobian matrix instead uses $O(m_t d)$ additional state and is not the on-the-fly JVP/VJP implementation analyzed by this operation count.

Krylov methods for Hessian spectral analysis have been run on networks with $> 10^8$ parameters under distributed training (Granziol and Juarev, 2026), which indicates that curvature–vector primitives of this general form are implementable at scale. We do *not* claim foundation-model feasibility from this: that work times Hessian–vector products, not the exact per-example GGN JVP/VJP used here, and we report no wall-clock benchmark of our primitive. Warm-starting CG from the previous round’s solution can reduce the iteration count in practice.

Cost accounting. Let $C_F(W_{\text{net}})$, $C_B(W_{\text{net}})$, $C_J(W_{\text{net}})$, and $C_V(W_{\text{net}})$ denote the one-example forward, reverse, JVP, and VJP costs for model-compute scale W_{net} , and define

$$C_{\text{op},t} := m_t [C_J(W_{\text{net}}) + C_V(W_{\text{net}})] + O(d).$$

Parameter count d alone does not determine these costs when architecture, input length, or weight sharing changes. Let J_{t+1}^{opt} be the number of full-batch first-order steps after round t ; $q_{t+1}^{\text{opt}} \in \{0, 1\}$ records a separate full-gradient optimizer certificate, and $q_{t,a}^{\text{CG}} \in \{0, 1\}$ records an explicit final CG residual application. Table 3 separates the training history N_t from the curvature buffer m_t .

Table 3: Per-round arithmetic cost. $N_t = t - 1$ is the training-history size; $m_t = |\mathcal{S}_t|$ is the curvature-buffer size.

Component	Cost
Candidate means/features	$O(K[C_F + C_B])$
CG recurrences/checks	$C_{\text{op},t} \sum_{a=1}^K (I_{t,a} + q_{t,a}^{\text{CG}})$
Parameter update/certificate	$(J_{t+1}^{\text{opt}} + q_{t+1}^{\text{opt}}) \{(N_t + 1)[C_F + C_B] + O(d)\}$
Path statistics/scoring	$O(d + K)$

This is an operation count, not a wall-clock benchmark. For illustration, $K = 10$, $I = 25$, and $m_t = 256$ give 64,000 sample-CVPs for the recurrences and 66,560 when every solve receives one explicit final residual

application. Both numbers exclude candidate gradients, full-history model training, and optimizer certification. Backtracking adds full-history objective passes, and second-order optimization has additional Hessian costs. A linear sufficient-statistic update may substitute a lower special-case cost, but must still supply the theorem’s optimizer certificate.

Cost of the analyzed configuration. The full-history instantiation $\mathcal{C}_t = \mathbf{C}_t$, $\mathcal{S}_t = \{1, \dots, t-1\}$ with adaptive CG stopping has the certificate $\bar{\kappa}_t = 1 + (t-1)G^2/(\lambda\sigma^2)$, so maintaining $\max_a \varepsilon_{t,a} \leq \bar{\varepsilon}_t$ is certified by $I_t = O(\sqrt{t}G/(\sigma\sqrt{\lambda}) \log(1/\bar{\varepsilon}_t))$ iterations per solve (Eq. (76)), so the per-round CG cost is $K I_t (t-1) = O(K t^{3/2})$ sample-CVPs, i.e. $O(K T^{5/2})$ cumulatively—the worst-case width-solve sample-CVP count under the stated CG certificate. End-to-end work additionally contains $\sum_{t \leq T} (J_t^{\text{opt}} + q_t^{\text{opt}}) N_t$ full-data sample-gradient evaluations. This is $O((J+q)T^2)$ only when those counts are uniformly bounded, and curvature windowing does not reduce it. Table 3 and the numerical example describe fixed-buffer operators ($m_t \ll N_t$), which reduce width-solve work but are covered by Theorem 1 only under the corrected window/subsample conditions of §A.3: an unrescaled current-parameter buffer has the deterministic approximate-to-full factor $\kappa_{+,t}=1$ (Proposition 23)—the overall guarantee remaining conditional on the full-to-frozen transfer and the other certificates of Theorem 1—while a rescaled or stale buffer additionally requires the verified transfer factor of Theorem 24.

For the growing window of Theorem 3, one solve costs $O(m_t^{3/2} \log(1/\bar{\varepsilon}))$ sample-CVPs under the analytic condition-number bound. The following table uses the sufficient fixed-burn-in setting $\tau_w = O(1)$, $H_T = \tilde{O}(1)$, and $E_T = \tilde{O}(T^{1-q/2})$; $E_T = o(T)$ alone is insufficient. The sample-CVP column counts only width solves. With uniformly bounded full-batch optimizer/certificate passes, the final column adds the T^2 full-history training term. These are achieved upper bounds for this implementation, not a Pareto frontier, lower bound, or wall-clock measurement (logarithmic factors suppressed):

q	conditional regret	width-solve CVPs	full-policy sample work
$\frac{1}{2}$	$T^{\frac{3}{4}}$	$KT^{\frac{7}{4}}$	T^2
$\frac{2}{3}$	$T^{\frac{2}{3}}$	KT^2	T^2
1	$T^{\frac{1}{2}}$	$KT^{\frac{5}{2}}$	$T^{\frac{5}{2}}$

A.5 An Exact Log-Determinant Bookkeeping Identity

The regret bound is driven not by γ_T but by the realized *width complexity* of the widths the algorithm actually uses,

$$\Lambda_T^{\mathcal{C}} := \sum_{t=1}^T \log(1 + \sigma^{-2} s_{\mathcal{C},t}^2(a_t)). \quad (47)$$

Unlike $\bar{\mathbf{C}}_t$, the algorithmic curvature $\mathcal{C}_t$ need *not* be monotone in t : relinearization, subsampling, windowing, and periodic refresh can all *shrink* it, so the elementary determinant telescoping of (25) no longer applies. The following lemma is an exact pathwise bookkeeping identity for a general predictable SPD sequence. It is not a standalone rate result: its upper bound is informative only when the terminal log determinant and determinant-decreasing refresh charge are independently controlled, as in Lemma 15.

Lemma 14 (Dynamic log-determinant potential). *Let $\{\mathcal{C}_t\}_{t \geq 1}$ satisfy (38) with $\mathcal{C}_1 = \lambda \mathbf{I}$. For the played feature $\mathbf{h}_t := \mathbf{g}_t(\mathbf{x}_t, a_t)$ define the rank-one update $\mathcal{C}_t^+ := \mathcal{C}_t + \sigma^{-2} \mathbf{h}_t \mathbf{h}_t^\top$ and, using the symmetric (principal) root $(\mathcal{C}_t^+)^{-1/2}$, the normalized perturbation*

$$\Xi_t := (\mathcal{C}_t^+)^{-1/2} (\mathcal{C}_{t+1} - \mathcal{C}_t^+) (\mathcal{C}_t^+)^{-1/2}, \quad (48)$$

so that $\mathbf{I} + \Xi_t = (\mathcal{C}_t^+)^{-1/2} \mathcal{C}_{t+1} (\mathcal{C}_t^+)^{-1/2}$ is SPD; equivalently, every eigenvalue of Ξ_t exceeds -1 and every eigenvalue of $\mathbf{I} + \Xi_t$ is positive (with no smallness condition on the jump $\mathcal{C}_{t+1} - \mathcal{C}_t^+$). Fix any end-of-round- T terminal operator $\mathcal{C}_{T+1}$ constructed only from data and randomness available through round T ($\mathcal{F}_T$ -measurable, hence not a function of $\mathbf{x}_{T+1}$, $\mathcal{A}_{T+1}$, or future rewards; e.g. the canonical rank-one update $\mathcal{C}_{T+1} = \mathcal{C}_T^+$, or the operator the algorithm commits for round $T+1$). The identity below is then an a posteriori statement about the completed sequence $\mathcal{C}_1, \dots, \mathcal{C}_{T+1}$; the quantities Ξ_T , $V_T^{\mathcal{C}}$, and Γ_T^{dyn} are defined relative to this choice, and the

bound $\Lambda_T^C \leq \Gamma_T^{\text{dyn}}$ holds for every such valid choice. Then

$$\Lambda_T^C = \log \frac{\det(\mathcal{C}_{T+1})}{\det(\lambda \mathbf{I})} - \sum_{t=1}^T \log \det(\mathbf{I} + \Xi_t). \quad (49)$$

With the determinant-destroying variation charge and the dynamic potential

$$V_T^C := \sum_{t=1}^T [-\log \det(\mathbf{I} + \Xi_t)]_+, \quad \Gamma_T^{\text{dyn}} := \log \frac{\det(\mathcal{C}_{T+1})}{\det(\lambda \mathbf{I})} + V_T^C, \quad (50)$$

we have $\Lambda_T^C \leq \Gamma_T^{\text{dyn}}$, and, using $s_{\mathcal{C},t}^2(a_t) \leq G^2/\lambda$ and $\log(1+x) \geq x/(1+x)$,

$$\sum_{t=1}^T s_{\mathcal{C},t}^2(a_t) \leq (\sigma^2 + G^2/\lambda) \Lambda_T^C \leq (\sigma^2 + G^2/\lambda) \Gamma_T^{\text{dyn}}. \quad (51)$$

Proof. By the matrix determinant lemma applied to $\mathcal{C}_t^+ = \mathcal{C}_t + \sigma^{-2} \mathbf{h}_t \mathbf{h}_t^\top$,

$$\frac{\det(\mathcal{C}_t^+)}{\det(\mathcal{C}_t)} = 1 + \sigma^{-2} \mathbf{h}_t^\top \mathcal{C}_t^{-1} \mathbf{h}_t = 1 + \sigma^{-2} s_{\mathcal{C},t}^2(a_t). \quad (52)$$

Writing $B_t := (\mathcal{C}_t^+)^{-1/2}$, we have $\mathbf{I} + \Xi_t = B_t \mathcal{C}_t^+ B_t + B_t (\mathcal{C}_{t+1} - \mathcal{C}_t^+) B_t = B_t \mathcal{C}_{t+1} B_t$, a congruence of the SPD matrix $\mathcal{C}_{t+1}$ by the invertible symmetric B_t , hence SPD (in particular its eigenvalues exceed -1 with no smallness condition on the jump $\mathcal{C}_{t+1} - \mathcal{C}_t^+$), and

$$\frac{\det(\mathcal{C}_{t+1})}{\det(\mathcal{C}_t^+)} = \det(B_t^{-1}(\mathbf{I} + \Xi_t)B_t^{-1}) / \det(\mathcal{C}_t^+) = \det(\mathbf{I} + \Xi_t), \quad (53)$$

using $\det(\mathcal{C}_{t+1}) = \det(\mathcal{C}_t^+) \det(\mathbf{I} + \Xi_t)$. Taking logs of (52)–(53) and adding,

$$\log(1 + \sigma^{-2} s_{\mathcal{C},t}^2(a_t)) = \log \det(\mathcal{C}_{t+1}) - \log \det(\mathcal{C}_t) - \log \det(\mathbf{I} + \Xi_t).$$

Summing over $t = 1, \dots, T$ telescopes the first difference to $\log \det(\mathcal{C}_{T+1}) - \log \det(\mathcal{C}_1)$; with $\mathcal{C}_1 = \lambda \mathbf{I}$ this is $\log[\det(\mathcal{C}_{T+1})/\det(\lambda \mathbf{I})]$, proving (49). Since $-\log \det(\mathbf{I} + \Xi_t) \leq [-\log \det(\mathbf{I} + \Xi_t)]_+$ term by term, $-\sum_t \log \det(\mathbf{I} + \Xi_t) \leq V_T^C$, so $\Lambda_T^C \leq \Gamma_T^{\text{dyn}}$. For (51): $\mathcal{C}_t \succeq \lambda \mathbf{I}$ gives $s_{\mathcal{C},t}^2(a_t) = \mathbf{h}_t^\top \mathcal{C}_t^{-1} \mathbf{h}_t \leq \|\mathbf{h}_t\|^2/\lambda \leq G^2/\lambda =: c_{\max} \sigma^2$. With $x_t := \sigma^{-2} s_{\mathcal{C},t}^2(a_t) \in [0, c_{\max}]$ and $\log(1+x) \geq x/(1+x) \geq x/(1+c_{\max})$, $\Lambda_T^C \geq (1+c_{\max})^{-1} \sum_t x_t$, i.e. $\sum_t s_{\mathcal{C},t}^2(a_t) \leq (\sigma^2 + G^2/\lambda) \Lambda_T^C$; the second inequality is $\Lambda_T^C \leq \Gamma_T^{\text{dyn}}$. $\square$

Remark (Interpretation and scope of the dynamic potential). *Identity (49) is pathwise. V_T^C charges only changes that reduce determinant relative to the rank-one update; determinant-increasing changes contribute zero. If $\mathcal{C}_1 \neq \lambda \mathbf{I}$, the denominator becomes $\det(\mathcal{C}_1)$. Neither V_T^C nor the endpoint term is generally small. Bounding both is sufficient, but not necessary because the exact identity may cancel them. Thus the identity does not imply sublinear regret; outside Corollary 9 and Theorem 3, controlling Λ_T^C for an adapting backbone remains open.*

Lemma 15 (Rank-sensitive refresh bound). *Let $\mathcal{R} = \{t : \mathcal{C}_{t+1} \neq \mathcal{C}_t^+\}$. Suppose that for every $t \in \mathcal{R}$, the negative part $(\Xi_t)_- = (-\Xi_t)_+$ has rank at most r_t and $\lambda_{\min}(\mathbf{I} + \Xi_t) \geq 1 - \nu_t$ for $0 \leq \nu_t < 1$. Then*

$$V_T^C \leq \sum_{t \in \mathcal{R}} r_t \log \frac{1}{1 - \nu_t}. \quad (54)$$

If additionally $\mathcal{C}_{T+1} = \lambda \mathbf{I} + A$, where $A \succeq 0$, $\text{rank}(A) \leq r_{\text{end}}$, and $\text{tr}(A) \leq W_{\text{end}} G^2/\sigma^2$ for a specified total endpoint weight $W_{\text{end}} \geq 0$, then, for $r_{\text{end}} > 0$,

$$\Lambda_T^C \leq r_{\text{end}} \log \left(1 + \frac{W_{\text{end}} G^2}{r_{\text{end}} \lambda \sigma^2} \right) + \sum_{t \in \mathcal{R}} r_t \log \frac{1}{1 - \nu_t}. \quad (55)$$

For $r_{\text{end}} = 0$, the first term is defined as zero.

Proof. Let $\xi_{t,i}$ be the eigenvalues of Ξ_t . At most r_t are negative, each is at least $-\nu_t$, and every nonnegative eigenvalue makes a nonpositive contribution to $-\sum_i \log(1 + \xi_{t,i})$. Hence $[-\log \det(\mathbf{I} + \Xi_t)]_+ \leq r_t \log(1/(1 - \nu_t))$, proving (54). List the nonzero eigenvalues of A and pad with zeros to obtain $a_1, \dots, a_{r_{\text{end}}}$. Concavity gives

$$\sum_i \log(1 + a_i/\lambda) \leq r_{\text{end}} \log\left(1 + \frac{\sum_i a_i}{r_{\text{end}}\lambda}\right),$$

and the trace premise plus Lemma 14 proves (55). $\square$

Consequences. Sequential Gram updates have $\Xi_t = 0$. If canonical rank-one updates are used outside J_T refresh rounds, and each refresh has $r_t \leq r$, $\nu_t \leq \nu < 1$, then $V_T^C \leq J_T r \log(1/(1 - \nu))$; a geometric schedule is therefore polylogarithmic only when these rank and spectral-floor bounds hold. For a replacement $\mathcal{C}_{t+1} - \mathcal{C}_t^+ = A_t^{\text{new}} - A_t^{\text{old}}$ with both terms PSD and $\text{rank}(A_t^{\text{old}}) \leq r$, Sylvester inertia gives $r_t \leq r$, but a spectral floor is still required.

For a fixed-feature window of length m , after warm-up each deletion has rank one. Writing $N_{\text{drop}} = \max\{0, T - m\}$, Sherman–Morrison gives the following exact charge for each $t > m$. Let $v_t^{\text{drop}} = \sigma^{-1} \mathbf{g}_{t-m}(\mathbf{x}_{t-m}, a_{t-m})$ and let $D_t = \mathcal{C}_t^+ - v_t^{\text{drop}}(v_t^{\text{drop}})^\top = \mathcal{C}_{t+1}$ be the remaining-window matrix. Then $-\log \det(\mathbf{I} + \Xi_t) = \log[1 + (v_t^{\text{drop}})^\top D_t^{-1} v_t^{\text{drop}}]$, and therefore

$$V_T^C \leq N_{\text{drop}} \log\left(1 + \frac{G^2}{\lambda \sigma^2}\right).$$

This can be linear in T . A relinearized window need not be a rank-one replacement. Likewise, merely holding a stale operator fixed does not make non-refresh transitions canonical rank-one updates, so geometric refresh timing alone does not imply a logarithmic charge.

Corollary 16 (Observable CG upper bound on the width complexity). *If the played-action CG solve satisfies $\varepsilon_{t,a} \leq \bar{\varepsilon}_t < 1$ so that $\tilde{s}_t^2(a_t) \geq (1 - \bar{\varepsilon}_t) s_{\mathcal{C},t}^2(a_t)$ (Lemma 17), then*

$$\Lambda_T^C \leq \widehat{\Lambda}_T^C := \sum_{t=1}^T \log\left(1 + \frac{\tilde{s}_t^2(a_t)}{\sigma^2(1 - \bar{\varepsilon}_t)}\right). \quad (56)$$

Proof. $s_{\mathcal{C},t}^2(a_t) \leq \tilde{s}_t^2(a_t)/(1 - \bar{\varepsilon}_t)$ and $x \mapsto \log(1 + \sigma^{-2}x)$ is increasing, so each summand of Λ_T^C is dominated term by term. $\square$

This gives an observable upper bound on the dynamic width-complexity term, conditional on the certified CG tolerances; V_T^C itself is *not* observable in general (it requires determinant estimation of the curvature sequence).

A.6 CG Approximation Lemma

Lemma 17 (CG multiplicative accuracy). *Fix a round t and an action a with right-hand side $\mathbf{g} := \mathbf{g}_t(a) \neq \mathbf{0}$. Let $\mathbf{u} = \mathcal{C}_t^{-1} \mathbf{g}$ and let $\tilde{\mathbf{u}}$ be the iterate after $I_{t,a}$ steps of CG on the SPD system $\mathcal{C}_t \mathbf{u} = \mathbf{g}$ starting from $\tilde{\mathbf{u}}_0 = \mathbf{0}$, with $\mathcal{C}_t$ held fixed across all iterations. The relative energy-norm error is the per-action quantity $\varepsilon_{t,a}$ of (9), and*

$$\varepsilon_{t,a} = \frac{\|\mathbf{u} - \tilde{\mathbf{u}}\|_{\mathcal{C}_t}}{\|\mathbf{u}\|_{\mathcal{C}_t}} \leq 2 \left(\frac{\sqrt{\bar{\kappa}_t} - 1}{\sqrt{\bar{\kappa}_t} + 1} \right)^{I_{t,a}}, \quad (57)$$

where $\kappa_t := \lambda_{\max}(\mathcal{C}_t)/\lambda_{\min}(\mathcal{C}_t)$ is the true condition number and $\bar{\kappa}_t \geq \kappa_t$ is a certified $\mathcal{H}_t$ -measurable upper bound. The convergence rate in (57) is right-hand-side independent, so it holds for every action with the same $\bar{\kappa}_t$. For a nonnegative weighted outer-product operator $\mathcal{C}_t = \lambda \mathbf{I} + \sigma^{-2} \sum_{s \in \mathcal{S}_t} w_s \mathbf{g}_t(\mathbf{x}_s, a_s) \mathbf{g}_t(\mathbf{x}_s, a_s)^\top$ ($w_s \geq 0$, so $\mathcal{C}_t \succeq \lambda \mathbf{I} \succ 0$ and CG is well-posed),

$$\kappa_t \leq \frac{\lambda_{\max}(\mathcal{C}_t)}{\lambda} \leq 1 + \frac{W_t G^2}{\lambda \sigma^2} =: \bar{\kappa}_t,$$

with $W_t = \sum_{s \in \mathcal{S}_t} w_s$ and $\|\mathbf{g}_t(\mathbf{x}_s, a_s)\| \leq G$ from Assumption 2; here $W_t = n$ for unrescaled windowed/refresh buffers, giving $\bar{\kappa}_t$ and α_t independent of T , and $W_t = t-1$ for full history and for rescaled uniform subsampling with $w_s = (t-1)/n$. For a general SPD operator that is not such an outer product, $\bar{\kappa}_t$ must be certified separately. Write $\tilde{s}_t^2(a) = \mathbf{g}^\top \tilde{\mathbf{u}}$ and $s_{\mathcal{C},t}^2(a) = \mathbf{g}^\top \mathcal{C}_t^{-1} \mathbf{g}$.

Assume $\varepsilon_{t,a} < 1$ (see Appendix E for the minimum $I_{t,a}$ ensuring this as a function of $\bar{\kappa}_t$; Algorithm 1 enforces a known per-round tolerance $\bar{\varepsilon}_t$ uniformly over the scored actions, $\max_a \varepsilon_{t,a} \leq \bar{\varepsilon}_t < 1$, via adaptive residual stopping or a fixed budget under a certified condition-number bound; see §3). Then

$$(1 - \varepsilon_{t,a}) s_{\mathcal{C},t}^2(a) \leq \tilde{s}_t^2(a) \leq (1 + \varepsilon_{t,a}) s_{\mathcal{C},t}^2(a). \quad (58)$$

For $\mathbf{g}_t(a) = \mathbf{0}$ the algorithm sets $\tilde{s}_t^2(a) = 0 = s_{\mathcal{C},t}^2(a)$ and (58) holds trivially.

Proof. The CG minimax property gives $\|e_I\|_{\mathcal{C}_t}/\|e_0\|_{\mathcal{C}_t} \leq \min_{p \in \mathcal{P}_I, p(0)=1} \max_{x \in [\lambda_{\min}, \lambda_{\max}]} |p(x)|$. Choosing the scaled Chebyshev polynomial bounds this maximum by $2[(\sqrt{\kappa_t} - 1)/(\sqrt{\kappa_t} + 1)]^I$; this ratio is increasing in κ_t , so $\bar{\kappa}_t \geq \kappa_t$ proves (57). $\tilde{s}_t^2(a) - s_{\mathcal{C},t}^2(a) = \mathbf{g}^\top(\tilde{\mathbf{u}} - \mathbf{u}) = (\mathcal{C}_t \mathbf{u})^\top(\tilde{\mathbf{u}} - \mathbf{u}) = \mathbf{u}^\top \mathcal{C}_t(\tilde{\mathbf{u}} - \mathbf{u})$. By Cauchy-Schwarz in the $\mathcal{C}_t$ -norm, $|\mathbf{u}^\top \mathcal{C}_t(\tilde{\mathbf{u}} - \mathbf{u})| \leq \|\mathbf{u}\|_{\mathcal{C}_t} \|\tilde{\mathbf{u}} - \mathbf{u}\|_{\mathcal{C}_t} \leq \varepsilon_{t,a} \|\mathbf{u}\|_{\mathcal{C}_t}^2 = \varepsilon_{t,a} s_{\mathcal{C},t}^2(a)$. For the outer-product special case the condition-number bound follows from $\lambda_{\max}(\mathcal{C}_t) \leq \lambda + \sigma^{-2} \sum_{s \in \mathcal{S}_t} w_s \|\mathbf{g}_t(\mathbf{x}_s, a_s)\|^2 \leq \lambda + W_t G^2 / \sigma^2$. $\square$

The algorithmic bonus. With the *per-round* certified tolerance $\bar{\varepsilon}_t \geq \max_a \varepsilon_{t,a}$ and the one-sided transfer factor $u_t(a) \geq 1$ of Assumption 5, Algorithm 1 plays the bonus

$$w_t(a) := \omega_t \sqrt{\frac{u_t(a)}{1 - \bar{\varepsilon}_t}} \sqrt{\tilde{s}_t^2(a)}, \quad (59)$$

where $\bar{\beta}_t \geq \beta_t$ is the predictable confidence upper bound, $\bar{\psi}_t$ the centering certificate (Assumption 4), and

$$\omega_t := \bar{\beta}_t + \bar{\psi}_t \quad (60)$$

is the combined confidence-plus-centering width factor. Define the *per-round* CG inflation factor $\alpha_t := \sqrt{(1 + \bar{\varepsilon}_t)/(1 - \bar{\varepsilon}_t)}$.

Lemma 18 (Bonus sandwiching). *Under Assumption 5 and $\max_{a \in \mathcal{A}_t} \varepsilon_{t,a} \leq \bar{\varepsilon}_t < 1$, the bonus (59) satisfies, for every $a \in \mathcal{A}_t$,*

$$\omega_t \bar{s}_t(a) \leq w_t(a) \leq \alpha_t \omega_t \sqrt{u_t(a)} s_{\mathcal{C},t}(a). \quad (61)$$

Proof. For $\mathbf{g}_t(a) = \mathbf{0}$ all three terms vanish and the bound is trivial. Otherwise Lemma 17 with $\varepsilon_{t,a} \leq \bar{\varepsilon}_t$ gives $(1 - \bar{\varepsilon}_t) s_{\mathcal{C},t}^2(a) \leq \tilde{s}_t^2(a) \leq (1 + \bar{\varepsilon}_t) s_{\mathcal{C},t}^2(a)$. The lower bound and $\omega_t \geq 0$ give $\sqrt{\tilde{s}_t^2(a)} \geq \sqrt{1 - \bar{\varepsilon}_t} s_{\mathcal{C},t}(a)$, so $w_t(a) \geq \omega_t \sqrt{u_t(a)} s_{\mathcal{C},t}(a)$; then (40) gives $\sqrt{u_t(a)} s_{\mathcal{C},t}(a) \geq \bar{s}_t(a)$, i.e. $w_t(a) \geq \omega_t \bar{s}_t(a)$. The upper bound gives $\sqrt{\tilde{s}_t^2(a)} \leq \sqrt{1 + \bar{\varepsilon}_t} s_{\mathcal{C},t}(a)$, whence $w_t(a) \leq \omega_t \sqrt{u_t(a)/(1 - \bar{\varepsilon}_t)} \sqrt{1 + \bar{\varepsilon}_t} s_{\mathcal{C},t}(a) = \alpha_t \omega_t \sqrt{u_t(a)} s_{\mathcal{C},t}(a)$. Both bounds are per-round; no horizon maximum is taken before summing. $\square$

A.7 Confidence and Discrepancy Lemmas

Lemma 19 (Confidence for the linearized ridge estimator). *Suppose Assumptions 1 to 3 hold and $\mu^* = \mu_{\theta^*}$ with $\|\theta^*\| \leq S$. Define the linearized ridge estimator $\hat{\theta}_t^{\text{lin}}$ by (71), the frozen-feature variance $\bar{s}_t(a) = \sqrt{\mathbf{g}_t(a)^\top \bar{\mathbf{C}}_t^{-1} \mathbf{g}_t(a)}$, and the cumulative squared linearization error $F_t := \sum_{s=1}^{t-1} \varepsilon_{\text{lin}}(s)^2$. Then with probability at least $1 - \delta$, for all $t \geq 1$ and all $a \in \mathcal{A}_t$ simultaneously,*

$$|\mathbf{g}_t(a)^\top (\hat{\theta}_t^{\text{lin}} - \theta^*)| \leq \bar{s}_t(a) \beta_t, \quad (62)$$

where

$$\beta_t := \sqrt{\gamma_{t-1} + 2 \log(1/\delta)} + \sqrt{\lambda} S + \frac{1}{\sigma} \sqrt{F_t}. \quad (63)$$

(Here $\gamma_{t-1} = \log \det(\bar{\mathbf{C}}_t) / \det(\lambda \mathbf{I})$ is the information gain through round $t-1$.)

The quantity β_t in (63) is a theoretical radius. The algorithm may use any predictable upper bound $\bar{\beta}_t \geq \beta_t$; all results below remain valid with $\bar{\beta}_t$ in place of β_t . This is necessary because β_t depends on the unknown cumulative linearization term F_t .

Proof. From (70), $y_s = \mathbf{g}_s^\top \boldsymbol{\theta}^* + \rho_s + \eta_s$ with $|\rho_s| \leq \varepsilon_{\text{lin}}(s)$. The closed form of $\hat{\boldsymbol{\theta}}_t^{\text{lin}}$ gives

$$\hat{\boldsymbol{\theta}}_t^{\text{lin}} - \boldsymbol{\theta}^* = -\lambda \bar{\mathbf{C}}_t^{-1} \boldsymbol{\theta}^* + \bar{\mathbf{C}}_t^{-1} \sigma^{-2} \sum_{s < t} \mathbf{g}_s \rho_s + \bar{\mathbf{C}}_t^{-1} \sigma^{-2} \sum_{s < t} \mathbf{g}_s \eta_s.$$

Taking the inner product with $\mathbf{g}_t(a)$ and bounding in absolute value:

- (i) *Prior bias.* $|\lambda \mathbf{g}_t(a)^\top \bar{\mathbf{C}}_t^{-1} \boldsymbol{\theta}^*| \leq \lambda \bar{s}_t(a) \|\boldsymbol{\theta}^*\|_{\bar{\mathbf{C}}_t^{-1}} \leq \bar{s}_t(a) \sqrt{\lambda} S$. (Cauchy–Schwarz in the $\bar{\mathbf{C}}_t$ -norm; $\lambda \|\boldsymbol{\theta}^*\|_{\bar{\mathbf{C}}_t^{-1}} \leq \lambda \|\boldsymbol{\theta}^*\| / \sqrt{\lambda_{\min}(\bar{\mathbf{C}}_t)} \leq \sqrt{\lambda} S$ since $\bar{\mathbf{C}}_t \succeq \lambda \mathbf{I}$.)
- (ii) *Linearization bias.* Define $\Phi = [\mathbf{g}_1/\sigma, \dots, \mathbf{g}_{t-1}/\sigma]$ (the $d \times (t-1)$ design matrix) and $\hat{\rho} = (\rho_1, \dots, \rho_{t-1})^\top$. Then $\sigma^{-2} \sum \mathbf{g}_s \rho_s = \sigma^{-1} \Phi \hat{\rho}$. Since $\bar{\mathbf{C}}_t \succeq \Phi \Phi^\top$, we have $\Phi^\top \bar{\mathbf{C}}_t^{-1} \Phi \preceq \mathbf{I}$, so $\|\Phi \hat{\rho}\|_{\bar{\mathbf{C}}_t^{-1}} = \|\hat{\rho}\|_{\Phi^\top \bar{\mathbf{C}}_t^{-1} \Phi} \leq \|\hat{\rho}\| = \sqrt{F_t}$. Therefore $|\mathbf{g}_t(a)^\top \bar{\mathbf{C}}_t^{-1} \sigma^{-1} \Phi \hat{\rho}| \leq \bar{s}_t(a) \sigma^{-1} \sqrt{F_t}$.
- (iii) *Noise.*

$$\left\| \sum_{s < t} \frac{\mathbf{g}_s}{\sigma} \frac{\eta_s}{\sigma} \right\|_{\bar{\mathbf{C}}_t^{-1}} \leq \sqrt{\gamma_{t-1} + 2 \log(1/\delta)}$$

holds for all $t \geq 1$ with probability at least $1 - \delta$ by the time-uniform vector self-normalized martingale bound (Abbasi-Yadkori et al., 2011). Here $x_s = \mathbf{g}_s/\sigma$ is predictable with respect to the pre-reward sigma-algebra $\mathcal{G}_s$, and $\xi_s = \eta_s/\sigma$ is conditionally 1-sub-Gaussian given $\mathcal{G}_s$. For any candidate action a , Cauchy–Schwarz in the $\bar{\mathbf{C}}_t$ -norm gives

$$\left| \mathbf{g}_t(a)^\top \bar{\mathbf{C}}_t^{-1} \sum_{s < t} \frac{\mathbf{g}_s \eta_s}{\sigma^2} \right| \leq \bar{s}_t(a) \sqrt{\gamma_{t-1} + 2 \log(1/\delta)}.$$

Summing gives (62)–(63). $\square$

Corollary 20 (Predictable confidence schedule). *Suppose a nonnegative $\mathcal{H}_t$ -measurable sequence $\{\bar{F}_t\}_{t \geq 1}$ satisfies $F_t \leq \bar{F}_t$ for all t . Then Algorithm 1 may set*

$$\bar{\beta}_t := \sqrt{\gamma_{t-1} + 2 \log(1/\delta)} + \sqrt{\lambda} S + \sigma^{-1} \sqrt{\bar{F}_t},$$

which is predictable and satisfies $\bar{\beta}_t \geq \beta_t$.

Proof. $\bar{F}_t \geq F_t$ term-by-term implies each summand in (63) is dominated, so $\bar{\beta}_t \geq \beta_t$. Predictability holds because γ_{t-1} , S , $\bar{F}_t$, λ , σ , and δ are all determined before round t . $\square$

Lemma 7 supplies the computable choice $\bar{F}_t = \sum_{s < t} (\bar{\epsilon}_s^{\text{lin}})^2$; a tighter externally certified predictable sequence may replace it.

Corollary 21 (Observable information-gain surrogate via the one-sided transfer factor). *Run Algorithm 1 with a predictable algorithmic curvature sequence $\mathcal{C}_s$, per-round CG tolerance with $\varepsilon_{s, a_s} \leq \bar{\varepsilon}_s < 1$ for the played action, and an $\mathcal{H}_s$ -measurable one-sided optimism-transfer factor $u_s(a) \geq 1$ satisfying $\bar{s}_s^2(a) \leq u_s(a) s_{\mathcal{C}, s}^2(a)$ (Assumption 5). Suppose also that the nonnegative predictable sequence $\bar{F}_t$ satisfies $\bar{F}_t \geq F_t$. Define the observable information-gain surrogate*

$$\hat{\gamma}_{t-1} := \sum_{s=1}^{t-1} \log \left(1 + \sigma^{-2} \frac{u_s(a_s)}{1 - \bar{\varepsilon}_s} \bar{s}_s^2(a_s) \right),$$

where $\bar{s}_s^2(a_s) = \mathbf{g}_s(a_s)^\top \hat{\mathbf{u}}_s$ is the CG variance proxy computed at round s for the played action. Then $\hat{\gamma}_{t-1}$ is $\mathcal{H}_t$ -measurable and $\hat{\gamma}_{t-1} \geq \gamma_{t-1}$, so

$$\hat{\beta}_t := \sqrt{\hat{\gamma}_{t-1} + 2 \log(1/\delta)} + \sqrt{\lambda} S + \sigma^{-1} \sqrt{\bar{F}_t}$$

is a predictable schedule with $\hat{\beta}_t \geq \beta_t$. Played-action CG accuracy is sufficient for this information comparison. If the same tolerance also holds uniformly for every candidate action before selection, then Theorem 1 holds with $\bar{\beta}_t = \hat{\beta}_t$.

Proof. The transfer factor gives $\bar{s}_s^2(a_s) \leq u_s(a_s) s_{\mathcal{C},s}^2(a_s)$, and Lemma 17 with $\varepsilon_{s,a_s} \leq \bar{\varepsilon}_s$ gives $s_{\mathcal{C},s}^2(a_s) \leq \tilde{s}_s^2(a_s)/(1 - \bar{\varepsilon}_s)$; hence $\bar{s}_s^2(a_s) \leq u_s(a_s) \tilde{s}_s^2(a_s)/(1 - \bar{\varepsilon}_s)$. By monotonicity of $x \mapsto \log(1 + \sigma^{-2}x)$, each summand of γ_{t-1} in (25) is dominated by the corresponding summand of $\hat{\gamma}_{t-1}$. Measurability holds because $\tilde{s}_s^2(a_s)$, $u_s(a_s)$, and $\bar{\varepsilon}_s$ are all $\mathcal{H}_s$ -measurable. The schedule property then follows term by term as in Corollary 20. $\square$

This makes the information-gain component observable when $u_s(a_s)$ is certified. Together with Lemma 7, it renders the full confidence and centering schedules operational; it does not make them tight.

Remark (Computability and dimension dependence of β_t). *Two caveats about β_t deserve emphasis. First, γ_{t-1} is defined through the frozen-feature Gram matrix $\bar{\mathbf{C}}_t$, whose per-sample Jacobians the matrix-free implementation deliberately does not store; γ_{t-1} is therefore not directly computable in the stated memory model. Second, the worst-case substitute $\gamma_{t-1} \leq d \log(1 + (t-1)G^2/(d\lambda\sigma^2))$ does not exploit low realized information gain and can be order t when $d \gg t$, rendering the resulting bonus and theorem bound vacuous. Corollary 21 addresses both issues with an observable surrogate built from quantities the algorithm already computes; the resulting guarantee is data-dependent, and its tightness still depends on the realized surrogate together with the other theorem factors.*

Lemma 22 (Action-scaled centering discrepancy). *Under Assumption 4, $|\mathbf{g}_t(a)^\top(\boldsymbol{\theta}_t - \hat{\boldsymbol{\theta}}_t^{\text{lin}})| \leq \bar{\psi}_t \bar{s}_t(a)$ for all $a \in \mathcal{A}_t$.*

Proof. Immediate from (32). $\square$

Remark (status of the centering certificate $\bar{\psi}_t$). Lemma 7 computes a predictable $\bar{\psi}_t \geq \psi_t$ from the certified optimizer residual, Welford path moments, observed collection residuals, and analytic smoothness constants. The schedule is operational but can be loose; the corrected-center variant (Corollary 5) removes it at the cost of frozen-feature access.

In the convex last-layer regime, under exact optimization, $\boldsymbol{\theta}_t = \hat{\boldsymbol{\theta}}_t^{\text{lin}}$ and $\bar{\psi}_t = 0$. The full prediction error at $(\mathbf{x}_t, a)$ is bounded by the triangle inequality, with the centering discrepancy *action-scaled*:

$$|\mu_{\boldsymbol{\theta}^*}(\mathbf{x}_t, a) - \mu_{\boldsymbol{\theta}_t}(\mathbf{x}_t, a)| \leq (\beta_t + \bar{\psi}_t) \bar{s}_t(a) + \varepsilon_{\text{lin}}(t) \leq \omega_t \bar{s}_t(a) + \varepsilon_{\text{lin}}(t), \quad (64)$$

using (62), Lemma 22, the per-round linearization remainder $|\rho_t| \leq \varepsilon_{\text{lin}}(t)$, and $\bar{\beta}_t \geq \beta_t$.

A.8 Cost of Approximate Curvature

Proposition 23 (Unrescaled current-parameter subset). *Let $\mathcal{S}_t \subseteq \{1, \dots, t-1\}$ and $0 \leq w_s \leq 1$, and define the current-parameter operator*

$$\hat{\mathbf{C}}_t = \lambda \mathbf{I} + \sigma^{-2} \sum_{s \in \mathcal{S}_t} w_s \mathbf{g}_t(\mathbf{x}_s, a_s) \mathbf{g}_t(\mathbf{x}_s, a_s)^\top,$$

using the same relinearized Jacobians $\mathbf{g}_t(\mathbf{x}_s, a_s) = \nabla_{\boldsymbol{\theta}} \mu_{\boldsymbol{\theta}_t}(\mathbf{x}_s, a_s)$ that define $\mathbf{C}_t$ (5). Then $\hat{\mathbf{C}}_t \preceq \mathbf{C}_t$, so Theorem 24 applies with $\kappa_{+,t} = 1$ and, in the exact-transfer case, $u_t = (1 + \bar{\chi}_t)^2$ (Lemma 11).

Proof. $\mathbf{C}_t - \hat{\mathbf{C}}_t = \sigma^{-2} \sum_{s \in \mathcal{S}_t} (1 - w_s) \mathbf{g}_t \mathbf{g}_t^\top + \sigma^{-2} \sum_{s \notin \mathcal{S}_t} \mathbf{g}_t \mathbf{g}_t^\top$, a sum of PSD outer products (each $1 - w_s \geq 0$), hence $\mathbf{C}_t - \hat{\mathbf{C}}_t \succeq 0$. With $\hat{\mathbf{C}}_t \preceq \mathbf{C}_t$ (so $\kappa_{+,t} = 1$ in (65)) and $\hat{\mathbf{C}}_t \succeq \lambda \mathbf{I}$, the hypotheses of Theorem 24 hold. $\square$

This covers an unrescaled current-parameter sliding window, an unrescaled current-parameter subsample, and any current-parameter downweighted subset with weights in $[0, 1]$. It does *not* cover rescaled subsampling with weights $(t-1)/n > 1$ (Lemma 13), a stale periodic-refresh operator evaluated at an older parameter, or an arbitrary randomized sketch (e.g. an eigenspace Lanczos surrogate), for which a factor $\kappa_{+,t}$ must be certified separately.

Theorem 24 (Approximate curvature via one-sided transfer). *Adopt the hypotheses of Theorem 1 with the algorithmic curvature $\mathbf{C}_t = \hat{\mathbf{C}}_t$, an approximate operator that is symmetric, $\mathcal{H}_t$ -measurable, and satisfies $\hat{\mathbf{C}}_t \succeq \lambda \mathbf{I}$. Suppose the two one-sided bounds*

$$\hat{\mathbf{C}}_t \preceq \kappa_{+,t} \mathbf{C}_t, \quad \mathbf{C}_t \preceq (1 + \bar{\chi}_t)^2 \bar{\mathbf{C}}_t \quad (65)$$

hold for every $t \leq T$, where $\kappa_{+,t} \geq 1$ is $\mathcal{H}_t$ -measurable and $\bar{\chi}_t \geq \chi_t$ is the whitened-drift certificate of Lemma 11. Then, for every action a ,

$$\bar{s}_t^2(a) \leq \kappa_{+,t} (1 + \bar{\chi}_t)^2 \hat{s}_t^2(a), \quad (66)$$

so Assumption 5 holds with the uniform factor $u_t = \kappa_{+,t}(1 + \bar{\chi}_t)^2$ and Theorem 1 applies verbatim with $\mathcal{C}_t = \hat{\mathbf{C}}_t$ and $s_{\mathcal{C},t} = \hat{s}_t$:

$$R_T \leq 2 \sqrt{(\sigma^2 + G^2/\lambda) \Lambda_T^{\mathcal{C}} \sum_{t=1}^T \alpha_t^2 u_t(a_t) \omega_t^2} + 2E_T.$$

No lower spectral factor on $\hat{\mathbf{C}}_t$ is required for optimism; the cost of an excessively small, stale, or nonmonotone approximate operator is charged through its realized width complexity $\Lambda_T^{\mathcal{C}}$ (and its variation $V_T^{\mathcal{C}}$ inside Γ_T^{dyn}), not through a lower Loewner bound.

Proof. $\hat{\mathbf{C}}_t \preceq \kappa_{+,t} \mathbf{C}_t$ with both sides SPD inverts to $\hat{\mathbf{C}}_t^{-1} \succeq \kappa_{+,t}^{-1} \mathbf{C}_t^{-1}$, so $\hat{s}_t^2(a) \geq \kappa_{+,t}^{-1} s_t^2(a)$; and (42) gives $\bar{s}_t^2(a) \leq (1 + \bar{\chi}_t)^2 s_t^2(a)$. Chaining, $\bar{s}_t^2(a) \leq (1 + \bar{\chi}_t)^2 \kappa_{+,t} \hat{s}_t^2(a)$, which is (66) with $s_{\mathcal{C},t} = \hat{s}_t$. Theorem 1 applies with $\mathcal{C}_t = \hat{\mathbf{C}}_t$. The two-sided $\sqrt{\rho_+/\rho_-}$ inflation—which instead assumes a lower spectral factor—is retained for comparison in Appendix F; the two results trade different assumptions (a lower spectral bound versus dynamic width complexity), and neither numerically dominates the other in general. $\square$

Proposition 25 (Roundwise scalar-width invariance). *If two width functions satisfy $\hat{s}_t(a) = c_t s_t(a)$ for every candidate action, where $c_t > 0$ is action independent, then replacing a scalar bonus coefficient b_t by b_t/c_t gives identical UCB scores and therefore the same selected action under the same tie rule.*

Proof. $\hat{\mu}_t(a) + (b_t/c_t) \hat{s}_t(a) = \hat{\mu}_t(a) + b_t s_t(a)$ for every a . $\square$

This is a roundwise statement. If two policies also have identical centers, histories, optimization, and coupled exogenous randomness, if c_t is predictable, and if proportionality holds at every subsequent common history, induction gives identical actions and rewards after the coefficient adjustment. Absent those conditions the proposition makes no regret claim. Accordingly, regret differences can reflect anisotropic width distortion or a difference in damping, center, adaptive path, optimization, calibration, or solve accuracy.

Proposition 26 (Off-diagonal linear-Gram witness). *Let $\sigma = 1$, $\mathbf{u} = (1, 1)/\sqrt{2}$, $\mathbf{v} = (1, -1)/\sqrt{2}$, the action set be $\{\mathbf{u}, \mathbf{v}\}$, and $\boldsymbol{\theta}^* = \epsilon \mathbf{u} + \Delta \mathbf{v}$ with $\Delta > \epsilon > 0$. Rewards are noiseless, the ridge is $\lambda \mathbf{I}$, all policies use the same full-ridge mean center, and ties select $\mathbf{u}$. If*

$$\max \left\{ \sqrt{\lambda(\epsilon^2 + \Delta^2)}, \epsilon \sqrt{\lambda} \left(1 + \sqrt{\frac{\lambda}{\lambda+1}} \right) \right\} < b \leq \left(\Delta - \frac{\epsilon}{\lambda+1} \right) \sqrt{\lambda+1}, \quad (67)$$

then full-Gram UCB plays $\mathbf{u}$ once and $\mathbf{v}$ thereafter, so $R_T^{\text{full}} = \Delta - \epsilon$. The raw diagonal policy and the policy inflated by the uniform analytic factor $\kappa_n = (\lambda + n/2)/\lambda$ play $\mathbf{u}$ forever and have $R_T^{\text{diag}} = T(\Delta - \epsilon)$.

Proof. After n pulls of $\mathbf{u}$ and no pull of $\mathbf{v}$,

$$C_n = \lambda \mathbf{I} + n \mathbf{u} \mathbf{u}^\top, \quad \hat{\boldsymbol{\theta}}_n = \frac{n\epsilon}{\lambda+n} \mathbf{u}.$$

Thus the full scores are $n\epsilon/(\lambda+n) + b/\sqrt{\lambda+n}$ and $b/\sqrt{\lambda}$. The strict lower inequality in (67) makes the second score larger at $n = 1$. The confidence lower bound also implies $b > \Delta\sqrt{\lambda}$; after any number of $\mathbf{v}$ pulls its score is strictly larger than Δ , whereas the upper inequality makes the fixed $\mathbf{u}$ score at most Δ . Hence there is no return. In contrast, $D_n = \text{diag}(C_n) = (\lambda + n/2)\mathbf{I}$, so both diagonal widths are equal and the center gives $\mathbf{u}$ a strict advantage after round one. Moreover $D_n \preceq \kappa_n C_n$ analytically, but multiplying both diagonal widths by $\sqrt{\kappa_n}$ leaves them equal; uniform certified inflation therefore does not change the diagonal action. For example, $(\lambda, \epsilon, \Delta, b) = (1, 0.1, 1, 1.1)$ satisfies the interval. $\square$

This is a two-dimensional linear-Gram existence witness. For a linear predictor with Gaussian squared loss, this Gram is exactly the GGN/Fisher operator, so the construction isolates a GGN-relevant geometric mechanism. It is neither a nonlinear-network result nor a uniform dominance result. A dense action-dependent factor recovers the full widths, but it uses the reference geometry that the diagonal approximation was meant to avoid.

Composing subsampling with feature drift. Lemma 13 guarantees a sandwich for the *relinearized* $\mathbf{C}_t$: $(1-\varepsilon)\mathbf{C}_t \preceq \hat{\mathbf{C}}_t \preceq (1+\varepsilon)\mathbf{C}_t$. Only the *upper* half is needed here: $\hat{\mathbf{C}}_t \preceq (1+\varepsilon)\mathbf{C}_t$ gives $\kappa_{+,t} = 1 + \varepsilon$, and composing with the one-sided whitened bound $\mathbf{C}_t \preceq (1 + \bar{\chi}_t)^2 \hat{\mathbf{C}}_t$ (Lemma 11) yields $u_t = (1 + \varepsilon)(1 + \bar{\chi}_t)^2$ in Theorem 24. A union bound over $t \leq T$ (replacing δ' by δ'/T in Lemma 13) makes the subsampling event time-uniform and contributes to δ_{cert} .

For an *unrescaled current-parameter window* (or any current-parameter downweighted subset with weights in $[0, 1]$), Proposition 23 gives the *deterministic* upper factor $\kappa_{+,t} = 1$ with *no* distributional assumption, so Theorem 24 applies directly with $u_t = (1 + \bar{\chi}_t)^2$. Only the *stale periodic-refresh* operator (§A.3(iii)), which reuses Jacobians from an older parameter, lacks a deterministic $\kappa_{+,t}$; there Theorem 24 provides a guarantee only when a one-sided bound $\hat{\mathbf{C}}_t \preceq \kappa_{+,t}\mathbf{C}_t$ is separately verified, which we evaluate empirically.

What remains open. The path schedules close the computability gap, but their constants can be extremely loose. Beyond the bounded low-rank and window-excitation regimes, useful control of Λ_T^C for a genuinely adapting backbone, tighter smoothness/residual certificates, and verified numerical enclosures for floating-point solver and spectral calculations remain open.

B Related Work

Neural and linear contextual bandits. LinUCB (Abbasi-Yadkori et al., 2011) and Thompson sampling (Thompson, 1933; Agrawal and Goyal, 2013) provide the analytical backbone for contextual bandit exploration. Neural extensions—NeuralUCB (Zhou et al., 2020), NeuralTS (Zhang et al., 2021), neural-linear models (Riquelme et al., 2018)—use different neural feature covariances. NeuralUCB and NeuralTS accumulate full neural-gradient Grams in their stated algorithms (and diagonalize them in experiments), while NeuralLinear maintains Bayesian linear uncertainty over learned last-layer features. CC-UCB instead re-evaluates every replayed Jacobian at the current parameter and accesses the resulting full operator implicitly. Neural-linear designs that train a deep representation but explore only in the last layer (Xu et al., 2022a) occupy the opposite design point. Recent alternatives sharpen different parts of this landscape. Salgia (2023) analyze overparameterized smooth-activation networks through non-asymptotic network–NTK approximation and give an efficient sublinear-regret algorithm. Deb et al. (2024) instead reduce contextual bandits to online neural regression, using perturbed predictions to obtain their regression guarantees. Bui et al. (2025) combine a learned deep representation with a variance-aware linear-UCB exploration layer. These are, respectively, an NTK analysis, a regression-oracle route, and a learned last-layer geometry; none analyzes the dynamically relinearized full GGN and its CG error. Our low-rank/excitation result is correspondingly narrower than generic neural regression, but controls that current-parameter operator.

Online Bayesian and EKF-style neural bandits. Durán-Martín et al. (2022) apply extended Kalman filter (EKF) updates in a learned or random low-dimensional affine parameter subspace. They maintain a dense or diagonal covariance in that subspace, inducing a low-rank covariance over parameters spanning all network layers, with memory independent of the number of rounds. This differs from applying exact curvature–vector products in the full parameter space. More directly, Chang et al. (2023) maintain a diagonal-plus-low-rank approximation to the full-parameter posterior precision through recursive EKF and online-SVD updates, and use it for Thompson sampling in an MNIST contextual bandit. Their LO-FI method is a one-pass compressed-posterior approach with fixed-rank cost linear in parameter count and no replay of past data. CC-UCB instead relinearizes the full replay history at the current parameter and accesses the uncompressed GGN through residual-checked CG. Their bandit study is empirical; our conditional UCB analysis explicitly accounts for CG truncation, but does not establish a practical advantage over LO-FI.

Last-layer and structured Laplace approximations. Last-layer Laplace fixes the backbone and restricts the approximate posterior to the final layer (Kristiadi et al., 2020), omitting backbone-weight uncertainty by construction. This is only one scalable option: all-layer block-diagonal Kronecker curvature has also been used (Ritter et al., 2018), and Daxberger et al. (2021) systematize all-weight, subnetwork, and last-layer variants with full, low-rank, KFAC, or diagonal curvature. CC-UCB uses the uncompressed current-parameter GGN as its reference operator without materializing the covariance matrix.

Sampling-based full-parameter exploration. Langevin Monte Carlo Thompson sampling (LMC-TS) (Xu et al., 2022b) explores with approximate samples of a loss-defined parameter posterior, obtained by noisy gradient steps on the regularized loss; like CC-UCB it never materializes a covariance matrix. Its regret guarantee is for linear contextual bandits; its neural-bandit evidence is empirical. The two routes pay for exploration differently: LMC-TS through the mixing of a per-round sampling chain, CC-UCB through CG truncation and spectral distortion of an explicit curvature metric, which yields deterministic widths, an optimism-style analysis, and a fixed curvature operator reused across separate per-action CG solves.

Efficient nonlinear UCB. Kassraie and Krause (2022) analyze NTK-UCB and propose NN-UCB and its convolutional variant CNN-UCB. NN-UCB combines a trained-network mean with confidence widths from the empirical NTK feature map at initialization. They prove sublinear regret for the SupNN-UCB variant under NTK-RKHS, sufficient-exploration, and network-width assumptions. Their fixed-feature uncertainty differs from CC-UCB’s replayed GGN/Fisher curvature at the current parameter θ_t , accessed matrix-free via CG. CC-UCB thus targets the regime where the practitioner has a specific pretrained model and wants to use its curvature as-is, rather than constructing a surrogate NTK-based estimator.

Spectral sketching for linear bandits. Kuzborskij et al. (2019) analyze linear bandits where the Gram matrix is replaced by a Frequent Directions sketch, deriving regret bounds that depend on the spectral approximation quality. Wen et al. (2026) show that fixed-size sketch regret bounds can degenerate to the trivial linear rate under heavy spectral tails, and propose an adaptive dyadic block sketch. Our spectral-distortion theorem (Theorem 24) instantiates the same conceptual lever in the matrix-free GGN/CVP setting by exposing transfer and realized-width terms along the realized trajectory. We do not claim novelty for the sandwich-to-regret reduction itself; the contribution is adapting it to the CG/GGN context with explicit CG truncation and feature-drift factors.

Laplace approximation in deep learning. The Laplace approximation (MacKay, 1992; Ritter et al., 2018) provides a Gaussian posterior centered at a MAP estimate with covariance given by the inverse curvature. Scalable variants use diagonal, KFAC (Martens and Grosse, 2015), or subnetwork approximations (Daxberger et al., 2021). Matrix-free full-Jacobian/Fisher predictive variance using implicit products and CG predates this work (Maddox et al., 2021); CC-UCB adapts that computational route to an online, current-relinearized GGN operator and supplies bandit-specific transfer and solver-error accounting.

GGN–vector products, Hessian spectral analysis, and influence functions. Pearlmutter’s algorithm (Pearlmutter, 1994) computes exact Hessian–vector products; the analogous GGN–vector product has the same Jacobian-transpose-times-Jacobian-times-vector structure (Schraudolph, 2002). Recent work on Krylov methods for Hessian spectral analysis (Granzol and Juarev, 2026) demonstrates related distributed HVP/Krylov infrastructure at $> 10^8$ -parameter scale, but does not benchmark the exact per-example GGN operator or end-to-end CC-UCB latency. Influence functions (Koh and Liang, 2017) use inverse-curvature–vector products to attribute predictions to training points; the CG-based implicit solve we employ is the same computational primitive.

Optimal experimental design. Information-theoretic exploration objectives—such as maximizing $\log \det(\bar{\mathbf{C}}_t)$ or minimizing the trace of $\bar{\mathbf{C}}_t^{-1}$ —connect bandit exploration to the classical D-optimal and A-optimal design criteria (Valko et al., 2013). CC-UCB’s use of the full GGN/Fisher curvature enables stochastic estimates of these quantities via Hutchinson-type trace estimators (Ubaru et al., 2017), without forming the matrix.

Filtration. We separate *pre-action* randomness, revealed before the agent commits to a_t , from *post-reward* randomness, drawn only after r_t is observed. Let Ω_t collect all pre-action algorithmic randomness used at round t : curvature subsampling indices, randomized sketches, and stochastic solver seeds. The randomized sketch or subsample that defines the curvature operator is drawn into Ω_t and *fixed before action selection*; it is not re-randomized during any conjugate-gradient (CG) solve at round t . (Standard CG requires a fixed symmetric positive definite matrix–vector oracle across its iterations and does not apply to a stochastic oracle that changes between iterations.) Let U_{t+1} collect the *post-reward* randomness used to form the next parameter estimate θ_{t+1} (minibatch order, dropout, perturbed warm starts). Define $\mathcal{F}_0 = \sigma(\theta_1)$ and, for $t \geq 1$, the recursively built σ -algebra

$$\mathcal{F}_t := \mathcal{F}_{t-1} \vee \sigma(\mathbf{x}_t, \mathcal{A}_t, \Omega_t, a_t, r_t, U_{t+1}), \quad (68)$$

so that $\boldsymbol{\theta}_{t+1}$ —computed from the data through round t together with the post-reward randomness U_{t+1} —is $\mathcal{F}_t$ -measurable. Define the *pre-action history* σ -algebra and the pre-reward σ -algebra

$$\mathcal{H}_t^- := \mathcal{F}_{t-1} \vee \sigma(\mathbf{x}_t, \mathcal{A}_t), \quad \mathcal{H}_t := \mathcal{H}_t^- \vee \sigma(\Omega_t), \quad \mathcal{G}_t := \mathcal{H}_t \vee \sigma(a_t). \quad (69)$$

At round t the context $\mathbf{x}_t$, the finite action set $\mathcal{A}_t$, and the pre-action randomness Ω_t are revealed; the agent selects a_t ; then the reward $r_t = \mu^*(\mathbf{x}_t, a_t) + \eta_t$ is observed; finally U_{t+1} is drawn and $\boldsymbol{\theta}_{t+1}$ is computed. $\mathcal{H}_t^-$ is the history immediately before the randomized certificate draw; after that draw, its realized operator and schedules are $\mathcal{H}_t$ -measurable. By construction of Algorithm 1 the selection rule is a deterministic function of $\mathcal{H}_t$ -measurable quantities, so the chosen action a_t is itself $\mathcal{H}_t$ -measurable and hence $\mathcal{G}_t = \mathcal{H}_t$; we retain the symbol $\mathcal{G}_t$ where the played action is used. The following are all $\mathcal{H}_t$ -measurable (i.e. determined before the reward r_t): the candidate action set $\mathcal{A}_t$ and every candidate feature $\mathbf{g}_t(\mathbf{x}_t, a)$, $a \in \mathcal{A}_t$; the chosen action a_t ; the algorithmic curvature operator $\mathcal{C}_t$ (Section 3); the confidence schedule $\bar{\beta}_t$; the centering certificate $\bar{\psi}_t$ (Assumption 4); the per-round CG tolerance $\bar{\varepsilon}_t$ together with any condition-number certificate used by the CG stopping rule; and the one-sided optimism-transfer factor $u_t(a)$ (Assumption 5). Since $\boldsymbol{\theta}_t$ is $\mathcal{F}_{t-1} \subseteq \mathcal{H}_t$ -measurable, each candidate design vector $\mathbf{g}_t(\mathbf{x}_t, a)$ is $\mathcal{H}_t$ -measurable and the realized design vector $\mathbf{g}_t(\mathbf{x}_t, a_t)$ is $\mathcal{G}_t$ -measurable. We assume $\{U_s\}_{s \geq 1}$ is independent of the reward noise $\{\eta_s\}_{s \geq 1}$, so that revealing U_{t+1} and forming $\boldsymbol{\theta}_{t+1}$ after r_t preserves the conditional sub-Gaussian property of Assumption 1 for the future noise $\{\eta_s\}_{s > t}$. Index every randomized certificate source (operator sketch, transfer bound, condition bound, or other randomized oracle) by $j \in \mathcal{J}$. For its round- t validity event $\mathcal{E}_{j,t}$ choose in advance $\delta_{j,t} \geq 0$ such that its stated conditional guarantee gives $\Pr(\mathcal{E}_{j,t}^c \mid \mathcal{H}_t^-) \leq \delta_{j,t}$ and

$$\sum_{j \in \mathcal{J}} \sum_{t=1}^T \delta_{j,t} \leq \delta_{\text{cert}}, \quad \mathcal{E}_{\text{cert}} := \bigcap_{j \in \mathcal{J}} \bigcap_{t=1}^T \mathcal{E}_{j,t}.$$

The tower property and union bound yield $\Pr(\mathcal{E}_{\text{cert}}) \geq 1 - \delta_{\text{cert}}$. Deterministic or almost-sure certificates use $\delta_{j,t} = 0$. The self-normalized confidence event is separate and receives probability δ .

Auxiliary linearized ridge estimator. For the analysis we define a *linearized ridge estimator* based on frozen features. Set the predictable intercept $b_s := \mu_{\boldsymbol{\theta}_s}(\mathbf{x}_s, a_s) - \mathbf{g}_s(\mathbf{x}_s, a_s)^\top \boldsymbol{\theta}_s$ and the pseudo-response $y_s := r_s - b_s$. Under Assumption 3 and realizability ($\mu^* = \mu_{\boldsymbol{\theta}^*}$),

$$y_s = \mathbf{g}_s(\mathbf{x}_s, a_s)^\top \boldsymbol{\theta}^* + \rho_s(\mathbf{x}_s, a_s, \boldsymbol{\theta}^*) + \eta_s, \quad (70)$$

where $|\rho_s| \leq \varepsilon_{\text{lin}}(s)$. The linearized estimator is

$$\hat{\boldsymbol{\theta}}_t^{\text{lin}} := \arg \min_{\boldsymbol{\theta}} \left[\frac{1}{2\sigma^2} \sum_{s=1}^{t-1} (y_s - \mathbf{g}_s(\mathbf{x}_s, a_s)^\top \boldsymbol{\theta})^2 + \frac{\lambda}{2} \|\boldsymbol{\theta}\|^2 \right] = \bar{\mathbf{C}}_t^{-1} \frac{1}{\sigma^2} \sum_{s=1}^{t-1} \mathbf{g}_s(\mathbf{x}_s, a_s) y_s. \quad (71)$$

This is a standard linear ridge regression with predictable design vectors $\mathbf{g}_s/\sigma$ and Gram matrix $\bar{\mathbf{C}}_t$, so the self-normalized martingale bound (Abbasi-Yadkori et al., 2011) applies to it directly (Lemma 19).

GGN versus Hessian (context only). The regret theorem uses the GGN/Fisher curvature as the exploration metric. It does not require the GGN to be an accurate Hessian approximation. For squared loss (3) the full Hessian of the data-fit term is

$$\mathbf{H}_t = \frac{1}{\sigma^2} \sum_{s=1}^{t-1} \left[\mathbf{g}_t(\mathbf{x}_s, a_s) \mathbf{g}_t(\mathbf{x}_s, a_s)^\top - (r_s - \mu_{\boldsymbol{\theta}_t}(\mathbf{x}_s, a_s)) \nabla^2 \mu_{\boldsymbol{\theta}_t}(\mathbf{x}_s, a_s) \right] + \lambda \mathbf{I}.$$

The curvature matrix $\mathbf{C}_t$ drops the residual-times-second-derivative term, retaining only the outer-product part that is guaranteed PSD. The dropped residual-times-second-derivative term $D_t = \sigma^{-2} \sum_{s < t} (r_s - \mu_{\boldsymbol{\theta}_t}) \nabla^2 \mu_{\boldsymbol{\theta}_t}$ has operator norm at most $(t-1)\epsilon_r H/\sigma^2$, where ϵ_r bounds the residuals and H the Hessian norms. When this quantity is small relative to λ , $\mathbf{C}_t$ approximates the full Hessian in the Loewner sense (a formal statement with proof is given in Appendix G). This comparison is *not* invoked in the regret analysis (Theorem 1 uses $\mathbf{C}_t$ directly as the exploration metric); it serves only to contextualize $\mathbf{C}_t$ as a proxy for the full Hessian. Outside the small-residual regime, the GGN is not an exact Hessian approximation; it is the PSD Fisher/GGN metric used by the algorithm.

Remark (Fisher, empirical Fisher, and GGN). Three curvature objects appear in the literature and must be distinguished. The *empirical Fisher* is the realized score outer product $\sum_s \nabla_{\theta} \log p(r_s | \mathbf{x}_s, a_s; \theta) \nabla_{\theta} \log p(r_s | \mathbf{x}_s, a_s; \theta)^{\top}$; it depends on observed rewards and is *not* the operator used by CC-UCB. The *true (model) Fisher* is the model expectation of the score outer product, $F(\theta) = \mathbb{E}_{r \sim p(\cdot | \mathbf{x}, a; \theta)} [\nabla_{\theta} \log p \nabla_{\theta} \log p^{\top}]$. The *generalized Gauss–Newton* (GGN) drops the residual-times-Hessian term from the full Hessian, retaining only the outer-product Jacobian part that is guaranteed PSD (Martens, 2020; Schraudolph, 2002). For Gaussian regression with fixed variance (the squared-loss setting of this paper), the true Fisher and GGN are identical, and both reduce to the outer-product curvature $\sigma^{-2} \sum_s \mathbf{g}_s \mathbf{g}_s^{\top}$ used in (5). More generally, for exponential-family negative log-likelihoods parameterized by natural parameters $z_{\theta}(\mathbf{x}, a)$, the true Fisher equals the GGN: $J_{\theta}^{\top} \nabla^2 A(z) J_{\theta}$, where $J_{\theta} = \nabla_{\theta} z_{\theta}$ and $\nabla^2 A$ is the Hessian of the log-partition function (Martens, 2020; Pascanu and Bengio, 2013). Bernoulli rewards yield scalar weight $p(1-p)$; categorical rewards yield $\text{Diag}(\mathbf{p}) - \mathbf{p}\mathbf{p}^{\top}$. See Appendix H for the extension.

We access $\mathbf{C}_t$ only through curvature–vector products $\mathbf{v} \mapsto \mathbf{C}_t \mathbf{v}$, computed via forward-mode Jacobian–vector products composed with reverse-mode vector–Jacobian products (Pearlmutter, 1994; Schraudolph, 2002).

Extension to exponential-family likelihoods. The same matrix-free CG/CVP machinery extends algorithmically to exponential-family likelihoods (e.g. Bernoulli or categorical rewards) via the Fisher/GGN curvature $J_t^{\top} (\nabla^2 A) J_t$, but no regret guarantee beyond Gaussian squared loss is claimed in this draft. Details of the curvature definitions are in Appendix H.

Per-round CG tolerance $\bar{\varepsilon}_t$. The algorithm uses a *known* per-round tolerance $\bar{\varepsilon}_t \in (0, 1)$ in the UCB bonus (line 8), not the true per-action energy-norm errors $\varepsilon_{t,a}$ (which are unobservable without the exact solves). It suffices that $\max_{a \in \mathcal{A}_t} \varepsilon_{t,a} \leq \bar{\varepsilon}_t$ at every round ((10); zero-gradient actions have $\varepsilon_{t,a} = 0$ by convention and are trivially within tolerance); the bonus is then conservative. Two strategies ensure this:

1. *Adaptive residual stopping (recommended).* In exact arithmetic, for each action monitor the computable Euclidean relative residual $\|r_I\|/\|\mathbf{g}_t(a)\|$ at each CG step (skipping actions with $\mathbf{g}_t(a) = \mathbf{0}$). Since $\varepsilon_{t,a} \leq \sqrt{\bar{\kappa}_t} \cdot \|r_I\|/\|\mathbf{g}_t(a)\|$ (standard CG bound, with $\bar{\kappa}_t$ a certified upper bound on the condition number of $\mathcal{C}_t$), stopping that solve when $\|r_I\|/\|\mathbf{g}_t(a)\| \leq \bar{\varepsilon}_t/\sqrt{\bar{\kappa}_t}$ guarantees $\varepsilon_{t,a} \leq \bar{\varepsilon}_t$, hence (10). This is conservative by the factor $\sqrt{\bar{\kappa}_t}$ but requires no access to $\mathcal{C}_t^{-1}$ and adapts automatically to the per-round condition number. A floating-point implementation needs a verified residual enclosure to inherit this certification literally.
2. *Fixed budget (requires a uniform condition-number bound).* If an a priori bound $\bar{\kappa}_t \leq \kappa_{\max}$ holds for *all* rounds $t \leq T$, choose I via Eq. (76) at $\kappa_{\max}$ and a fixed target $\bar{\varepsilon}_t \equiv \bar{\varepsilon}$. For a *nonnegative weighted outer-product* operator $\mathcal{C}_t = \lambda \mathbf{I} + \sigma^{-2} \sum_{s \in \mathcal{S}_t} w_s \mathbf{g}_t(\mathbf{x}_s, a_s) \mathbf{g}_t(\mathbf{x}_s, a_s)^{\top}$ the condition number is bounded in closed form, $\bar{\kappa}_t \leq 1 + W_t G^2 / (\lambda \sigma^2)$ with $W_t = \sum_{s \in \mathcal{S}_t} w_s$; a fixed buffer of size n ($W_t = n$, unrescaled) makes this independent of T . For the full-history operator ($W_t = t-1$), and for rescaled uniform subsampling with $w_s = (t-1)/n$ (so $W_t = \sum_{s \in \mathcal{S}_t} w_s = t-1$), the bound grows with t . For a known finite horizon, a fixed budget at $\kappa_{\max} = 1 + (T-1)G^2/(\lambda \sigma^2)$ is valid but scales as $O(\sqrt{T} \log(1/\bar{\varepsilon}))$; a horizon-independent fixed budget is not uniform, so adaptive stopping or an increasing I_t is preferable (see Appendix E). For a *general* SPD operator that is not a nonnegative weighted outer product (e.g. a randomized sketch), the closed-form bound $1 + W_t G^2 / (\lambda \sigma^2)$ does not apply, and a separately certified $\mathcal{H}_t$ -measurable condition-number upper bound $\bar{\kappa}_t$ must be supplied.

In the regret bound (Theorem 1), the CG inflation enters through the per-round factor $\alpha_t = \sqrt{(1+\bar{\varepsilon}_t)/(1-\bar{\varepsilon}_t)}$, inside the sum of squares $\sum_t \alpha_t^2 u_t(a_t) \omega_t^2$; no horizon maximum over $\bar{\varepsilon}_t$ is taken.

Action screening (practical heuristic). When the action set $\mathcal{A}_t$ is large, computing the CG solve for every action is prohibitive ($|\mathcal{A}_t| \times I$ CG steps per round). A practical heuristic is to screen the top- K actions by mean prediction $\hat{\mu}_t(a)$ (forward passes only), then compute UCB widths for the K candidates. This reduces cost to $K \times I$ CG solves. *The regret guarantee in Theorem 1 analyzes the full-enumeration case ($K = |\mathcal{A}_t|$); screening can drop the optimal arm and is not covered by the theory unless one adds an assumption that $a_t^* \in \mathcal{A}_t^K$ for all t .*

Algorithm 2 Operational CC-UCB with path certificates

Require: Reward predictor μ_θ , damping $\lambda > 0$, noise proxy σ , confidence level δ , radius S , analytic constants G, L_g, L_μ ; algorithmic curvature mode $\mathcal{C}_t$ (exact full-history $\mathbf{C}_t$ or approximate $\hat{\mathbf{C}}_t$) with certified one-sided factor $\kappa_{+,t}$ and condition upper bound $\bar{\kappa}_t$; per-round CG tolerance $\bar{\varepsilon}_t < 1$

- 1: Initialize θ_1 , certified $\zeta_1 \geq \|\nabla \mathcal{L}_1(\theta_1)\|$, $n \leftarrow 0$, $\bar{\theta} \leftarrow \mathbf{0}$, $J \leftarrow 0$, $R^{\text{col}} \leftarrow 0$, $\bar{F} \leftarrow 0$, $\hat{\gamma} \leftarrow 0$, and $\mathcal{D}_0 \leftarrow \emptyset$
- 2: **for** $t = 1, 2, \dots, T$ **do**
- 3: Observe context $\mathbf{x}_t$ and action set $\mathcal{A}_t$; draw pre-action randomness Ω_t { $\mathcal{H}_t$ of (69)}
- 4: $Q_t \leftarrow J + n\|\theta_t - \bar{\theta}\|^2$; $\bar{\chi}_t \leftarrow L_g\sqrt{Q_t}/(\sigma\sqrt{\lambda})$; $u_t \leftarrow \kappa_{+,t}(1 + \bar{\chi}_t)^2$
- 5: $\bar{M}_t \leftarrow \sigma^{-2}[L_g\sqrt{R^{\text{col}}Q_t} + (3GL_g/2)Q_t + (L_g^2/2)Q_t^{3/2}]$; $\bar{\psi}_t \leftarrow (\zeta_t + \bar{M}_t)/\sqrt{\lambda}$ {in a radius- R region use $L_g^2RQ_t$ for the last term}
- 6: $\bar{\epsilon}_t^{\text{lin}} \leftarrow (L_\mu/2)(S + \|\theta_t\|)^2$; $\bar{\beta}_t \leftarrow \sqrt{\hat{\gamma} + 2\log(1/\delta)} + \sqrt{\lambda}S + \sigma^{-1}\sqrt{\bar{F}}$; $\omega_t \leftarrow \bar{\beta}_t + \bar{\psi}_t$ {corrected center uses $\omega_t = \bar{\beta}_t$ }
- 7: Form or receive a fixed $\mathcal{H}_t$ -measurable SPD matrix-vector oracle $\mathbf{v} \mapsto \mathcal{C}_t\mathbf{v}$, $\mathcal{C}_t \succeq \lambda\mathbf{I}$, held fixed across all CG solves this round {exact/buffered special case $\mathcal{C}_t\mathbf{v} = \lambda\mathbf{v} + \sigma^{-2}\sum_{s \in \mathcal{S}_t} w_s \mathbf{g}_t(\mathbf{x}_s, a_s)[\mathbf{g}_t(\mathbf{x}_s, a_s)^\top \mathbf{v}]$, weights w_s per §A.3}
- 8: **for** each $a \in \mathcal{A}_t$ **do**
- 9: $\hat{\mu}_t(a) \leftarrow \mu_{\theta_t}(\mathbf{x}_t, a)$ {mean prediction (or corrected center (28))}
- 10: $\mathbf{g}_t(a) \leftarrow \nabla_{\theta} \mu_{\theta_t}(\mathbf{x}_t, a)$ {gradient feature}
- 11: **if** $\mathbf{g}_t(a) = \mathbf{0}$ **then**
- 12: $\tilde{s}_t^2(a) \leftarrow 0$ {skip CG; no residual ratio}
- 13: **else**
- 14: $\tilde{\mathbf{u}} \leftarrow \text{CG}(\mathcal{C}_t, \mathbf{g}_t(a))$; stop only once $\|r_I\|/\|\mathbf{g}_t(a)\| \leq \bar{\varepsilon}_t/\sqrt{\bar{\kappa}_t}$, or use a certified budget I_t from (76) {recompute the original residual in floating point; a point check is not an enclosure}
- 15: $\tilde{s}_t^2(a) \leftarrow \mathbf{g}_t(a)^\top \tilde{\mathbf{u}}$ {approx. predictive variance}
- 16: **end if**
- 17: **end for**
- 18: $a_t \leftarrow \arg \max_{a \in \mathcal{A}_t} \left[\hat{\mu}_t(a) + \omega_t \sqrt{\frac{u_t(a)}{1 - \bar{\varepsilon}_t}} \sqrt{\tilde{s}_t^2(a)} \right]$ {bonus (59); ties broken by a fixed rule (e.g. lowest index) or by Ω_t }
- 19: Play a_t , observe r_t
- 20: $\mathcal{D}_t \leftarrow \mathcal{D}_{t-1} \cup \{(\mathbf{x}_t, a_t, r_t)\}$
- 21: $c_t \leftarrow \mu_{\theta_t}(\mathbf{x}_t, a_t) - r_t$; $R^{\text{col}} \leftarrow R^{\text{col}} + c_t^2$; $\bar{F} \leftarrow \bar{F} + (\bar{\epsilon}_t^{\text{lin}})^2$; $\hat{\gamma} \leftarrow \hat{\gamma} + \log[1 + \sigma^{-2}u_t\tilde{s}_t^2(a_t)/(1 - \bar{\varepsilon}_t)]$
- 22: $n \leftarrow n + 1$; $d \leftarrow \theta_t - \bar{\theta}$; $\bar{\theta} \leftarrow \bar{\theta} + d/n$; $J \leftarrow J + d^\top(\theta_t - \bar{\theta})$ {vector Welford update}
- 23: Run J_{t+1}^{opt} warm-started optimizer steps on the *full* data set $\mathcal{D}_t$ to obtain θ_{t+1} , projecting to the certified region if required
- 24: Certify $\zeta_{t+1} \geq \|\nabla \mathcal{L}_{t+1}(\theta_{t+1})\|$; unless the terminal optimizer evaluation supplies this quantity, this requires one additional full-data gradient evaluation
- 25: **end for**

For the exact-curvature theorem, Algorithm 1 is run with $\mathcal{S}_t = \{1, \dots, t-1\}$, $w_s = 1$, full action enumeration, $\mathcal{C}_t = \mathbf{C}_t$, and the transfer factor $u_t = (1 + \bar{\chi}_t)^2$ from Lemma 11. For an approximate operator $\mathcal{C}_t = \hat{\mathbf{C}}_t$, use $u_t = \kappa_{+,t}(1 + \bar{\chi}_t)^2$ from Theorem 24. The selection rule needs only the *one-sided* upper transfer factor $u_t(a)$ for optimism; no lower spectral factor is required (the cost of the operator is charged through the realized dynamic complexity $\Lambda_T^{\mathcal{C}}$). Algorithm 2 computes u_t , $\bar{\psi}_t$, and $\bar{\beta}_t$ from pre-action state. The operator-specific inputs are the proved factor $\kappa_{+,t}$ and condition upper bound $\bar{\kappa}_t$; exact full and unrescaled-subset operators use $\kappa_{+,t} = 1$. Windowed/refresh curvature and action screening are practical heuristics unless the corresponding one-sided transfer factor or optimal-arm-survival condition is separately verified.

Corrected-center remark. Replacing the center $\hat{\mu}_t(a)$ by the corrected center $\hat{\mu}_t^{\text{corr}}(a)$ of (28) sets $\bar{\psi}_t = 0$ (so $\omega_t = \bar{\beta}_t$) and removes the centering-stability assumption exactly (Corollary 5), at the cost of frozen-feature access for $\hat{\theta}_t^{\text{lin}}$; this departs from the $O(d)$ additional curvature-state-memory, no-stored-feature design unless a frozen-feature oracle, replay mechanism, or certified sketch is supplied (see the remark after Corollary 5).

CC-UCB is the analyzed method throughout.

How to read the theorem. The theorem is a transfer result for a general predictable SPD curvature sequence. Lemma 7 and Corollary 2 instantiate its nonlinear confidence, centering, and feature-transfer inputs from $O(d)$ path state, observed collection residuals, analytic smoothness constants, and a certified optimizer residual. These schedules are executable but can be very loose. A sublinear guarantee from this bound still requires control of the realized dynamic width complexity and a restricted parameter path; unrestricted end-to-end fine-tuning is not covered. No horizon-uniform lower transfer is required by Theorem 1; the lower sandwich appears only in the explicit near-linear corollary.

Discussion of terms.

- *Exploration term.* Write $S_T := \sum_{t=1}^T \alpha_t^2 u_t(a_t) \omega_t^2$ for the sum of squares of the per-round factors. The exploration term is $2\sqrt{(\sigma^2 + G^2/\lambda) \Lambda_T^C S_T}$. A sufficient condition for sublinear regret is

$$\Lambda_T^C S_T = o(T^2), \quad E_T = o(T).$$

Concretely this holds whenever the per-round factors are polylogarithmically bounded ($\sup_t \alpha_t^2 u_t(a_t) \omega_t^2 = O(\text{polylog } T)$, so $S_T = O(T \text{polylog } T)$) and the realized dynamic complexity is small ($\Lambda_T^C = O(\text{polylog } T)$, e.g. via $\Lambda_T^C \leq \Gamma_T^{\text{dyn}}$ with both the variation charge V_T^C and the endpoint $\log \det(\mathcal{C}_{T+1}/\lambda \mathbf{I})$ polylogarithmically bounded). Note ω_t^2 contains the cumulative squared linearization error through $\bar{F}_t$ (via β_t) and the centering certificate $\bar{\psi}_t$, so a small Λ_T^C alone is insufficient. α_t is bounded whenever the CG tolerance $\bar{\varepsilon}_t$ is bounded away from 1; in the exact-curvature case $u_t = (1 + \bar{\chi}_t)^2$ is bounded whenever the whitened-drift certificate $\bar{\chi}_t$ is bounded (Lemma 11); the whitened ℓ_2 accumulation $\chi_t \leq \|\mathbf{D}_t\|_F$ makes this a near-stationarity condition on replayed-Jacobian drift. The unstable-curvature cost is isolated in Λ_T^C (through V_T^C), not hidden in a spectral constant. The factor $\sqrt{\sigma^2 + G^2/\lambda}$ arises from the width-sum bound of the dynamic width-complexity lemma (Lemma 14, Eq. (51)); when $G^2 \leq \lambda \sigma^2$ it is at most $\sigma\sqrt{2}$.

- *Linearization remainder.* The bound involves $E_T = \sum_t \varepsilon_{\text{lin}}(t)$, not $T\varepsilon_{\text{lin}}$. Near-linear regimes also require the squared-error sum $F_T = \sum_t \varepsilon_{\text{lin}}(t)^2$ (through $\bar{F}_t$, hence ω_t) to be controlled. $\|\boldsymbol{\theta}^* - \boldsymbol{\theta}_t\| = O(1/t)$ gives $E_T = O(1)$; $\|\boldsymbol{\theta}^* - \boldsymbol{\theta}_t\| = O(t^{-1/4})$ gives $E_T = O(\sqrt{T})$. The additive role of E_T mirrors misspecified linear bandits, where a uniformly bounded per-round model error commonly contributes an additive regret term that scales linearly in the cumulative approximation error (Ghosh et al., 2017; Lattimore et al., 2020). The displayed bound requires $E_T/T \rightarrow 0$; a nonvanishing uniform per-round error bound alone yields an $O(T)$ additive term.
- *Centering.* The centering certificate enters *multiplicatively*, through $\omega_t = \bar{\beta}_t + \bar{\psi}_t$ inside the sum of squares S_T , rather than as a separate additive term. It vanishes ($\bar{\psi}_t = 0$) when $\boldsymbol{\theta}_t = \hat{\boldsymbol{\theta}}_t^{\text{lin}}$ (convex last-layer models with exact optimization), and the corrected-center variant (Corollary 5) removes it in general at the cost of a frozen-feature solve.
- *Regret improvement over diagonal/last-layer surrogates.* The bound itself does not prove that full curvature yields strictly lower regret than diagonal or last-layer methods in any specific regime; it provides an upper bound with explicit per-round factors. *Demonstrating regret improvement is an empirical hypothesis* (Section 5). The theorem exposes the transfer factor u_t and realized width complexity Λ_T^C as operator-dependent terms along the realized trajectory; it does not identify a causal regret cost because the adaptive path and other factors may also change.
- *Scope.* The bound is proved for Gaussian/squared-loss rewards. Extending it to exponential-family likelihoods (Bernoulli, categorical) would require a weighted frozen-feature analysis accounting for the data-dependent Fisher weights $\nabla^2 A(z)$ and is left as future work (Appendix H).

Corollary 27 (Sanity-check: frozen features recover linear UCB). *Suppose the backbone is frozen and the reward head is linear in its parameters, the ridge objective is optimized exactly, and take the algorithmic curvature to be the sequential frozen Gram matrix, $\mathcal{C}_t = \bar{\mathcal{C}}_t$. Then: $\varepsilon_{\text{lin}}(t) = 0$ for all t ($L = 0$ in Assumption 3), so $E_T = F_T = 0$; $\bar{\psi}_t = 0$ (under exact optimization the MAP and linearized estimators coincide, so Assumption 4 holds with equality), giving $\omega_t = \bar{\beta}_t$; $u_t(a) = 1$ for all a (indeed $\bar{s}_t^2(a) = s_{\bar{\mathcal{C}}_t}^2(a)$, so Assumption 5 holds with equality); and, since $\bar{\mathcal{C}}_{t+1} = \bar{\mathcal{C}}_t + \sigma^{-2} \mathbf{g}_t(\mathbf{x}_t, a_t) \mathbf{g}_t(\mathbf{x}_t, a_t)^\top$ by (7), the update is exactly the rank-one step, i.e. $\mathcal{C}_{t+1} = \mathcal{C}_t^+$, so $\Xi_t = 0$, $V_T^C = 0$, and $\Lambda_T^C = \gamma_T$. Then Theorem 1 gives*

$$R_T \leq 2\sqrt{(\sigma^2 + G^2/\lambda) \gamma_T \sum_{t=1}^T \alpha_t^2 \bar{\beta}_t^2}.$$

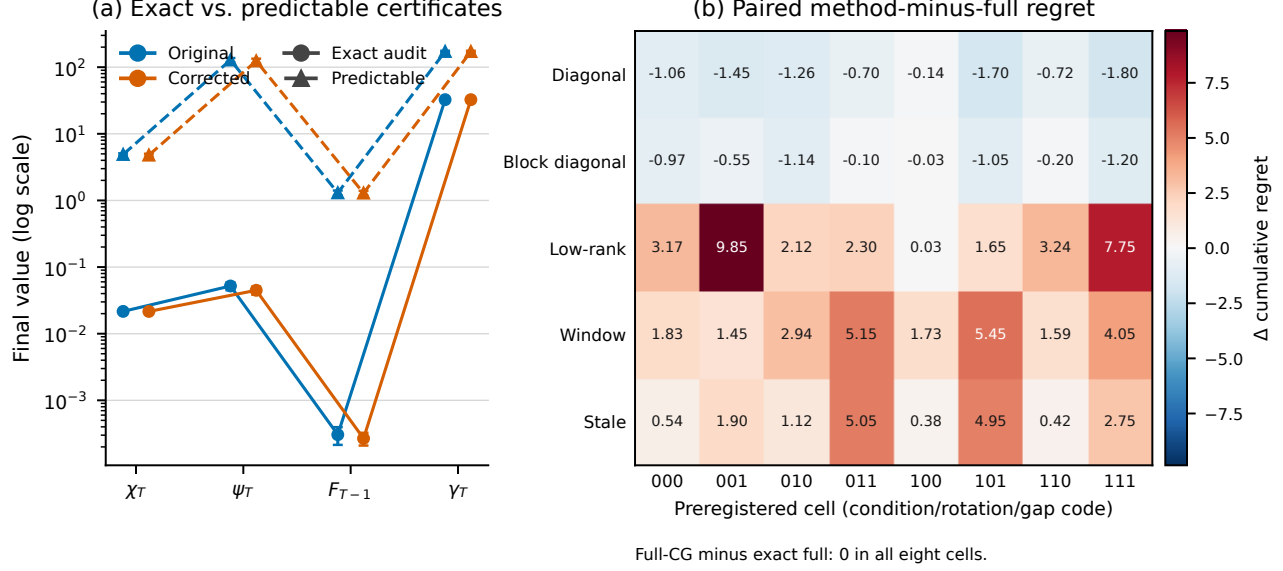


Figure 2: Supplementary certificate-tightness and curvature-mechanism overview. Panel (a) compares final-round float64 audit quantities with predictable schedules in the 50-seed tanh execution. Panel (b) gives paired regret differences in the preregistered curvature phase map. These are descriptive point audits.

If in addition $\bar{\varepsilon}_t \leq \bar{\varepsilon}$ so that $\alpha_t \leq \alpha_I := \sqrt{(1 + \bar{\varepsilon})/(1 - \bar{\varepsilon})}$, the looser Corollary 28 recovers the familiar LinUCB form $R_T \leq 2\alpha_I \bar{\beta}_{\max, T} \sqrt{(\sigma^2 + G^2/\lambda)T\gamma_T}$, where $\bar{\beta}_{\max, T} := \max_{t \leq T} \bar{\beta}_t$.

Proof. Substitute $\varepsilon_{\text{lin}}(t) = 0$, $\bar{\psi}_t = 0$, $u_t(a) = 1$, and $\Lambda_T^C = \gamma_T$, $V_T^C = 0$ into (15); the frozen Gram is monotone, so $\Xi_t = 0$ and the dynamic potential collapses to the information gain γ_T . The LinUCB form is Corollary 28 with $\alpha_t \leq \alpha_I$, $u_t = 1$, $\omega_t = \bar{\beta}_t \leq \bar{\beta}_{\max, T}$. $\square$

C Executed-Policy Audit Protocols

Interpretation and statistics. Primary regret results are generated by online policies executed from scratch. Synthetic evaluators know the teacher only to compute pseudo-regret, optimism failures, and post-hoc theorem-event checks; the action rule never receives it. Hyperparameter selection, where used, is performed on the declared tuning seeds before evaluation. Common random numbers pair methods within an evaluation seed. We report arithmetic means, marginal 95% Student- t intervals, and paired Student- t intervals for method differences. No multiplicity adjustment is applied, so the intervals are descriptive. Scaling-curve bands instead use pointwise percentile bootstrap intervals. Their slope intervals resample whole seed trajectories, reaverage the same five nested prefixes, and refit the specified unweighted OLS model; they measure seed sensitivity conditional on that finite grid, not whether a power law is appropriate. An offline common-trajectory diagnostic uses one logged action and checkpoint sequence for all operators and is never mixed with online regret.

Predictable tanh execution. The analytically bounded tanh instance is $\mu_\theta(x, a) = \tanh(\phi(x, a)^\top \theta)$ with $d = 6$, four actions, $\|\phi\| \leq 0.75$, Gaussian noise $\sigma = 0.20$, teacher radius $S = 0.35$, and an optimizer projected to radius 0.50. We use the global constants $G = 0.75$ and $L_g = L_\mu = 4(0.75)^2/(3\sqrt{3})$, $\lambda = 1$, $\delta = 0.05$, eight clipped gradient steps per round, and a CG energy target 0.05 with the analytic condition bound $1 + (t - 1)G^2/(\lambda\sigma^2)$. At every round the policy records Q_t , R_t^{col} , $\bar{F}_t$, $\hat{\gamma}_t$, $\bar{\chi}_t$, $\bar{M}_t$, $\bar{\psi}_t$, $\bar{\beta}_t$, the optimizer gradient norm, all candidate residuals, and whether any post-hoc exact check fails. No post-hoc field is read by action selection. Seeds 60–69 run a preregistered eight-cell one-factor sensitivity grid; the fixed reference configuration is rerun from scratch on evaluation seeds 160–209 for both centers. The 50-seed means and intervals in Section 5.1 come only from those evaluation runs.

r	Method	Regret slope [95% interval]	R^2	Max. $ \log R - \widehat{\log R} $	Λ slope [95% interval]
4	Exact current	-0.000 [-0.000, 0.000]	1.0000	2.220e-16	0.106 [0.105, 0.107]
4	Full CG	-0.000 [-0.000, 0.000]	1.0000	2.220e-16	0.105 [0.104, 0.106]
4	Window $q = 1/2$	0.851 [0.843, 0.859]	0.9937	8.445e-02	0.465 [0.464, 0.465]
4	Window $q = 2/3$	0.309 [0.292, 0.326]	0.8529	1.550e-01	0.334 [0.333, 0.334]
4	Window $q = 1$	-0.000 [-0.000, 0.000]	1.0000	2.220e-16	0.106 [0.105, 0.107]
4	Frozen	1.000 [1.000, 1.000]	1.0000	4.441e-15	0.106 [0.106, 0.106]
4	Diagonal	1.000 [1.000, 1.000]	1.0000	4.441e-15	0.673 [0.673, 0.673]
4	Greedy	0.000 [-0.000, 0.000]	1.0000	5.551e-17	0.115 [0.114, 0.115]
8	Exact current	0.134 [0.115, 0.153]	0.6887	1.217e-01	0.099 [0.098, 0.100]
8	Full CG	0.141 [0.122, 0.161]	0.6969	1.247e-01	0.099 [0.098, 0.099]
8	Window $q = 1/2$	1.000 [1.000, 1.000]	1.0000	1.066e-14	0.383 [0.383, 0.383]
8	Window $q = 2/3$	0.994 [0.993, 0.995]	1.0000	7.659e-03	0.304 [0.304, 0.304]
8	Window $q = 1$	0.134 [0.116, 0.153]	0.6887	1.217e-01	0.099 [0.098, 0.100]
8	Frozen	1.000 [1.000, 1.000]	1.0000	1.066e-14	0.115 [0.115, 0.115]
8	Diagonal	1.000 [1.000, 1.000]	1.0000	1.066e-14	0.671 [0.671, 0.671]
8	Greedy	0.037 [0.018, 0.057]	0.5542	4.606e-02	0.125 [0.124, 0.126]
16	Exact current	0.818 [0.804, 0.831]	0.9827	1.452e-01	0.117 [0.117, 0.118]
16	Full CG	0.829 [0.815, 0.842]	0.9841	1.418e-01	0.118 [0.117, 0.118]
16	Window $q = 1/2$	1.000 [1.000, 1.000]	1.0000	2.665e-15	0.406 [0.406, 0.406]
16	Window $q = 2/3$	1.000 [1.000, 1.000]	1.0000	2.665e-15	0.278 [0.278, 0.278]
16	Window $q = 1$	0.818 [0.804, 0.831]	0.9827	1.452e-01	0.117 [0.117, 0.118]
16	Frozen	1.000 [1.000, 1.000]	1.0000	2.665e-15	0.125 [0.125, 0.125]
16	Diagonal	1.000 [1.000, 1.000]	1.0000	2.665e-15	0.669 [0.669, 0.669]
16	Greedy	0.273 [0.229, 0.315]	0.9024	1.086e-01	0.140 [0.138, 0.141]

Table 4: Five-point OLS diagnostics for every method at $d = 2048$. Slope brackets are obtained from 2,000 paired resamples of whole seed trajectories. R^2 and maximum absolute log residual describe the fit to the five selected prefix means; they are not tests of a power-law model or asymptotic exponents.

Linear theorem-event audit. The context is $x \in \{-1, +1\}^8/\sqrt{8}$, there are five actions, and $\phi(x, a) = [x, e_a, x \otimes e_a]$ has dimension 53 and norm $\sqrt{3}$. The fixed teacher contains shared, arm-intercept, and context-arm interaction blocks; exact enumeration of all 2^8 contexts confirms that multiple arms are optimal. Noise is $\mathcal{N}(0, 0.25^2)$. Evaluation uses seeds 100–119 after tuning on seeds 0–9. The tuning grid is $\lambda \in \{0.5, 1, 2\}$ and $c_{\text{bonus}} \in \{1, 1.25\}$; a candidate is eligible only if every tuning run passes the recorded confidence and small-scale numerical checks. Fresh evaluation policies use dense full Gram, CG full Gram, diagonal, window 64, rescaled subsample 64, rank-16 Lanczos–Ritz, and refresh period 20. The CG target is 0.05 with at most 106 iterations. Fixed-reference and independently tuned comparisons are both saved.

Linear policy checks and bound scale. The implemented base radius is

$$\beta_t^{\text{base}} = \sqrt{d \log \left(1 + \frac{(t-1)G^2}{d\lambda\sigma^2} \right) + 2 \log(1/\delta) + \sqrt{\lambda} S}, \quad \bar{\beta}_t = c_{\text{bonus}} \beta_t^{\text{base}}, \quad c_{\text{bonus}} \geq 1,$$

with $d = 53$, $G = \sqrt{3}$, $S = \|\theta^*\|_2$, and $\delta = 0.05$; hence tuning $c_{\text{bonus}} \in \{1, 1.25\}$ can only inflate the valid radius. Because $\bar{\psi}_t = 0$, the policy uses $\omega_t = \bar{\beta}_t$. Dense, CG, unrescaled window, and stale-refresh policies use the analytic factor $u_t = 1$. Diagonal, rescaled-subsample, and Lanczos policies materialize dense float64 small matrices before action selection and use $u_t = \max\{1, (1 + 32\epsilon_{\text{mach}})\lambda_{\max}(\bar{C}_t^{-1/2}\mathcal{C}_t\bar{C}_t^{-1/2})\}$. The multiplicative $1 + 32\epsilon_{\text{mach}}$ is a heuristic inflation, not a verified rounding-error enclosure. The CG policy likewise sets $\bar{\kappa}_t$ to a float64 point estimate of $\text{cond}(\mathcal{C}_t)$ before action selection and enforces $\bar{\epsilon}_t = 0.05$ using relative residual target $\bar{\epsilon}_t/\sqrt{\bar{\kappa}_t}$; the other policies have $\bar{\epsilon}_t = 0$. These pre-action computations are deterministic conditional on each seeded operator draw, but predictability does not turn an unenclosed point estimate into an upper certificate. The analytic $u_t = 1$ transfer statements for dense full Gram, unrescaled window, and stale-refresh are unaffected; diagonal, rescaled-subsample, Lanczos, and full-Gram CG remain numerical audits of theorem semantics rather than certified executions. Tuning-run event checks are a selection gate, not the mathematical basis of the evaluation checks. The machine-readable per-policy trace is `results/derived/certification_audit.json`.

Nonlinear drift stress test. The teacher and student are one-hidden-layer width-four tanh networks with four inputs, five outputs, and 45 displacement parameters. They share the initialized backbone, making the exact frozen-head ridge solve a 25-parameter linear regime. Full-network regimes use respectively (learning rate, updates, step cap) = (0.002, 2, 0.005), (0.004, 3, 0.020), (0.008, 4, 0.080). Each original- or corrected-center policy executes for 100 rounds on seeds 110–129 with a fixed, time-only schedule; no evaluation-seed

Study	Policy	Regret [95% CI]	Measured run time (s)	Event fail.	Status
Tanh fixed	Original center	7.86 [7.66, 8.07]	0.094	0/50	Post-hoc verified
Tanh fixed	Corrected center	5.69 [5.46, 5.92]	0.095	0/50	Post-hoc verified
Balanced tuned	CC-UCB (full GGN-CG)	21.94 [20.64, 23.23]	0.966	–	Uncertified
Balanced tuned	Diagonal full network	16.11 [14.62, 17.61]	0.024	–	Uncertified
Balanced tuned	LinUCB	10.58 [9.82, 11.33]	0.015	–	Uncertified
Balanced tuned	LinearTS	12.67 [12.02, 13.31]	0.017	–	Uncertified
Balanced tuned	NeuralLinear	10.01 [9.33, 10.68]	0.014	–	Uncertified
Balanced tuned	NeuralUCB	23.04 [21.81, 24.27]	0.027	–	Uncertified
Balanced tuned	NeuralTS	25.07 [23.40, 26.75]	0.032	–	Uncertified
Balanced tuned	Frozen last layer	7.44 [6.90, 7.99]	0.012	–	Uncertified
Balanced tuned	Greedy	16.26 [14.44, 18.09]	0.017	–	Uncertified
Balanced tuned	UCB1	90.35 [87.95, 92.75]	0.003	–	Uncertified
Balanced tuned	Context-free TS	89.40 [86.53, 92.26]	0.004	–	Uncertified

Table 5: Supplementary executed-policy summary. Tanh rows use one fixed theoretical configuration; balanced rows are selected only on tuning seeds and rerun on disjoint evaluation seeds. Float64 event checks are not verified enclosures, and balanced rows are uncertified performance comparisons. Measured run times are resource accounting under each executed implementation, not a controlled speed or matched-wall-clock comparison.

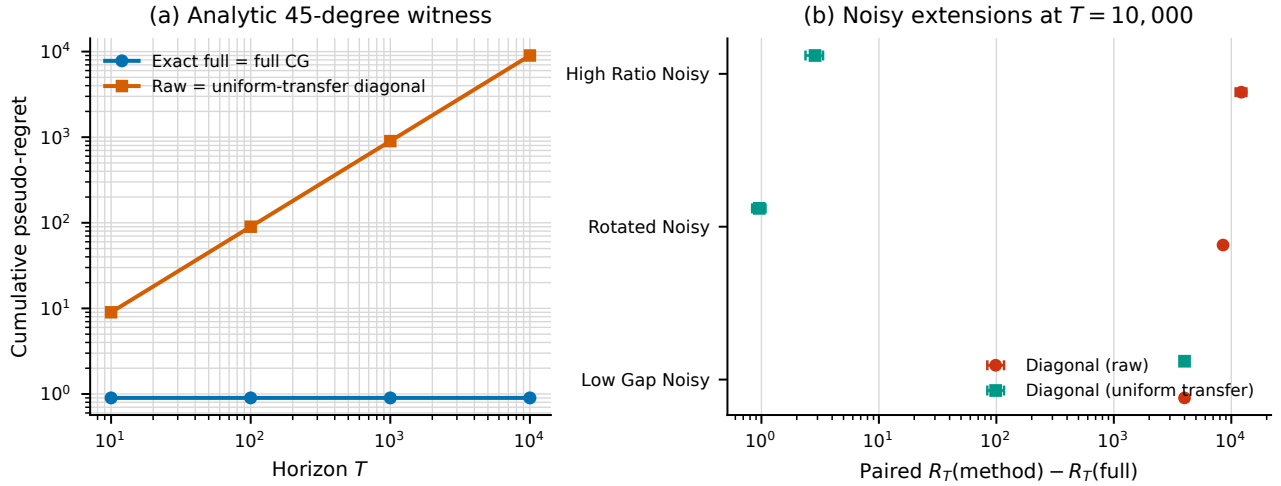


Figure 3: Off-diagonal linear-Gram witness. The analytic cell is constructive; the noisy cells are uncertified extensions. Paired differences are descriptive and establish no uniform ordering of curvature surrogates.

tuning occurs. Dense float64 replay computes E_T, F_T , the dense-reference centering ratio and primitive ψ_t, χ_t , generalized transfer factors, condition numbers, dense-reference and residual-based CG errors, Λ_T^C , endpoint log determinant, $V_T^C, \Gamma_T^{\text{dyn}}$, and all-action optimism failures. Policy-available and post-hoc fields are separated in every JSONL record. The corrected-center identity is checked directly.

A secondary post-hoc analysis across the eight dependent regime-center cell means gives descriptive Spearman correlation 0.83 between the post-hoc decomposition score and regret, versus 0.57 for relative curvature change. With $n = 8$, reused seeds, and no seed-level sensitivity or inferential analysis, this is neither a primary result nor a causal or inferential claim. This stress test is secondary to the predictable tanh execution because its schedules fail.

Preregistered curvature-mechanism map. The fixed-gap bounded-linear family has $d = 10$, four actions, $T = 30$, and Gaussian noise 0.2. Its eight-cell fractional design crosses active-spectrum condition number $\{3, 50\}$, rotation $\{0, 60^\circ\}$, action gap $\{0.15, 0.50\}$, nuisance strength $\{0.4, 1.2\}$, effective rank $\{3, 8\}$, and damping $\{0.5, 2\}$ according to the generators stored in the configuration; all cells are fixed before execution and no tuning is run. Exact full, residual-checked full CG, diagonal, block-diagonal, rank-three Lanczos, window-eight, and refresh-period-five policies execute independently on seeds 200–229. The exact-full trajectory is additionally

Policy	R_{10}	R_{100}	R_{1000}	R_{10000}	Log-log slope
Exact full	0.90	0.90	0.90	0.90	0.000
Residual-checked full CG	0.90	0.90	0.90	0.90	0.000
Diagonal (raw)	9.00	90.00	900.00	9000.00	1.000
Diagonal (uniform transfer)	9.00	90.00	900.00	9000.00	1.000
Diagonal (actionwise reference)	0.90	0.90	0.90	0.90	0.000
Greedy	9.00	90.00	900.00	9000.00	1.000

Table 6: Executed analytic off-diagonal linear-Gram witness by horizon. Every value is generated from the provenance-validated derived artifact; fitted slopes are finite-horizon diagnostics.

Method	$T = 250$	$T = 500$	$T = 1000$
full CG	375.7	438.1	479.3
full dense	361.5	422.4	461.7
diagonal	4120.8	7710.5	13456.0
Lanczos–Ritz	495.7	496.4	509.3
subsample 64	339.7	373.1	396.6
refresh 20	370.3	465.1	526.7
window 64	252.0	255.3	260.2

Table 7: Fixed-reference linear RHS-to-regret ratio RHS/R_T . The JSON artifact also records R_T , RHS, R_T/T , and RHS/T at every horizon for both fixed and tuned configurations. Every displayed RHS exceeds T times the exact maximum one-round pseudo-regret 0.8118 on the finite context support, so the reported RHS values are numerically vacuous. For the four affected policies they are not rigorously certified bounds because their spectral point estimates lack verified upper enclosures.

replayed through every operator. These common-trajectory records contain per-action width ratios, cross-action coefficient of variation, Spearman and Kendall ranks, raw and scalar-rescaled top-score disagreement, normalized decision-margin distortion, candidate-gradient alignment with retained and discarded eigenspaces, global and action-set transfer, effective rank, and condition number. Paired intervals are method-minus-exact-full; they are descriptive and unadjusted across cells.

Balanced contextual benchmark. Normalized Rademacher contexts have dimension four, five available actions, and Gaussian noise 0.10; exact enumeration of the 16 contexts verifies that every arm is optimal somewhere. Each method selects $(\lambda, c_{\text{bonus}})$ from $\{0.5, 1\} \times \{0.5, 1, 2\}$ by mean pseudo-regret over tuning seeds 70–79 at $T = 80$, then runs from scratch for $T = 200$ on seeds 260–289 with shared contexts and reward noise. Full-network methods use the same width-four tanh backbone, one clipped float64 update per round, learning rate 0.01, and model ridge 0.01. The CC-UCB row relinearizes all selected historical samples at the current parameter and solves one fixed full-GGN operator per action with explicit residual checks. The local NeuralUCB row is the historical-time full-network gradient-Gram baseline: it stores each selected gradient at its collection parameter rather than relinearizing it. NeuralLinear is a Bayesian Gaussian head on the frozen initialized representation; the frozen-last-layer row uses the same posterior with a UCB rule. NeuralUCB and NeuralTS are identified as local matched implementations rather than exact reproductions of every published training detail. The one-sample clipped updates do not optimize (3) or certify its full-gradient residual, so these benchmark rows are practical comparisons rather than theorem instantiations. LinUCB, linear Thompson sampling, diagonal full-network UCB, greedy, Gaussian UCB1, and Gaussian context-free Thompson sampling complete the comparison. Arithmetic means, marginal intervals, paired differences, selected hyperparameters, run times, and all seed-level records are retained.

Secondary operator and CG audits. The operator factorial varies windows 16/64, uniform resamples 16/64, Lanczos ranks 4/16, and refresh periods 5/20 in addition to full, frozen, and diagonal operators. On both the bounded linear task and the medium-drift, corrected-center tanh task, twenty online evaluation seeds (120–139) and matching offline common trajectories record exact global and action-set transfer, widths, disagreement, determinant terms, regret, runtime, and memory. The nonlinear policies use the fixed schedule and remain post-hoc audits rather than certified executions. The CG audit uses $d = 64$, $n = 256$, $K = 5$, condition numbers 10/100/1000, five target errors, two initializations, and two preconditioners on seeds 130–139. Its solver audit contains 600,000 dense-reference action solves; its separate policy audit executes 200 full-enumeration linear-bandit

Policy	$T = 200$	$T = 500$	$T = 1000$	$T = 1500$
full GGN-CG	163.3 / 0.183	399.0 / 0.202	794.3 / 0.206	1189.7 / 0.207
frozen Gram	164.0 / 0.180	401.1 / 0.198	797.5 / 0.202	1193.0 / 0.205
diagonal full	166.4 / 0.168	407.7 / 0.185	807.2 / 0.193	1204.4 / 0.197
last-layer full	170.7 / 0.146	420.9 / 0.158	849.4 / 0.151	1273.1 / 0.151
last-layer diagonal	171.1 / 0.144	426.0 / 0.148	851.6 / 0.148	1274.9 / 0.150
greedy	162.3 / 0.189	393.9 / 0.212	781.9 / 0.218	1166.2 / 0.223
UCB1	145.4 / 0.273	332.5 / 0.335	616.2 / 0.384	895.6 / 0.403
Beta-Bernoulli TS	124.7 / 0.376	284.9 / 0.430	544.0 / 0.456	802.6 / 0.465

Table 8: Covertypes evaluation-seed means, shown as cumulative pseudo-regret / accuracy. Contextual hyperparameters are selected only on validation seeds; UCB1 and Thompson sampling are parameter-free. Per-horizon marginal and paired 95% Student- t intervals, runtimes, policy types, and certification status are in `results/derived/covertypes_horizon_results.json`.

policies for 200 rounds with the same five tolerances and start/preconditioner factorial. Unlike the one-step scaling construction, the rotated-SPD right-hand sides exercise nontrivial Krylov convergence. At target energy error 0.1 with zero start and no preconditioner, mean iteration counts are 6.026, 22.643, and 55.043 for condition numbers 10, 10^2 , and 10^3 ; mean measured energy errors are 2.23×10^{-2} , 6.87×10^{-3} , and 2.45×10^{-3} , and mean relative width errors are 2.54×10^{-4} , 2.43×10^{-5} , and 3.28×10^{-6} . The residual certificate is at most the prescribed 0.1 in every retained solve. These solver audits do not establish a method-speed advantage or a policy benefit.

Systems audit. Five evaluation seeds (140–144) cross $d \in \{32, 64, 128\}$, $n \in \{32, 128, 512\}$, $K \in \{5, 10\}$, and $I \in \{5, 15, 30\}$. We measure dense exact, separate full CG, diagonal, final-coordinate block, Lanczos–Ritz, row-batched independent CG, and row-batched Jacobi-PCG. Every action retains its own original-system residual; batched methods share only the operator evaluation. An additional matrix-free CPU grid uses $d \in \{512, 2048, 8192\}$, $n = 32$, $K = 4$, and $I = 16$ for the two batched solvers, with no dense allocation. Records include wall time, host memory, operator calls, equivalent sample CVPs, per-action iterations and residuals, and width error. These are synthetic float64 operator timings on one CPU host and use different computational primitives. The timed operator stores an $n \times d$ random-feature matrix and applies an averaged Gram; it does not time model training or on-the-fly network JVP/VJP. Timings are descriptive resource accounting under the executed implementations, not matched-wall-clock comparisons; no trained-network or accelerator extrapolation is made. The configured 131,841-parameter `torch.func` JVP/VJP benchmark was not run because its declared Buck PyTorch dependency failed to configure.

C.1 Covertypes real-data baseline audit

We use UCI Covertypes DOI 10.24432/C50K5N via scikit-learn 1.9.0; the cached dataset checksum is `eeab5db05bb3a9c00c62ced4c7d13112a00cd3487f59ac1943aaf5983d7c3ddf`. A PCG64 permutation with seed 20260717 creates non-stratified train/validation/test counts 348,607/116,202/116,203. Training means and scales standardize columns 0–9; binary columns 10–53 are unchanged. A width-four tanh network is initialized from seeded fan-in normals and optimized by one online SGD step (learning rate 0.01, ridge 1) using only selected-arm feedback. Full-GGN, frozen-Gram, diagonal, and greedy policies train all 255 parameters; last-layer policies train 35 head parameters. This one-sample update is not the full-history objective (3) and supplies no optimizer-residual certificate. Damping $\{1, 10\}$ and bonus $\{1, 2, 4\}$ are selected on validation seeds 50–59; greedy fixes bonus zero. Evaluation seeds 150–159 use test contexts only after selection. CG uses zero initialization, relative residual 10^{-3} , at most 80 iterations, and one fixed operator per solve. The selection artifact and its validation evidence are SHA-256 committed inside every evaluation manifest.

The exact test-split label counts are (42267, 56737, 7221, 536, 1902, 3444, 4096), giving a label-2 majority rate 0.488258. Uniform random has exact expected accuracy $1/7 = 0.142857$. The fixed majority arm is a whole-test-label oracle diagnostic, not a deployable policy. We add deployable context-free UCB1 (one forced pull per arm, then $\hat{\mu}_a + \sqrt{2 \log t / N_a}$) and independent Beta(1, 1)–Bernoulli Thompson sampling. All eight executed methods share evaluation seeds and context order.

Both non-contextual policies beat every contextual method at every horizon. At $T = 1500$, UCB1 and Thompson

sampling have regret 895.6 and 802.6 (accuracy 0.403 and 0.465), while full GGN-CG has 1189.7 (0.207). Their paired method-minus-full differences are -294.1 $[-388.1, -200.1]$ and -387.1 $[-479.3, -294.9]$. Full and diagonal last-layer policies have regret 1273.1 and 1274.9 (accuracy 0.151 and 0.150). Mean complete-run times at $T = 1500$ are 5.665 seconds for full GGN-CG, 0.602 for diagonal full-network, 0.007 for UCB1, and 0.017 for Thompson sampling. No method is theorem-certified: binary rewards are routed through Gaussian squared-loss curvature. The full-GGN CG calls reach their relative-residual rule, but no energy-error certificate is supplied, so they are not called solver-certified. This study is a failed baseline check, not external validation of curvature-aware exploration.

Environment and artifacts. All reported new runs used CPython 3.14.0 on one arm64 host with 16 logical CPUs, 64 GiB physical memory, Darwin 25.5.0, and no accelerator. Core versions are NumPy 2.5.1, SciPy 1.18.0, matplotlib 3.11.0, psutil 7.2.2, and scikit-learn 1.9.0. Each run directory contains the resolved configuration, seed, UTC timestamp, Git revision and dirty flag, package and hardware record, raw per-round JSONL, and a summary. Derived JSON, CSV, table, and figure artifacts carry SHA-256 sidecars linking them to the aggregate and raw inputs.

Corollary 28 (Looser worst-case reduction). *If $\alpha_t \leq \alpha_{\max}$, $u_t(a_t) \leq u_{\max}$, and $\omega_t \leq \omega_{\max}$ for all $t \leq T$, then $S_T \leq \alpha_{\max}^2 u_{\max} \omega_{\max}^2 T$ and*

$$R_T \leq 2 \alpha_{\max} \omega_{\max} \sqrt{u_{\max}} \sqrt{(\sigma^2 + G^2/\lambda) T \Lambda_T^C} + 2E_T.$$

This $\sqrt{T \Lambda_T^C}$ form recovers the familiar worst-case expression; it is weakly looser than (15) (no tighter, with equality when the per-round factors are constant) and is not the main result.

D Proof of Theorem 1

We recall the key objects: $\mathbf{g}_s(\mathbf{x}, a) = \nabla_{\theta} \mu_{\theta_s}(\mathbf{x}, a)$ (frozen features), $\bar{\mathbf{C}}_t = \lambda \mathbf{I} + \sigma^{-2} \sum_{s < t} \mathbf{g}_s(\mathbf{x}_s, a_s) \mathbf{g}_s(\mathbf{x}_s, a_s)^\top$ (frozen-feature Gram matrix), $\bar{s}_t(a) = \sqrt{\mathbf{g}_t(a)^\top \bar{\mathbf{C}}_t^{-1} \mathbf{g}_t(a)}$, the algorithmic curvature $\mathcal{C}_t$ with width $s_{\mathcal{C},t}(a) = \sqrt{\mathbf{g}_t(a)^\top \mathcal{C}_t^{-1} \mathbf{g}_t(a)}$, the linearized ridge estimator $\hat{\theta}_t^{\text{lin}}$ (Eq. 71), the combined width factor $\omega_t = \bar{\beta}_t + \bar{\psi}_t$, the per-round CG inflation $\alpha_t = \sqrt{(1 + \bar{\varepsilon}_t)/(1 - \bar{\varepsilon}_t)}$, and the realized dynamic complexity Λ_T^C (Eq. 47). Write $\hat{\mu}_t(a) = \mu_{\theta_t}(\mathbf{x}_t, a)$.

Step 1: Confidence, centering, and prediction-error bound. By Lemma 19, with probability $\geq 1 - \delta$,

$$|\mathbf{g}_t(a)^\top (\hat{\theta}_t^{\text{lin}} - \theta^*)| \leq \bar{s}_t(a) \beta_t, \quad \forall t, a, \quad (72)$$

where β_t is as in (63). The self-normalized martingale inequality (Abbasi-Yadkori et al., 2011) is applied to the 1-sub-Gaussian sequence $\xi_s = \eta_s/\sigma$ with the predictable design $\mathbf{g}_s/\sigma$ and Gram matrix $\bar{\mathbf{C}}_t$. Each realized design vector $\mathbf{g}_s(\mathbf{x}_s, a_s)$ is $\mathcal{G}_s$ -measurable (where $\mathcal{G}_s = \mathcal{H}_s \vee \sigma(a_s)$ is the pre-reward sigma-algebra of (69)), and the sequential update (7) ensures the hypotheses of the self-normalized bound are satisfied. By Lemma 22 (Assumption 4), $|\mathbf{g}_t(a)^\top (\theta_t - \hat{\theta}_t^{\text{lin}})| \leq \bar{\psi}_t \bar{s}_t(a)$. Combined with $|\rho_t| \leq \varepsilon_{\text{lin}}(t)$ and $\bar{\beta}_t \geq \beta_t$, the triangle inequality gives the prediction-error bound (64)

$$|\mu_{\theta^*}(\mathbf{x}_t, a) - \hat{\mu}_t(a)| \leq (\beta_t + \bar{\psi}_t) \bar{s}_t(a) + \varepsilon_{\text{lin}}(t) \leq \omega_t \bar{s}_t(a) + \varepsilon_{\text{lin}}(t), \quad \forall a. \quad (73)$$

(The corrected-center variant of Corollary 5 reaches (73) with $\omega_t = \bar{\beta}_t$ and without Assumption 4.)

Step 2: Bonus sandwiching. By Lemma 18 (which uses Lemma 17 with $\max_{a \in \mathcal{A}_t} \varepsilon_{t,a} \leq \bar{\varepsilon}_t$ and the one-sided transfer (40)), for all a ,

$$\omega_t \bar{s}_t(a) \leq w_t(a) \leq \alpha_t \omega_t \sqrt{u_t(a)} s_{\mathcal{C},t}(a). \quad (74)$$

The lower bound and (73) give $\mu^*(\mathbf{x}_t, a) \leq \hat{\mu}_t(a) + w_t(a) + \varepsilon_{\text{lin}}(t)$ for all a : the UCB covers μ^* up to the additive $\varepsilon_{\text{lin}}(t)$ model-mismatch term.

Step 3: Instantaneous regret. From the UCB rule $\hat{\mu}_t(a_t) + w_t(a_t) \geq \hat{\mu}_t(a_t^*) + w_t(a_t^*)$, and using $\mu^*(a_t^*) \leq \hat{\mu}_t(a_t^*) + w_t(a_t^*) + \varepsilon_{\text{lin}}(t)$ (Step 2 at a_t^* , valid since $a_t^* \in \mathcal{A}_t$),

$$\begin{aligned} \mu^*(a_t^*) - \mu^*(a_t) &\leq [\hat{\mu}_t(a_t^*) + w_t(a_t^*) + \varepsilon_{\text{lin}}(t)] - \mu^*(a_t) \\ &\leq [\hat{\mu}_t(a_t) + w_t(a_t) + \varepsilon_{\text{lin}}(t)] - \mu^*(a_t) \quad (\text{UCB selection}) \\ &= w_t(a_t) + [\hat{\mu}_t(a_t) - \mu^*(a_t)] + \varepsilon_{\text{lin}}(t) \\ &\leq w_t(a_t) + \omega_t \bar{s}_t(a_t) + 2\varepsilon_{\text{lin}}(t) \quad ((73) \text{ at } a_t) \\ &\leq 2w_t(a_t) + 2\varepsilon_{\text{lin}}(t) \quad (\text{lower bound in (74)}). \end{aligned}$$

Step 4: Upper bonus bound. The upper bound in (74) gives

$$\mu^*(a_t^*) - \mu^*(a_t) \leq 2\alpha_t \omega_t \sqrt{u_t(a_t)} s_{\mathcal{C},t}(a_t) + 2\varepsilon_{\text{lin}}(t). \quad (75)$$

Step 5: Cauchy–Schwarz after summing. Summing (75) over $t = 1, \dots, T$ and applying Cauchy–Schwarz to the sum (*not* per round),

$$R_T \leq 2 \sum_{t=1}^T \alpha_t \omega_t \sqrt{u_t(a_t)} s_{\mathcal{C},t}(a_t) + 2E_T \leq 2 \sqrt{\sum_{t=1}^T \alpha_t^2 \omega_t^2 u_t(a_t)} \sqrt{\sum_{t=1}^T s_{\mathcal{C},t}^2(a_t)} + 2E_T.$$

Step 6: Dynamic potential and collection. Lemma 14 gives $\sum_{t=1}^T s_{\mathcal{C},t}^2(a_t) \leq (\sigma^2 + G^2/\lambda) \Lambda_T^{\mathcal{C}}$. Substituting,

$$R_T \leq 2 \sqrt{(\sigma^2 + G^2/\lambda) \Lambda_T^{\mathcal{C}} \sum_{t=1}^T \alpha_t^2 u_t(a_t) \omega_t^2} + 2E_T,$$

which is (15). The interpretable and observable versions (27) follow from $\Lambda_T^{\mathcal{C}} \leq \Gamma_T^{\text{dyn}}$ (Lemma 14) and $\Lambda_T^{\mathcal{C}} \leq \widehat{\Lambda}_T^{\mathcal{C}}$ (Corollary 16). The Cauchy–Schwarz step is applied only after summing, so the time-varying factors $\alpha_t, u_t(a_t), \omega_t$ enter through a sum of squares; no horizon maximum is taken before Cauchy–Schwarz, and the usual $\sqrt{T}$ scaling is recovered when the per-round factors are uniformly bounded (Corollary 28). The stated probability follows from a union bound over the self-normalized confidence event $\mathcal{E}_{\text{conf}} (\geq 1 - \delta)$ and the certificate event $\mathcal{E}_{\text{cert}} (\geq 1 - \delta_{\text{cert}})$, giving $\geq 1 - \delta - \delta_{\text{cert}}$. $\square$

E CG Convergence Analysis

Fix a round t and an action a with $\mathbf{g} := \mathbf{g}_t(a) \neq \mathbf{0}$. After $I_{t,a}$ iterations of CG on the SPD system $\mathcal{C}_t \mathbf{u} = \mathbf{g}$ (starting from $\tilde{\mathbf{u}}_0 = \mathbf{0}$), the standard energy-norm bound gives

$$\varepsilon_{t,a} = \frac{\|\tilde{\mathbf{u}}_{I_{t,a}} - \mathcal{C}_t^{-1} \mathbf{g}\|_{\mathcal{C}_t}}{\|\mathcal{C}_t^{-1} \mathbf{g}\|_{\mathcal{C}_t}} \leq 2 \left(\frac{\sqrt{\kappa_t} - 1}{\sqrt{\kappa_t} + 1} \right)^{I_{t,a}},$$

where $\kappa_t = \lambda_{\max}(\mathcal{C}_t)/\lambda_{\min}(\mathcal{C}_t)$ is the true condition number. The bound is right-hand-side independent, so a common iteration budget bounds $\max_a \varepsilon_{t,a}$. This zero-start form is what the fixed-budget certificate (76) uses. For a *warm start* $\tilde{\mathbf{u}}_0 \neq \mathbf{0}$, energy-norm minimization over the shifted Krylov space instead gives

$$\frac{\|\tilde{\mathbf{u}}_{I_{t,a}} - \mathcal{C}_t^{-1} \mathbf{g}\|_{\mathcal{C}_t}}{\|\mathcal{C}_t^{-1} \mathbf{g}\|_{\mathcal{C}_t}} \leq 2q^{I_{t,a}} \frac{\|\tilde{\mathbf{u}}_0 - \mathcal{C}_t^{-1} \mathbf{g}\|_{\mathcal{C}_t}}{\|\mathcal{C}_t^{-1} \mathbf{g}\|_{\mathcal{C}_t}}, \quad q = \frac{\sqrt{\kappa_t} - 1}{\sqrt{\kappa_t} + 1},$$

i.e. the extra factor is the initial relative energy error $\|e_0\|_{\mathcal{C}_t}/\|\mathcal{C}_t^{-1} \mathbf{g}\|_{\mathcal{C}_t}$ (equal to one at $\tilde{\mathbf{u}}_0 = \mathbf{0}$, recovering the zero-start bound). Under the same certified condition-number bound, adaptive residual stopping (line 5) remains valid with a warm start because it certifies the realized residual directly; but the fixed-budget count (76) is valid *as written* only for zero initialization, and a fixed-budget warm-start certificate additionally requires a bound on that initial relative energy error (the zero-start iteration count does not by itself certify a warm-started

solve). Any certified upper bound $\bar{\kappa}_t \geq \kappa_t$ may replace κ_t ; for a nonnegative weighted outer-product operator, $\lambda_{\min}(\mathcal{C}_t) \geq \lambda$ and $\lambda_{\max}(\mathcal{C}_t) \leq \lambda + W_t G^2 / \sigma^2$ (where $W_t = \sum_{s \in \mathcal{S}_t} w_s$) give

$$\kappa_t \leq \frac{\lambda_{\max}(\mathcal{C}_t)}{\lambda} \leq 1 + \frac{W_t G^2}{\lambda \sigma^2} =: \bar{\kappa}_t,$$

whereas a general SPD operator requires a separately certified $\bar{\kappa}_t$.

Two cases at value one must be distinguished. The true condition number $\kappa_t = 1$ forces all eigenvalues equal, i.e. $\mathcal{C}_t = c_t \mathbf{I}$ for some scalar $c_t > 0$, and one CG iteration from $\tilde{\mathbf{u}}_0 = \mathbf{0}$ is *exact* ($\varepsilon_{t,a} = 0$) for every $\mathbf{g} \neq \mathbf{0}$; this does *not* by itself imply $c_t = \lambda$. With the specific certificate used here, $\bar{\kappa}_t = 1$ additionally forces $\lambda_{\max}(\mathcal{C}_t) = \lambda$, which together with $\mathcal{C}_t \succeq \lambda \mathbf{I}$ gives $\mathcal{C}_t = \lambda \mathbf{I}$ exactly. In either reading the single-step solve is exact, so we keep $\bar{\kappa}_t = 1$ as a separate branch and never evaluate the logarithmic budget below at its zero denominator. For $\bar{\kappa}_t > 1$, setting the right-hand side of the rate equal to $\bar{\varepsilon}_{\text{CG}}$ and requiring $\bar{\varepsilon}_{\text{CG}} < 1$ (where $\bar{\varepsilon}_{\text{CG}}$ denotes the round- t target tolerance $\bar{\varepsilon}_t$) gives the per-solve budget

$$I_{\min}(\bar{\kappa}_t, \bar{\varepsilon}_{\text{CG}}) = \left\lceil \frac{\log(2/\bar{\varepsilon}_{\text{CG}})}{\log((\sqrt{\bar{\kappa}_t} + 1)/(\sqrt{\bar{\kappa}_t} - 1))} \right\rceil = O(\sqrt{\bar{\kappa}_t} \log(1/\bar{\varepsilon}_{\text{CG}})), \quad (76)$$

which, applied to every scored action, ensures $\max_a \varepsilon_{t,a} \leq \bar{\varepsilon}_{\text{CG}} = \bar{\varepsilon}_t$. For $W_t = n$ (unrescaled fixed buffer size), $\bar{\kappa}_t \leq 1 + nG^2/(\lambda\sigma^2)$ is bounded independently of T , so $I_{\min}$ is constant. For full history, and for rescaled uniform subsampling with $w_s = (t-1)/n$, we have $W_t = t-1$, so the condition-number *bound* $\bar{\kappa}_t = O(tG^2/(\lambda\sigma^2))$ grows with t , and the worst-case certificate is then obtained with $I_{\min} = \Theta(G\sqrt{t}/(\lambda\sigma^2) \log(1/\bar{\varepsilon}_{\text{CG}}))$ iterations per solve—i.e. $I = O(\sqrt{t} \log(1/\bar{\varepsilon}_{\text{CG}}))$ suffices under this bound (this is an upper-bound certificate, not a lower bound on the number of iterations CG actually needs). CG may converge faster than this worst-case certificate suggests under favorable spectral clustering of $\mathcal{C}_t$; the realized iteration counts would be measured empirically.

F Two-Sided Spectral-Distortion Corollary

The main-text Theorem 24 uses only a one-sided upper transfer factor. For comparison we record the earlier two-sided result, which instead assumes both an upper *and* a lower spectral factor and yields the familiar $\sqrt{\rho_+/\rho_-}$ inflation. The two results trade different assumptions—the corollary below needs a lower spectral bound; the dynamic Theorem 1 needs the dynamic width complexity $\Lambda_T^{\mathcal{C}}$ instead—and neither numerically dominates the other in general.

Corollary 29 (Two-sided spectral-distortion inflation). *Adopt the hypotheses of Theorem 1 with $\mathcal{C}_t = \hat{\mathbf{C}}_t$, retaining the original nonlinear center (so $\omega_t = \bar{\beta}_t + \bar{\psi}_t$; in the convex last-layer regime under exact optimization, $\bar{\psi}_t = 0$ and $\omega_t = \bar{\beta}_t$), and a uniform CG tolerance $\bar{\varepsilon}_t \leq \bar{\varepsilon}$ so that $\alpha_t \leq \alpha_I := \sqrt{(1+\bar{\varepsilon})/(1-\bar{\varepsilon})}$. Suppose predictable factors $\rho_{-,t}, \rho_{+,t}$ are certified before action selection and the two-sided sandwich $\rho_{-,t} \bar{\mathbf{C}}_t \preceq \hat{\mathbf{C}}_t \preceq \rho_{+,t} \bar{\mathbf{C}}_t$ holds for all $t \leq T$ with $0 < \rho_{-,t} \leq \rho_{+,t} < \infty$; set $\rho_- = \min_t \rho_{-,t}$, $\rho_+ = \max_t \rho_{+,t}$, and (to satisfy the convention $u_t \geq 1$ of Assumption 5) take the transfer factor $u_t := \max\{1, \rho_{+,t}\} \leq \max\{1, \rho_+\}$. Write $\omega_{\max,T} := \max_{t \leq T} (\bar{\beta}_t + \bar{\psi}_t)$ and $\rho_+^* := \max\{1, \rho_+\}$. Then, on the intersection of the confidence event $\mathcal{E}_{\text{conf}}$ ($\geq 1 - \delta$) and the certificate event $\mathcal{E}_{\text{cert}}$ (which now also contains the two-sided sandwich event; $\geq 1 - \delta_{\text{cert}}$),*

$$R_T \leq 2\alpha_I \sqrt{\rho_+^*/\rho_-} \omega_{\max,T} \sqrt{(\sigma^2 + G^2/\lambda) T \gamma_T} + 2E_T$$

with probability at least $1 - \delta - \delta_{\text{cert}}$, where γ_T is the frozen-feature information gain (25). Under the standard normalization $\rho_+ \geq 1$ (achievable by rescaling the sandwich so its upper factor is at least one), $\rho_+^* = \rho_+$ and the inflation is the familiar $\sqrt{\rho_+/\rho_-}$.

Proof. This is a *direct* argument through the monotone frozen Gram $\bar{\mathbf{C}}_t$; it does *not* substitute $\hat{s}_t$ for the dynamic width in Corollary 28. *Step 1 (variance sandwich).* Inverting the two-sided sandwich gives $\rho_{+,t}^{-1} \bar{\mathbf{C}}_t^{-1} \preceq \hat{\mathbf{C}}_t^{-1} \preceq \rho_{-,t}^{-1} \bar{\mathbf{C}}_t^{-1}$, so, evaluating the quadratic form at $\mathbf{g}_t(a)$, $\rho_{+,t}^{-1/2} \bar{s}_t(a) \leq \hat{s}_t(a) \leq \rho_{-,t}^{-1/2} \bar{s}_t(a)$. The lower half gives $\bar{s}_t^2(a) \leq \rho_{+,t} \hat{s}_t^2(a) \leq u_t \hat{s}_t^2(a)$, so Assumption 5 holds with $u_t = \max\{1, \rho_{+,t}\}$. *Step 2 (per-round regret).* With $s_{\mathcal{C},t} = \hat{s}_t$, Theorem 1 Steps 1–4 give $\text{regret}_t \leq 2\alpha_t \omega_t \sqrt{u_t} \hat{s}_t(a_t) + 2\varepsilon_{\text{lin}}(t)$; using $\hat{s}_t(a_t) \leq \rho_{-,t}^{-1/2} \bar{s}_t(a_t)$, $u_t = \max\{1, \rho_{+,t}\} \leq \rho_+^*$, $\alpha_t \leq \alpha_I$, and $\omega_t \leq \omega_{\max,T}$,

$$\text{regret}_t \leq 2\alpha_I \omega_{\max,T} \sqrt{\rho_+^*/\rho_-} \bar{s}_t(a_t) + 2\varepsilon_{\text{lin}}(t) \leq 2\alpha_I \omega_{\max,T} \sqrt{\rho_+^*/\rho_-} \bar{s}_t(a_t) + 2\varepsilon_{\text{lin}}(t).$$

Step 3 (frozen-feature elliptic potential). Sum over t and apply Cauchy–Schwarz, then the classical elliptic-potential identity for the *monotone* Gram $\bar{\mathbf{C}}_t$ ((7)), $\sum_t \bar{s}_t^2(a_t) \leq (\sigma^2 + G^2/\lambda) \sum_t \log(1 + \sigma^{-2} \bar{s}_t^2(a_t)) = (\sigma^2 + G^2/\lambda) \gamma_T$:

$$R_T \leq 2\alpha_I \omega_{\max,T} \sqrt{\rho_+^*/\rho_-} \sqrt{T \sum_t \bar{s}_t^2(a_t)} + 2E_T \leq 2\alpha_I \omega_{\max,T} \sqrt{\rho_+^*/\rho_-} \sqrt{(\sigma^2 + G^2/\lambda) T \gamma_T} + 2E_T.$$

The probability statement is the union bound over $\mathcal{E}_{\text{conf}}$ and $\mathcal{E}_{\text{cert}}$. $\square$

For Lemma 13, allocate δ_{cert}/T per round and use the worst-case $m = T-1$ in (46) to obtain the required time-uniform sandwich.

G GGN Approximation Error

Lemma 30 (GGN approximation error). *Suppose $\mu_{\boldsymbol{\theta}}(\mathbf{x}, a)$ is twice differentiable in $\boldsymbol{\theta}$. Suppose $|r_s - \mu_{\boldsymbol{\theta}_t}(\mathbf{x}_s, a_s)| \leq \epsilon_r$ and $\|\nabla^2 \mu_{\boldsymbol{\theta}_t}(\mathbf{x}_s, a_s)\|_{\text{op}} \leq H$ for all $s < t$. Define $\zeta := (t-1)\epsilon_r H/(\sigma^2 \lambda)$. Then:*

- (i) $\|\mathbf{H}_t - \mathbf{C}_t\|_{\text{op}} \leq (t-1)\epsilon_r H/\sigma^2 = \zeta \lambda$.
- (ii) If $\zeta < 1$, then $(1-\zeta)\mathbf{C}_t \preceq \mathbf{H}_t \preceq (1+\zeta)\mathbf{C}_t$.

Proof. For $D_t = \sigma^{-2} \sum_{s < t} (r_s - \mu_{\boldsymbol{\theta}_t}) \nabla^2 \mu_{\boldsymbol{\theta}_t}$, the triangle inequality gives $\|D_t\|_{\text{op}} \leq \zeta \lambda$, proving (i). Combining $-\|D_t\|\mathbf{I} \preceq -D_t \preceq \|D_t\|\mathbf{I}$ with $\mathbf{C}_t \succeq \lambda \mathbf{I}$ proves (ii). $\square$

H Exponential-Family Curvature Definitions

For a regular exponential-family negative log-likelihood with natural parameter $z_{\boldsymbol{\theta}}(\mathbf{x}, a) \in \mathbb{R}^k$ (scalar for Bernoulli, k -dimensional for categorical) and log-partition function A , the GGN/Fisher curvature matrices are

$$\mathbf{C}_t^{\text{exp}} = \lambda \mathbf{I} + \sum_{s=1}^{t-1} J_t(\mathbf{x}_s, a_s)^\top \nabla^2 A(z_t(\mathbf{x}_s, a_s)) J_t(\mathbf{x}_s, a_s), \quad (77)$$

$$\bar{\mathbf{C}}_t^{\text{exp}} = \lambda \mathbf{I} + \sum_{s=1}^{t-1} J_s(\mathbf{x}_s, a_s)^\top \nabla^2 A(z_s(\mathbf{x}_s, a_s)) J_s(\mathbf{x}_s, a_s), \quad (78)$$

where $J_t(\mathbf{x}, a) = \nabla_{\boldsymbol{\theta}} z_{\boldsymbol{\theta}_t}(\mathbf{x}, a) \in \mathbb{R}^{k \times d}$ is the Jacobian of the natural parameter and $\nabla^2 A(z)$ is the output-space Hessian of the log-partition function. For Bernoulli rewards $\nabla^2 A(z) = p(1-p)$ (scalar); for categorical rewards $\nabla^2 A(z) = \text{Diag}(\mathbf{p}) - \mathbf{p}\mathbf{p}^\top$. Setting $k = 1$ and $\nabla^2 A = \sigma^{-2}$ recovers the Gaussian curvature (5)–(6).

I Broader Impact

Exploration can expose users to under-vetted actions and amplify feedback loops. Deployment requires safety constraints, action filters, offline evaluation, and online monitoring; the regret certificates here do not establish safety.

J Data and Code Availability

The repository contains the Buck pipeline, tests, artifact scripts, validated derived results, and retained $d = 128, r = 4$ slice. The 2.1 GB scaling tree and historical inputs outside Git are absent, so not every raw-derived chain is reproducible. Neither anonymous tier is built; the review builder lacks `experiments/data/sklearn/covertime/samples_py3` and `experiments/data/sklearn/covertime/targets_py3`. Once restored, the builder sanitizes identities and validates hashes/provenance. Covertime is CC BY 4.0 (DOI 10.24432/C50K5N) (Blackard, 1998); autodiff remains `not_run`. No current bundle or identifying URL is claimed.